Building Bioinformatics Resources

Make quantitative biological data accessible

Qingyao Huang | UZH BIO390 HS24

Learning objectives

- What comprises a bioinformatics resource?
- What types of data are typically included?
- How to use the resources?
- How to build / maintain a resource?
- What tools / skills are needed?

Primary databases

- Conducted a first, large-scale/break-through experiment to define the reference, standards or host a rich, comprehensive dataset of a topic
- Examples:
  - Ensembl
  - RefSeq
  - TCGA (The cancer genome atlas)
  - HPA (Human Protein Atlas)

Public data repositories

- Collects primary datasets conducted by individual researchers
- Often required by publication, as part of reproducibility, open data effort
- Examples:
  - Gene Expression Omnibus (GEO; NCBI)
  - ArrayExpress (EBI)
  - Proteomics Identification Database (PRIDE; EBI)

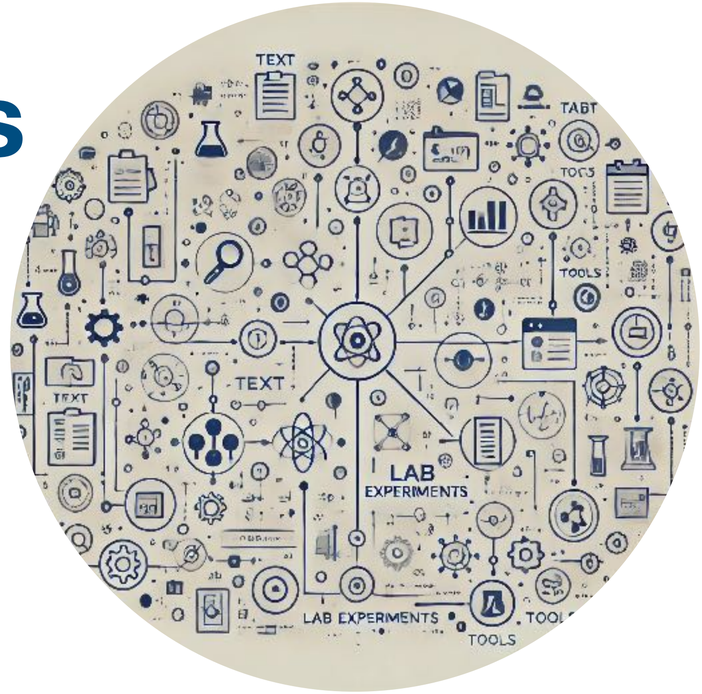
Curated databases

- Codify terms, classifications, based on existing knowledge derived from primary research
- Examples:
  - Kyoto Encyclopedia of Genes and Genomes (KEGG)
  - Gene Ontology (GO)
  - Reactome
  - Most ontologies, e.g.
    - Disease ontology
    - BRENDA tissue ontology
    - Cell Line Ontology …

Bioinformatics online tools

- Web service often with computation support, so the user can access the information through browsing or upload the data for analysis without necessarily possessing programming skills or set up computation locally
- Examples:
  - UCSC genome browser
  - NCBI BLAST (Basic Local Alignment Search Tool)
  - Galaxy (workflow) server
  - Enrichment services

Meta-databases / knowledge-bases



- Integrates many primary datasets and their metadata (type of study, experimental set-up/ replicates, sample conditions…) with text mining techniques
- Examples:
  - Progenetix
  - PaxDb
  - STRING
  - UniProtKB

Progenetix

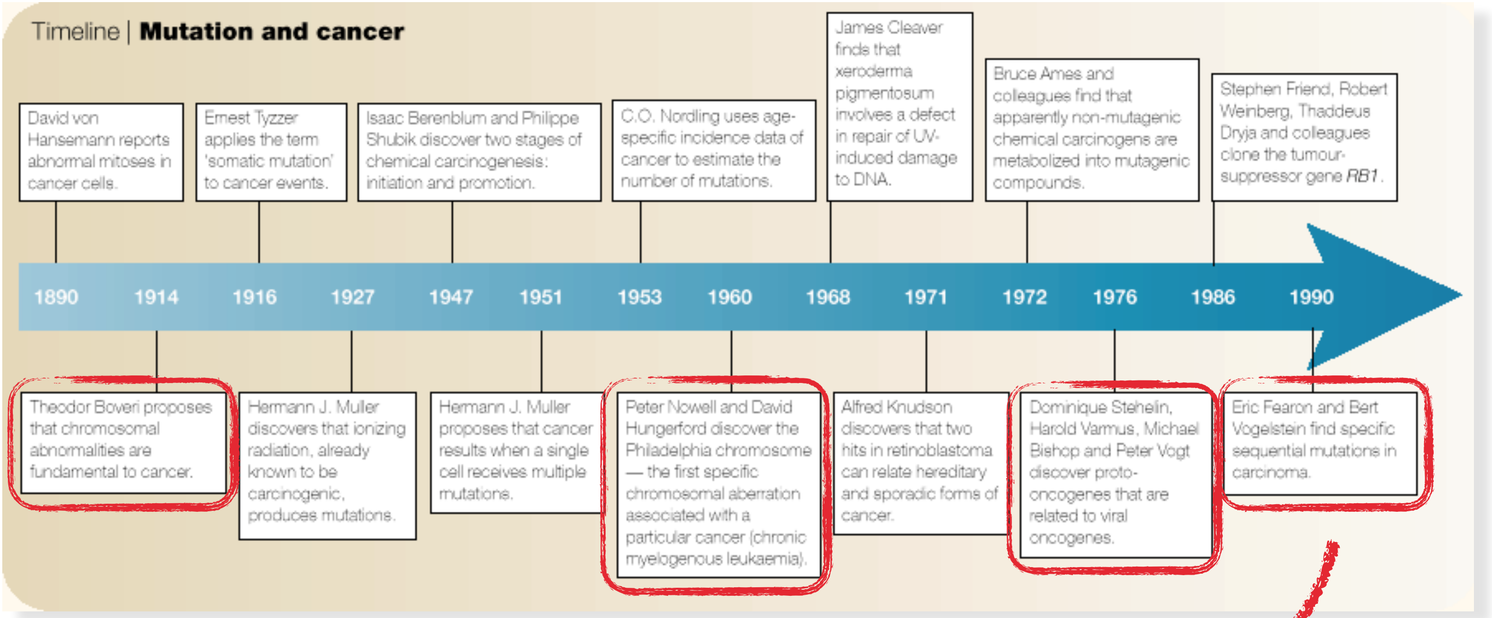
- Motivation for building a DB
  - History
  - Relevance
- What is the quantitative data and how is it represented?
  - Copy number variation
  - Techniques
- What is the metadata?
  - Codify cancer types, stage, patient information
  - Geographical information
- How to access the data safely?
  - Sensitive human data

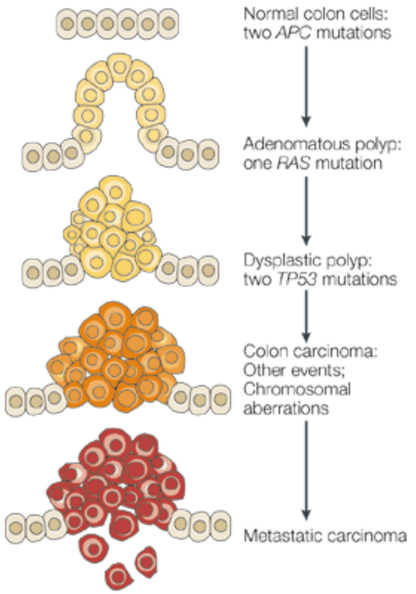
Building a Genomics Resource

journey through time...

- Genomic Copy Number Variations in cancer (CNA / CNV)
- Comparative Genomic Hybridization (CGH) as the original CNV screening technique
- CNVs differ between cancer (sub)types and may correlate to clinical outcome
- single studies are limited in understanding disease-specific changes - **let's build a database**
- databases should be accessible - **let's move online**
- **more data** - data parsers & text mining
- **visualization** - graphics libraries and data formatting
- large datasets - access through **APIs**

progenetix





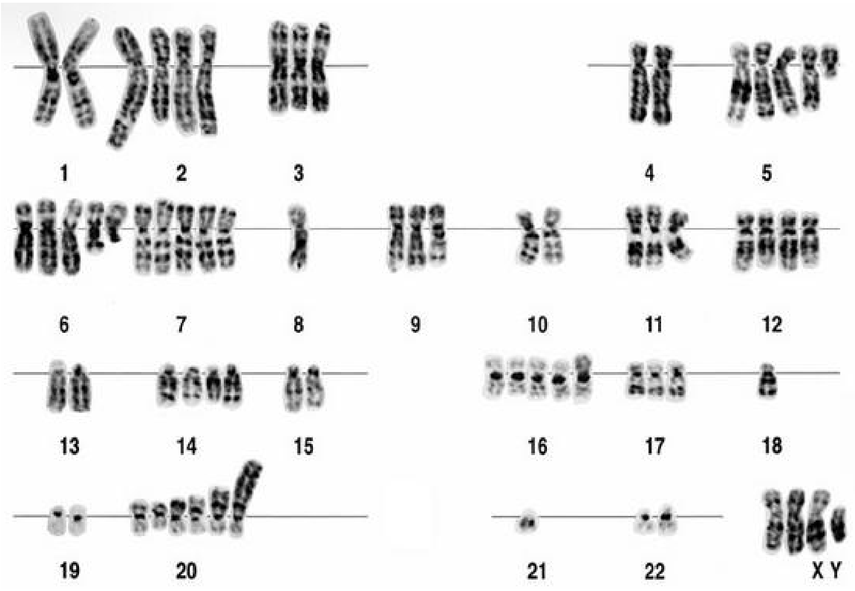
Cancers are based on acquired and inherited genomic mutations

Knudson, A. G. (2001). Two genetic hits (more or less) to cancer. Nature Reviews Cancer, 1(2), 157–162.

Genomic changes at the DNA level are hallmarks of cancer

We inherited 23 paternal and 23 maternal chromosomes, mostly identical.

Normal karyotype



Tumor karyotype

Our goal: identify CN changes to improve characterization, classification, and treatment of cancers

Janet Rowley (1972/73)

Chromosomal translocations in cancer

- Recurrent chromosomal translocations in leukemias /lymphomas
- "Philadelphia chromosome" in CML (Nowell & Hungerford, 1960) represents a reciprocal translocation between chromosomes 9 and 22
- 1972: t(8;21) ALL manuscript rejected by NEJM
- 1973: t(9;22) manuscript rejected by *Nature* "*with some reasonable comments and some truly wrong*"
- Clinical implications: **Tyrosine Kinase inhibitors** as standard first-line therapy in CML
  - first trials in 1998 (STI-571; Imatinib/Gleevec)
  - cf. Druker BJ, Lydon NB (2000). Lessons learned from the development of an Abl tyrosine kinase inhibitor… J Clin Invest 2000;105:3-7)

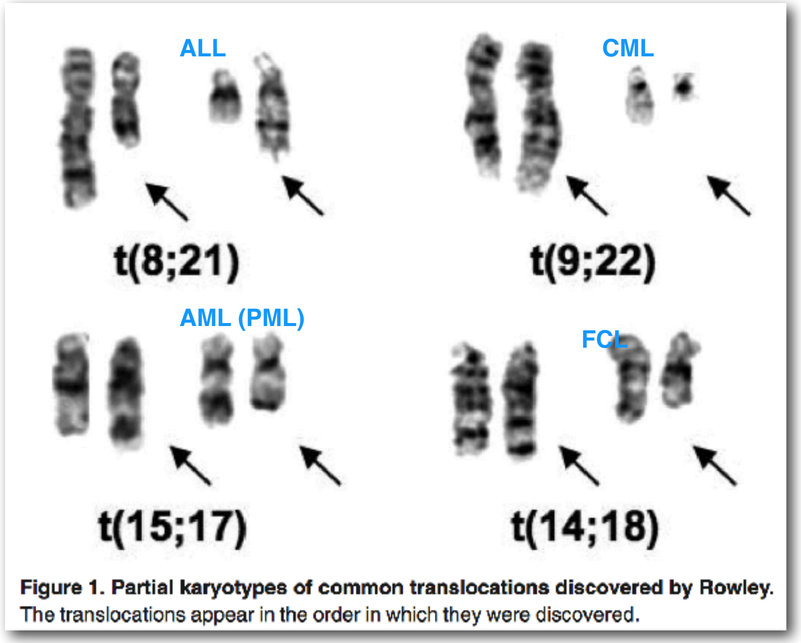
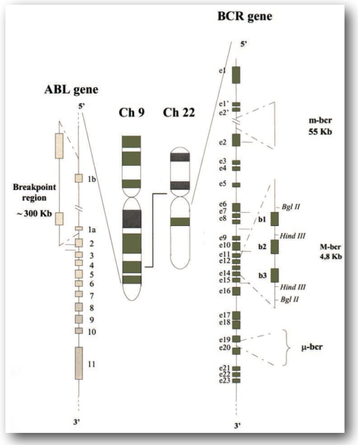
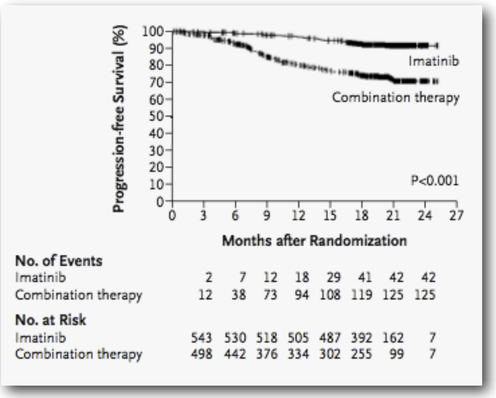


Figure 1. Partial karyotypes of common translocations discovered by Rowley. The translocations appear in the order in which they were discovered.

Janet D Rowley. Chromosomal translocations: revisited yet again Blood (2008), 112(6)



Pane et al. BCR/ABL genes …. Oncogene (2002), 21 (56)



Event free Survival in first large Imatinib Trials

O'Brien et al. Imatinib compared with interferon and low-dose cytarabine… NEJM (2003) vol. 348 (11)

University of Zurich[UZH]

Department of Molecular Life Sciences

Genome screening at the core of “Personalised Health”

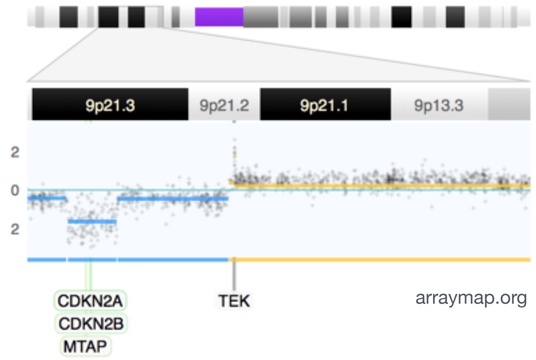
- **Genome analyses** (including transcriptome, metagenomics) are core technologies for Personalised Health™ applications
- The unexpectedly large amount of **sequence variants** in human genomes - germline and somatic/cancer - requires huge analysis efforts and creation of **reference repositories**
- **Standardized data formats** and **exchange protocols** are needed to connect these resources throughout the world, for reciprocal, international **data sharing** and **biocuration** efforts
- Our work @ UZH:
  - ***cancer*** genome repositories
  - biocuration
  - protocols & formats

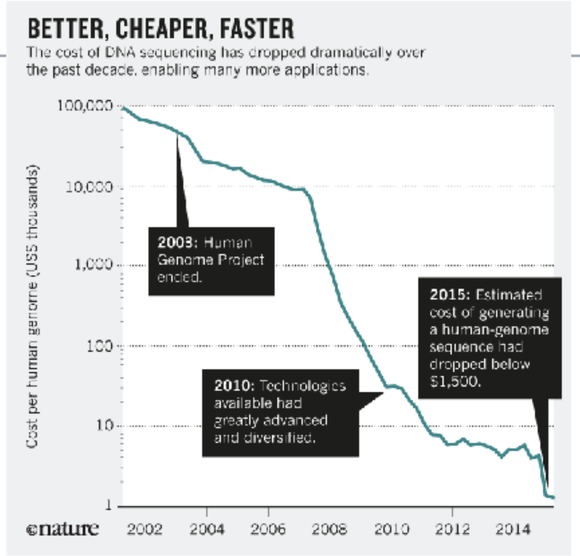
arrayMap

progenetix

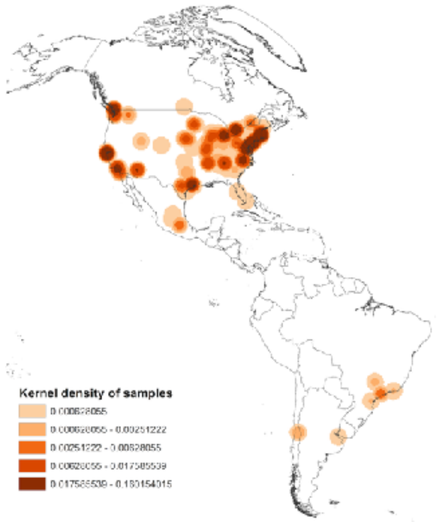
elixir

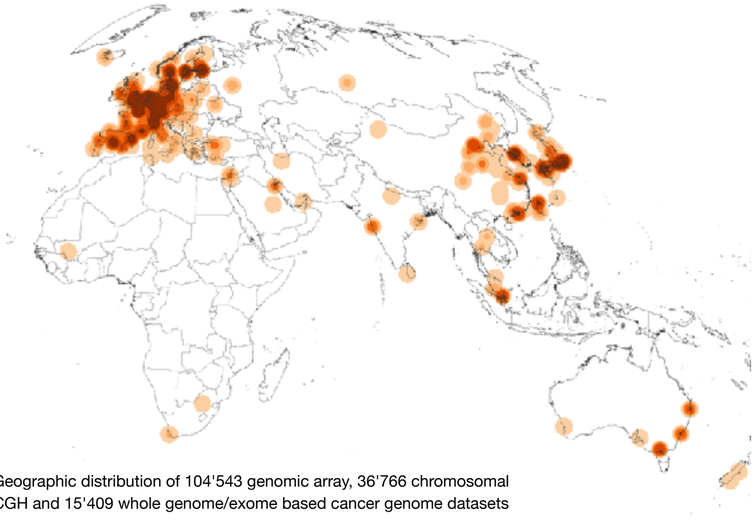
Global Alliance for Genomics & Health





The future of DNA sequencing. Eric D. Green, Edward M. Rubin & Maynard V. Olson. Nature; 11 October 2017 (News & Views)



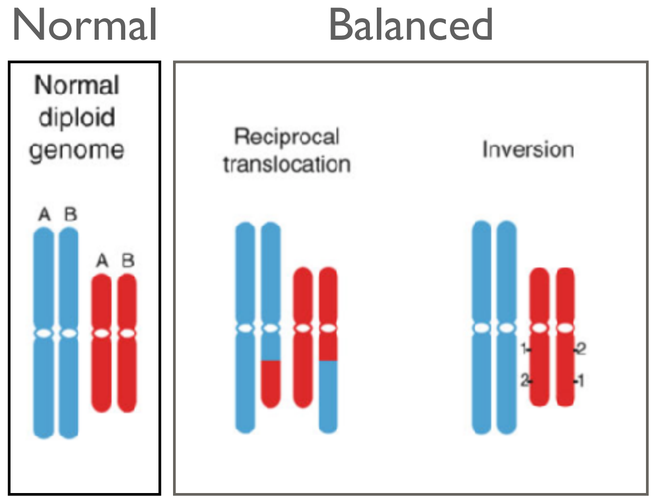


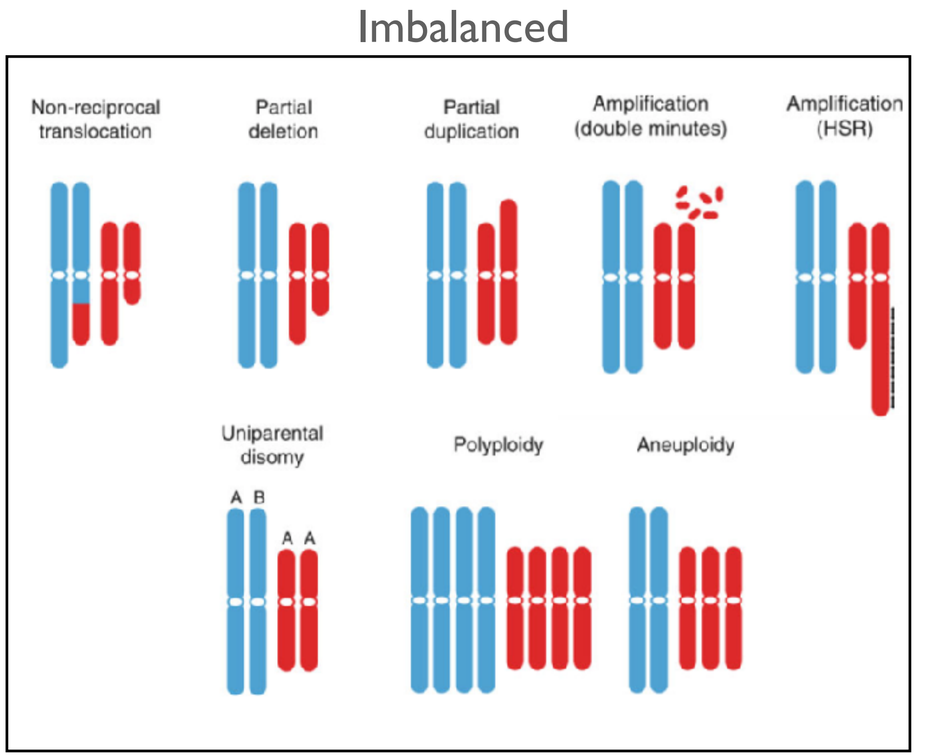
Geographic distribution of 104'543 genomic array, 36'766 chromosomal CGH and 15'409 whole genome/exome based cancer genome datasets

Types of genomic alterations in Cancer

Imbalanced Chromosomal Changes: CNV

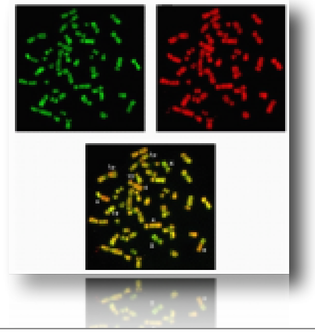
- Point mutations (insertions, deletions, substitutions)
- Chromosomal rearrangements
- Structural chromosomal Aberrations
  - ➡ **Regional Copy Number Alterations** (losses, gains)
- Epigenetic changes (e.g. DNA methylation abnormalities)

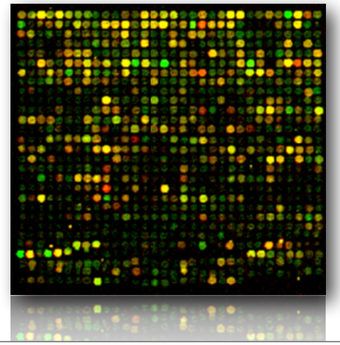


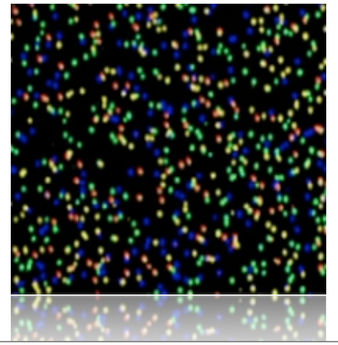


Grade et al., 2015 Recent Results Cancer Res

WHOLE GENOME SCREENING IN CANCER





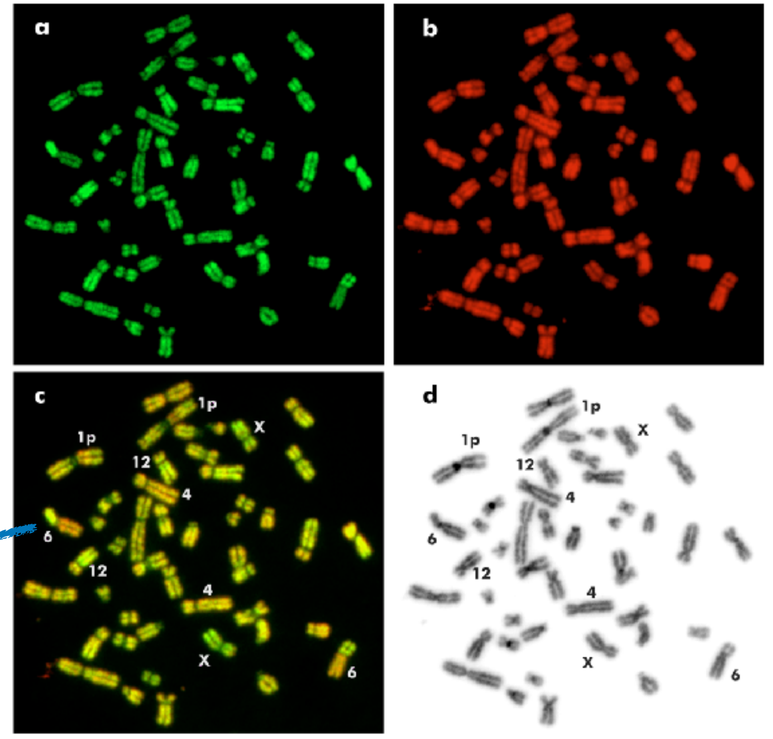


	Chromosomal CGH	Array CGH	"NGS" genome sequencing (WES, WGS)
1st application report	**1992**	**1997**	**2010**
source	DNA (paraffin, micro-dissected ...)	DNA (paraffin, micro-dissected ...)	DNA (paraffin, micro-dissected ...)
main source problems	mixed/degraded source tissue	mixed/degraded source tissue	mixed/degraded source tissue
resolution	chromosomal bands = few megabases	mostly in the 100kb range, but tiling possible	single bases
target identification	surrogate (position)	"semidirect" (segmentation spanning probes)	direct quantitative and qualitative
available data	>24,000 cases (57%) through **Progenetix**	raw data repositories (GEO, EMBL, SMD), **Progenetix**	Limited for raw data (BAMs ...); variant call data in dbgap, clinvar; selected studies with called CNV segments
predominant data format	ISCN = static	raw => depends on bioinformatics	mostly annotated variant calls or SNVs

Chromosomal Comparative Genomic Hybridization (CGH)

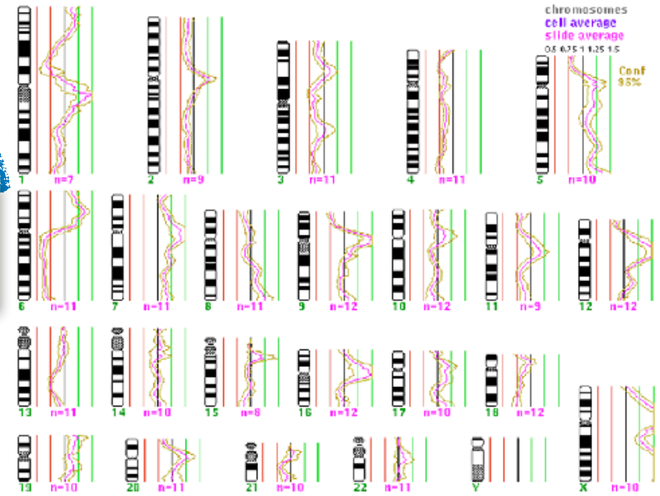
Molecular-Cytogenetic in situ hybridization

- Identify regional genomic copy number variations (CNV/CNA)
- **In situ hybridization** of **genomic** tumor and reference DNA against a karyotypically normal metaphase chromosomes
- analysis of relative fluorescence ratio allows **semi-quantitative copy number** read-out
- **indirect** attribution of involved genes through cytogenetic bands (**megabase resolution**)



CGH-Experiment: **a** Hybridisierung mit Tumor-DNA; **b** Hybridisierung mit normaler menschlicher DNA als Kontrolle; **c** Überlagerung der Signale; **d** Bänderungsfärbung zur Identifizierung der Chromosomen

+6p, -6q...



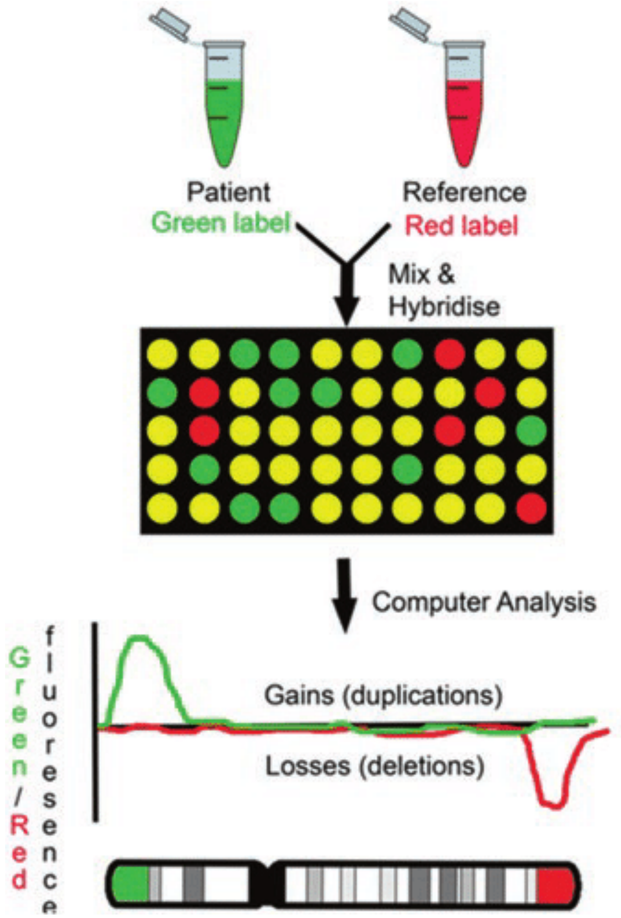
Auswertung: Summationsprofil der computergestützten Analyse mehrerer Metaphasen des dargestellten Falles; die Profilausschläge stehen für Zugewinne bzw. Verluste von chromosomalen Anteilen im Tumorgenom

- Kallioniemi A, Kallioniemi OP, Sudar D, Rutovitz D, Gray JW, Waldman F, Pinkel D. Comparative genomic hybridization for molecular cytogenetic analysis of solid tumors. Science. 1992;5083:818-821.
- Joos S, Scherthan H, Speicher MR, Schlegel J, Cremer T, Lichter P. Detection of amplified DNA sequences by reverse chromosome painting using genomic tumor DNA as probe. Hum Genet. 1993;90:584-589.

Array CGH

Fluorescent microarray with DNA probes

- Quantify ratio of probed DNA between patient and control samples
- Resolution ranges from **1 - 300 kb** on average depending on the platform
- Array probe design
  - cover clinically relevant locations
  - avoid repetitive sequences
  - distribute over whole range of genome



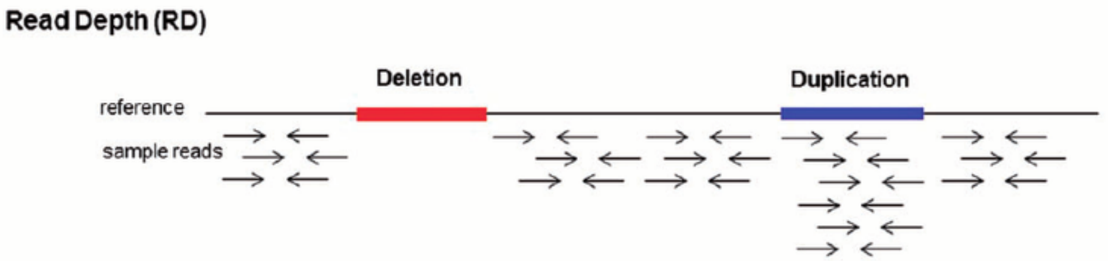
Spot	Patient	Control	Green : Red
	2 copies	2 copies	1.0 : 1.0
	3 copies	2 copies	1.5 : 1.0
	1 copy	2 copies	0.5 : 1.0

Techniques of Chromosomal Studies 2017 10.1007/978-81-322-3673-3_14

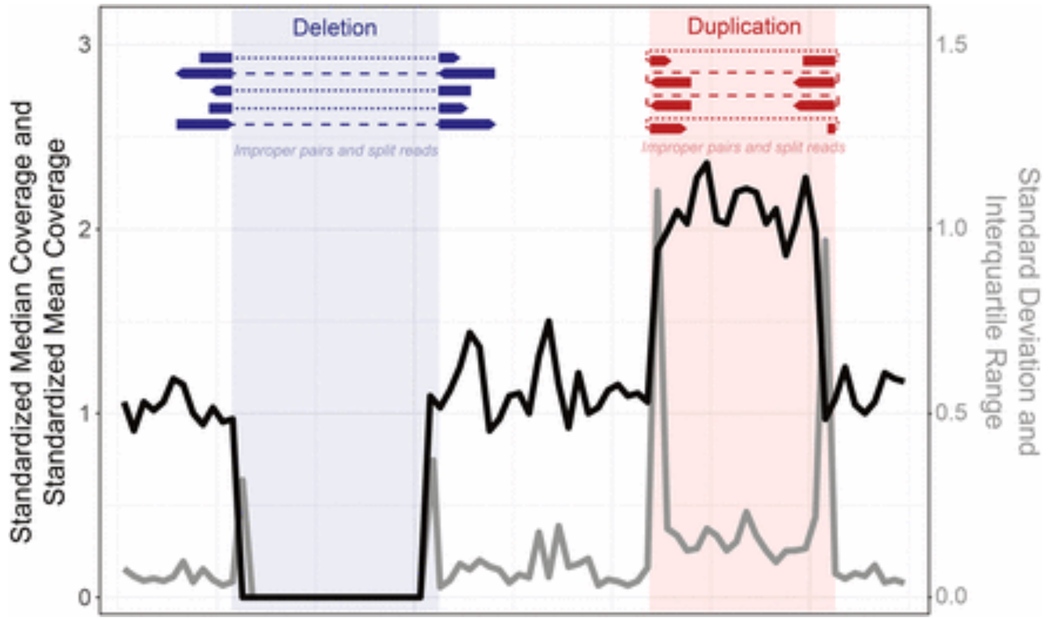
NGS-based method

WES, WGS

- Developed in 2010s
- Single base level detection
- Use read depth to quantify copy number change
- Possible to detect breakpoints
- Not directly standardized comparison, requires normalization

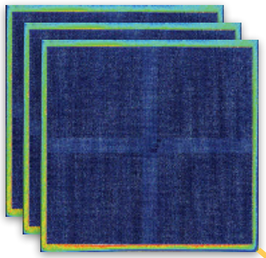


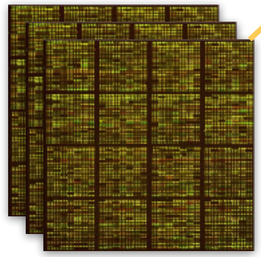
Sirbu et al., Apple Opt. 2016 10.1364/AO.55.006083

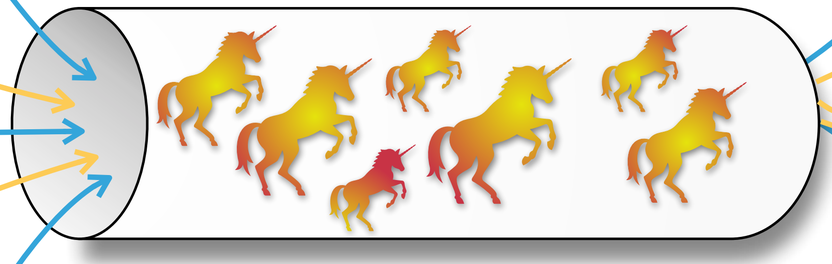


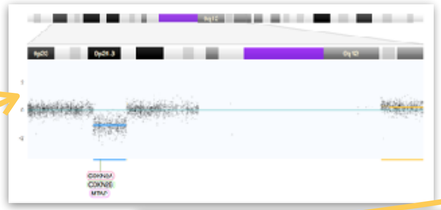
Hill and Uncles, 2019 G3 Genes|Genomes|Genetics 10.1534/g3.119.400596

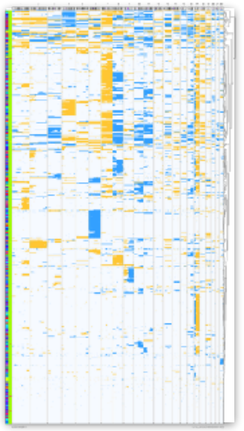
DATA PIPELINE

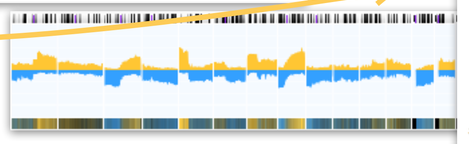


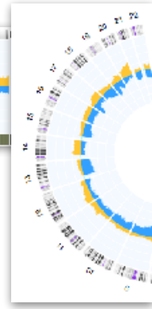


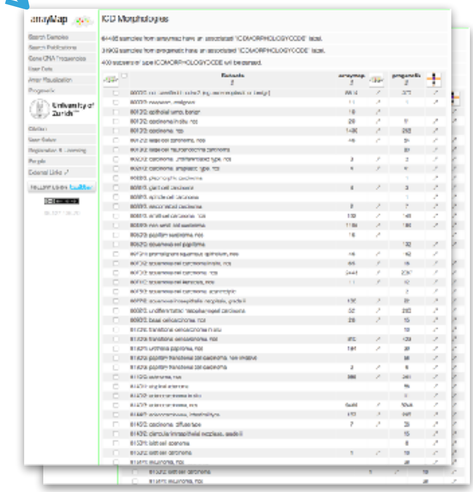








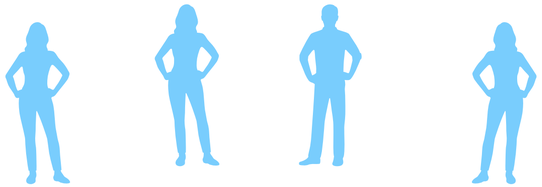




arrayMap

progenetix

DATA PIPELINE



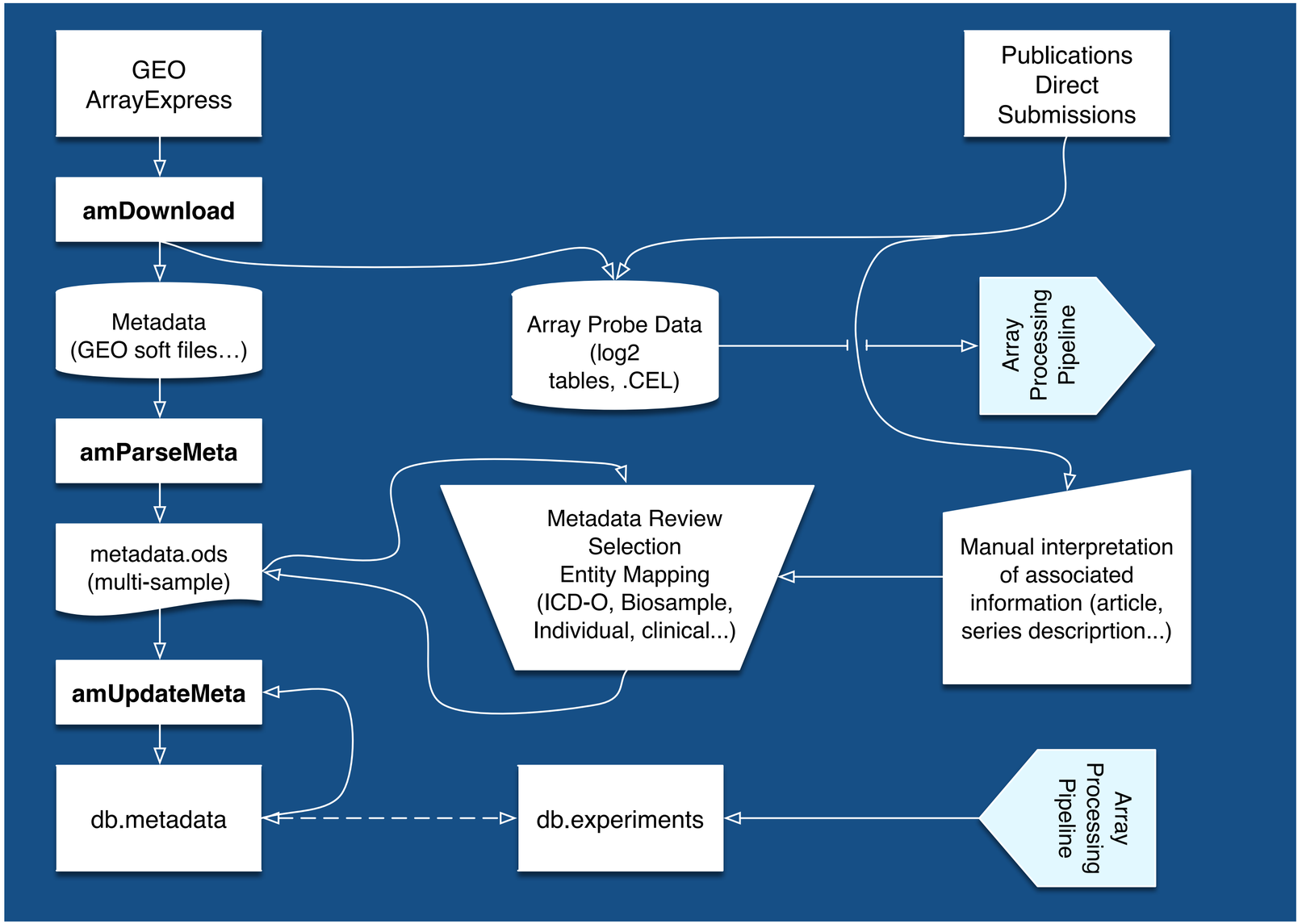
BIOCURATION

BIOINFORMATICS

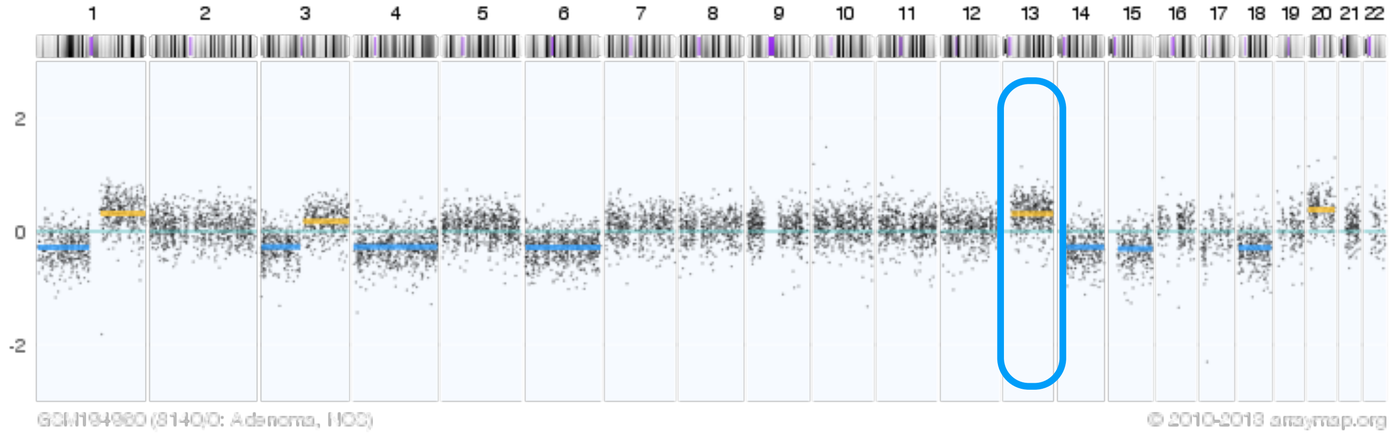
arrayMap

progenetix

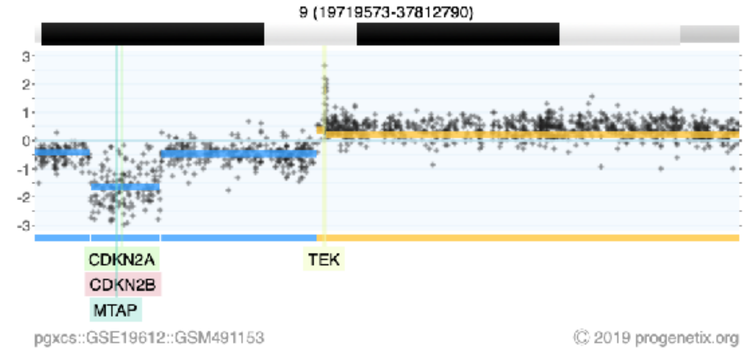
Bioinformatics & Data Curation - arrayMap data “Pipeline”



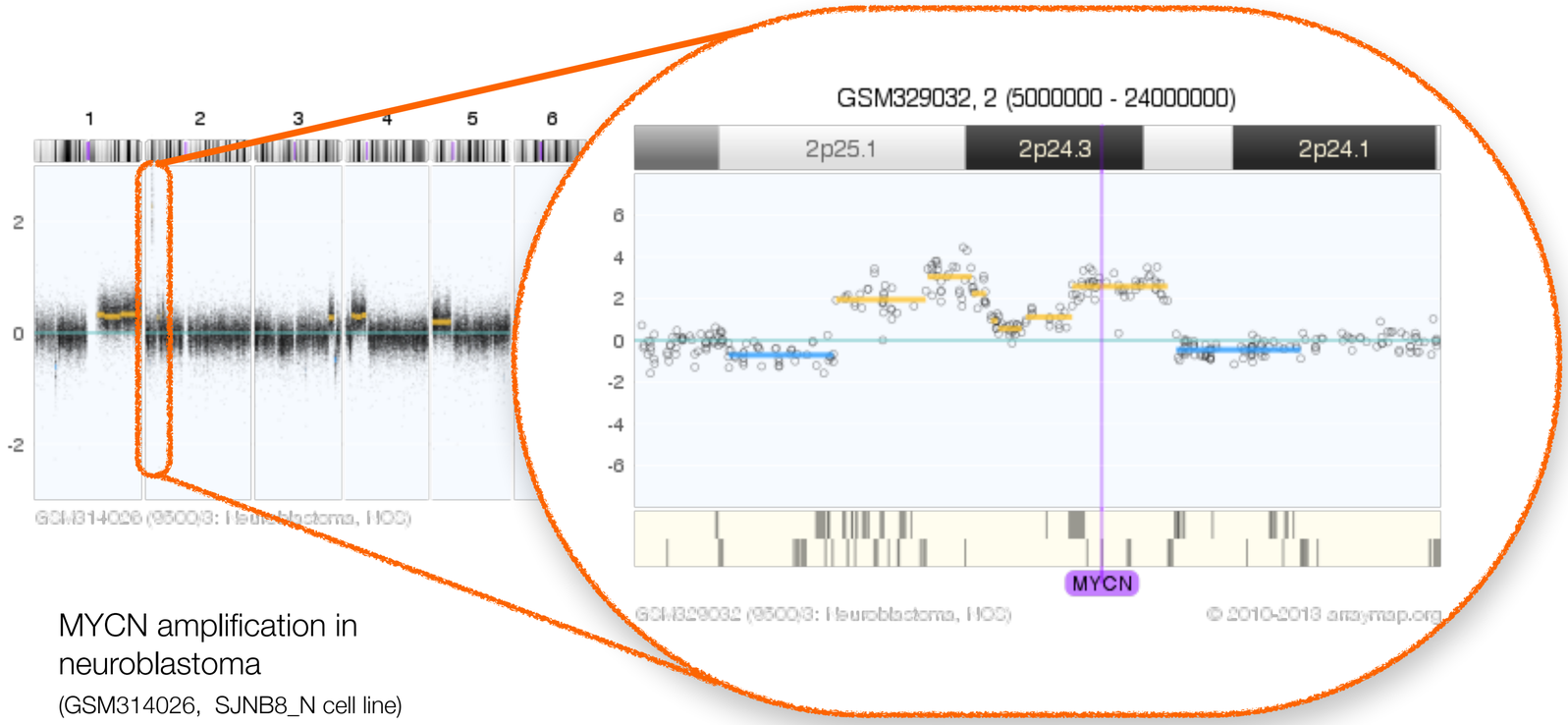
Somatic Copy Number Variations



Gain of chromosome arm 13q in colorectal carcinoma



2-event, homozygous deletion in a Glioblastoma



MYCN amplification in neuroblastoma

(GSM314026, SJNB8_N cell line)

low level/**high level** copy number alterations (CNAs)

arrayMap

What is Metadata?

- Summarize the data in a **structured, machine-readable** way.
- Describe the data using unique **identifiers**, and **controlled vocabularies**.
- **Searchable** in files, ontologies, websites and in registries.
- Essential to **Findable, Accessible, Interoperable and Reusable** (FAIR) bioinformatics.

Adapted from SIB course Metadata in Bioinformatics

Progenetix Metadata Scopes

Biomedical and procedural

- Diagnostic classification
  - mapping text-based cancer diagnoses to standard classification systems
- Provenance data
  - store identifier-based pointers
  - geographic attribution (individual, biosample, experiment)
- Clinical information
  - **core set** of typical cancer study values:

    ➡ stage, grade, followup time, survival status, genomic sex, age at diagnosis
  - balance between annotation effort and expected usability

progenetix

Data sets in tutorials

Data sets in the wild

Data Curation - Happy RegExing!

Extracting clinical and technical metadata from GEO SOFT file

^SAMPLE = GSM174832
!Sample_title = 9194
!Sample_geo_accession = GSM174832
!Sample_status = Public on May 01 2007
!Sample_submission_date = Mar 13 2007
!Sample_last_update_date = Mar 13 2007
!Sample_type = genomic
!Sample_channel_count = 1
!Sample_source_name_ch1 = Bone marrow with 96% blasts
!Sample_organism_ch1 = Homo sapiens
!Sample_taxid_ch1 = 9606
!Sample_characteristics_ch1 = Immunotype: common ALL; Age: 9.2 yrs; Gender: F
!Sample_molecule_ch1 = genomic DNA
!Sample_extract_protocol_ch1 = QiaAmp purification kit (Qiagen)
!Sample_label_ch1 = biotin
!Sample_label_protocol_ch1 = Biotinylated DNA was prepared according to the standard Affymetrix protocol from 250 ng genomic DNA (Genechip Mapping 500k assay manual 701930 Rev.3 or 100k assay manual 701684 Rev.3, Affymetrix).
!Sample_hyb_protocol = Hybridizations were performed according to the standard Affymetrix protocol from 250 ng genomic DNA (Genechip Mapping 500k assay manual 701930 Rev.3 or 100k assay manual 701684 Rev.3, Affymetrix) using an Affymetrix hybridisation oven 640 and an Affymetrix Fluidic station 450.
!Sample_scan_protocol = Scanning performed according to the standard Affymetrix protocol from 250 ng genomic DNA (Genechip Mapping 500k assay manual 701930 Rev.3 or 100k assay manual 701684 Rev.3, Affymetrix) using an Affymetrix scanner 3000.
!Sample_description = primary ALL diagnosis sample
!Sample_data_processing = copy number detection using CNAG2.0 software (http://www.genome.umin.jp/)
!Sample_platform_id = GPL3718
!Sample_contact_name = Roland,P.,Kuiper
!Sample_contact_email = r.kuiper@antrg.umcn.nl, e.verwiel@antrg.umcn.nl
!Sample_contact_phone = +31243610868
!Sample_contact_fax = +31243668752
!Sample_contact_department = Human Genetics
!Sample_contact_institute = Radboud University Nijmegen Medical Centre
!Sample_contact_address = Geert Grooteplein 10
!Sample_contact_city = Nijmegen
!Sample_contact_zip/postal_code = 6525GA
!Sample_contact_country = Netherlands
!Sample_supplementary_file = ftp://ftp.ncbi.nlm.nih.gov/geo/samples/GSM174nnn/GSM174832/suppl/GSM174832.CEL.gz
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progenetix

Data Curation - Happy RegExing!

Extracting clinical and technical metadata from GEO SOFT file

^SAMPLE = GSM174832
!Sample_title = 9194
!Sample_geo_accession = GSM174832
!Sample_status = Public on May 01 2007
!Sample_submission_date = Mar 13 2007
!Sample_last_update_date = Mar 13 2007
!Sample_type = genomic
!Sample_channel_count = 1
!Sample_source_name_ch1 = Bone marrow with 96% blasts
!Sample_organism_ch1 = Homo sapiens
!Sample_taxid_ch1 = 9606
!Sample_characteristics_ch1 = Immunotype: common ALL; Age: 9.2 yrs; Gender: F
!Sample_molecule_ch1 = genomic DNA
!Sample_extract_protocol_ch1 = QiaAmp purification kit (Qiagen)
!Sample_label_ch1 = biotin
!Sample_label_protocol_ch1 = Biotinylated DNA was prepared according to the standard Affymetrix protocol
701684 Rev.3, Affymetrix).
!Sample_hyb_protocol = Hybridizations were performed according to the standard Affymetrix protocol from 2
Rev.3, Affymetrix) using an Affymetrix hybridisation oven 640 and an Affymetrix Fluidic station 450.
!Sample_scan_protocol = Scanning performed according to the standard Affymetrix protocol from 250 ng ge
Affymetrix) using an Affymetrix scanner 3000.
!Sample_description = primary ALL diagnosis sample
!Sample_data_processing = copy number detection using CNAG2.0 software (http://www.genome.umin.jp/)
!Sample_platform_id = GPL3718
!Sample_contact_name = Roland,P.,Kuiper
!Sample_contact_email = r.kuiper@antrg.umcn.nl, e.verwiel@antrg.umcn.nl
!Sample_contact_phone = +31243610868
!Sample_contact_fax = +31243668752
!Sample_contact_department = Human Genetics
!Sample_contact_institute = Radboud University Nijmegen Medical Centre
!Sample_contact_address = Geert Grooteplein 10
!Sample_contact_city = Nijmegen
!Sample_contact_zip/postal_code = 6525GA
!Sample_contact_country = Netherlands
!Sample_supplementary_file = ftp://ftp.ncbi.nlm.nih.gov/geo/samples/GSM174nnn/GSM174832/suppl/GSM174832.CEL.gz
!Sample_supplementary_file = ftp://ftp.ncbi.nlm.nih.gov/geo/samples/GSM174nnn/GSM174832/suppl/GSM174832.CHP.gz

```
foreach (grep { ! /characteristics_ch\d/ } @in) {
    my ($key, $value) = split(' = ', $_);
    $key =~ s/[^\w]/_/g;
    if ($key =~ /submission_date/i) {
        $sample->{ YEAR } = $value;
        $sample->{ YEAR } =~ s/^.*?(\d\d\d\d)$/\1/;
    }
```

```
$mkey->{ samplekey } = 'AGE';
$mkey->{ matches } = [ qw( age )];

( $mkey->{ retv }, $mkey->{ retk } ) = _grepmeta( $mkey, $meta );

if ($mkey->{ retv } =~ /^(.+?)$/) {
    if ($mkey->{ retv } =~ /month/i) {
        $mkey->{ retk } .= '_months';
        $mkey->{ retv } =~ s/[^\d\.]//g;
    }

    $sample->{ $mkey->{ samplekey } } = _normNumber($mkey->{ retv });
    if ($mkey->{ retk } =~ /month/i) { $sample->{ $mkey->{ samplekey } } /= 12 }
    if ($sample->{ $mkey->{ samplekey } } == 0) { $sample->{ $mkey->{ samplekey } } = 'NA' }
    $sample->{ $mkey->{ samplekey } } = sprintf "%.2f", $sample->{ $mkey->{ samplekey } };
}
```

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Data Curation - Happy RegExing!

Extracting clinical and technical metadata from GEO SOFT file

^SAMPLE = GSM286922
!Sample_title = 481 - mAdbID:75320
!Sample_geo_accession = GSM286922
!Sample_status = Public on Sep 04 2008
!Sample_submission_date = May 06 2008
!Sample_last_update_date = Nov 26 2008
!Sample_type = genomic
!Sample_channel_count = 2
!Sample_source_name_ch1 = Normal Lymphocytes
!Sample_organism_ch1 = Homo sapiens
!Sample_taxid_ch1 = 9606
!Sample_characteristics_ch1 = Tissue: lymphocytes
!Sample_molecule_ch1 = genomic DNA
!Sample_extract_protocol_ch1 = Sample DNA Extraction Protocol
!Sample_extract_protocol_ch1 = Other: The DNA was isolated by Qiagen DNeasy Tissue kit accor
!Sample_label_ch1 = cy5
!Sample_label_protocol_ch1 = NimbleGen Cy5 Sample Labeling Protocol
!Sample_label_protocol_ch1 = Other: Proprietary protocol information available at http://www.nimblegen.com/technology/index.html
!Sample_source_name_ch2 = 481
!Sample_organism_ch2 = Homo sapiens
!Sample_taxid_ch2 = 9606
!Sample_characteristics_ch2 = Gender: male
!Sample_characteristics_ch2 = Age: 49
!Sample_characteristics_ch2 = Tissue: lymph node
!Sample_characteristics_ch2 = Disease state: Lymphoma
!Sample_characteristics_ch2 = Individual: 481
!Sample_characteristics_ch2 = Clinical info: Submitting diagnosis: DLBCL
!Sample_characteristics_ch2 = Clinical info: Final microarray diagnosis: ABC DLBCL
!Sample_characteristics_ch2 = Clinical info: Follow up status: ALIVE
!Sample_characteristics_ch2 = Clinical info: Follow up years: 10.75
!Sample_characteristics_ch2 = Clinical info: Chemotherapy: CHOP-Like Regimen
!Sample_characteristics_ch2 = Clinical info: ECOG performance status: 2
!Sample_characteristics_ch2 = Clinical info: Stage: 4
!Sample_characteristics_ch2 = Clinical info: LDH ratio: 0.82
!Sample_characteristics_ch2 = Clinical info: Number of extranodal sites: 1

Channel 1 is normal -> Cave value swap!

Gender or "chromosomal sex"?

context indicates years, but if it would be a medulloblastoma...

Not yet registered way to express "alive"!

```
$mkey->{ samplekey } = 'DEATH';
$mkey->{ matches } = [ (
    'death',
    'dead ',
    'vital_status',
    'dead_alive',
    'alive_dead',
) ];

( $mkey->{ retv }, $mkey->{ retk } ) = _grepmeta( $mkey, $meta );

if ($mkey->{ retv } =~ /^(.+?)$/) {
    $sample->{ $mkey->{ samplekey } } = _normDeath($mkey->{ retv }) }
```

Data Curation

Happy RegExing!

Source: https://xkcd.com/208/

```yaml
extraction_scopes:
  description: >-
    Detection and processing of clinical scopes goes through several stages:
    1. line cleanup - so far run for the input before processing the individual
    scopes
    2. line match using sme general pattern expected in all lines containing
    data for the current scope (`filter` pattern)
    3. finding and extracting the relevant data by looping over a list of
    specific patterns with memorized matches (`find`)
    4. post-processing using empirical cleanp replacements (`cleanup`)
    5. checking the correct structure (`final_check` - a global pattern can be
    used if other post-processing is performed)


  survival_status:
    filter: '(?i).*?(?:(?:dea(?:d|th))|alive|surviv|outcome|status)'
    preclean:
      - m: '(?i)days to death or last seen alive[^\w]+?\d+?(?:[^\w\.]|$)'
        s: ''
      - m: '[^\w]+?NA(?:[^\w\.]|$)'
        s: ' '
      - m: 'remission status past double induction .cr. complete remission. RD. refractory disease. ED. early death[^\w]+?ED'
        s: 'survival: dead'
      - m: 'remission status past double induction .cr. complete remission. RD. refractory disease. ED. early death[^\w]+?NA'
        s: ''
      - m: 'remission status past double induction .cr. complete remission. RD. refractory disease. ED. early death[^\w]+?CR'
        s: 'survival: alive'
      - m: 'remission status past double induction .cr. complete remission. RD. refractory disease. ED. early death[^\w]+?RD'
        s: '' # alive but not responding to therapy so removed?
      - m: 'Event Free Survival[^\w]+?no event'
        s: 'recurrence: no'
      - m: 'Event Free Survival.event'
        s: 'recurrence: yes'
      - m: 'Outcome[^\w]+?no event'
        s: 'survival: alive'
      - m: 'Outcome[^\w]+?event'
        s: 'survival: dead'
      - m: 'survival status[^\w]+?0'
        s: 'survival: dead'
      - m: 'survival status[^\w]+?1'
        s: 'survival: alive'
      - m: 'overall[^\w]+?survival[^\w]+?days[^\w]+?NA'
        s: ''
      - m: 'survival(?: time|from diagnosis)?[^\w]+?(days|months|years?)[^\w]+?(\d\d?\d?\d?\.?\d?\d?)'
        s: 'survival: \2\1'
```

From Classification to Hierarchical Ontology: ICD-O -> NCIt

example_dx	ICDMORPHOLOGY	ICDOM	ICDTOPOGRAPHY	ICDOT	NCIT:CODE
malignant melanoma [metastatic cell line MaMel19]	Malignant melanoma NOS	8720/3	skin	C44	C3224
malignant melanoma [vagina]	Malignant melanoma NOS	8720/3	vagina and labia	C510	C3224
malignant melanoma [uvea metastasized]	Malignant melanoma NOS	8720/3	retina	C692	C3224
meningioma	Meningioma NOS	9530/0	meninges cerebral spinal	C700	C3230
mesothelioma	Mesothelioma NOS	9050/3	lung and bronchus	C34	C3234
pleural mesothelioma	Mesothelioma NOS	9050/3	pleura	C384	C3234
mesothelioma	Mesothelioma NOS	9050/3	connective and soft tissue NOS	C499	C3234
multiple myeloma	Plasma cell myeloma	9732/3	hematopoietic and reticuloendothelial syste	C42	C3242
Mycosis fungoides	Mycosis fungoides	9700/3	skin	C44	C3246
Myelodysplastic syndrome	Myelodysplastic syndrome NOS	9989/3	hematopoietic and reticuloendothelial syste	C42	C3247
Acute myeloblastic leukemia with maturation [FAB M2]	Acute myeloblastic leukemia with maturation [FAB M2]	9874/3	hematopoietic and reticuloendothelial syste	C42	C3250
neuroblastoma	Neuroblastoma NOS	9500/3	peripheral nervs incl. autonomous	C47	C3270
Cerebral neuroblastoma [cerebral region midline frontal lobe]	Neuroblastoma NOS	9500/3	cerebrum	C710	C3270
neuroblastoma [adrenal gland cell line]	Neuroblastoma NOS	9500/3	adrenal gland	C76	C3270
Cutaneous neurofibroma	Neurofibroma NOS	9540/0	skin	C44	C3272
Plexiform neurofibroma	Neurofibroma NOS	9540/0	Nervous system NOS	C729	C3272
Oligodendroglioma [Supratentorial Frontal Lobe]	Oligodendroglioma NOS	9450/3	cerebrum	C710	C3288
oilgodendroglioma	Oligodendroglioma NOS	9450/3	Brain NOS	C719	C3288
oligodendroglioma	Oligodendroglioma NOS	9450/3	brain nos	c719	C3288
Paraganglioma	Paraganglioma NOS	8680/1	Nervous system NOS	C729	C3308
paraganglioma	paraganglioma NOS	8680/1	adrenal cortex	C740	C3308

- Historically classified using the 2 arms of the ICD-O system
  - morphology ~ histology
  - topography ~ organ/tissue)
- mappings between ICD-O code pairs and the NCIt "neoplasm" part of the NCI meta-thesaurus empower **hierarchical** data structures for search and analysis

Standardized Data

Data re-use depends on standardized, machine-readable metadata

- Multiple international initiatives (ELIXIR, GA4GH, MONARCH...) and resource providers (EBI, NCBI ...) work on the generation and implementation of data annotation standards
- emerging / established principles are the use of hierarchical coding systems where individual codes are represented as CURIEs
- other formats for non-categorical annotations based on international standards, e.g.
  - ISO (ISO 8601 time & period, ISO 3166 country codes ...)
  - IETF (GeoJSON ...)
  - W3C (CURIE ...)
- these standards become pervasive throughout GA4GH's ecosystem (e.g. Phenopackets ...)

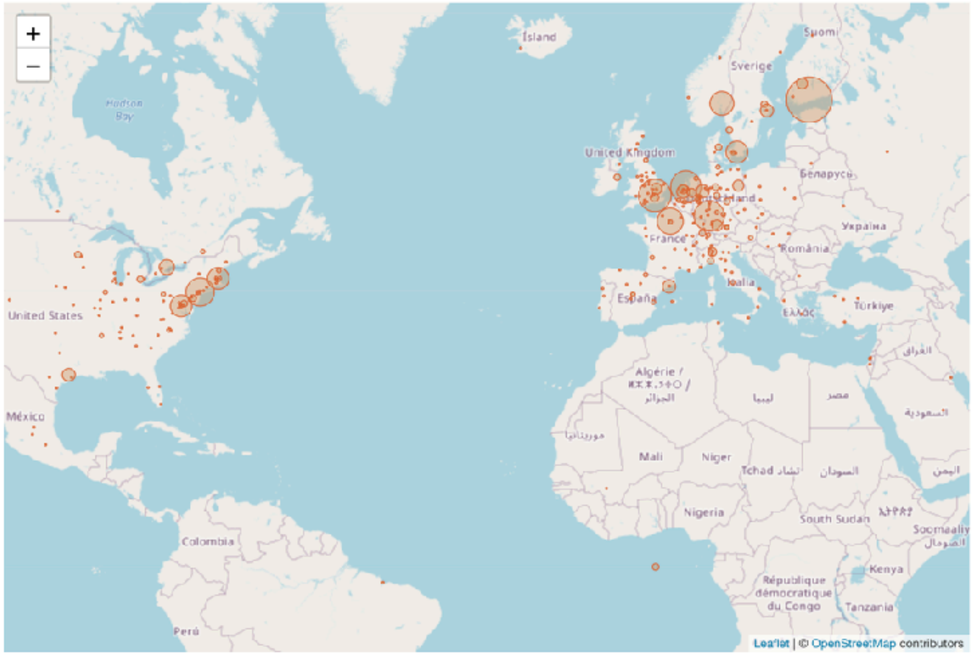

```
"data_use_conditions" : {
    "label" : "no restriction",
    "id" : "DUO:0000004"
},

"provenance" : {
    "material" : {
        "type" : {
            "id" : "EFO:0009656",
            "label" : "neoplastic sample"
        }
    },
    "geo" : {
        "label" : "Zurich, Switzerland",
        "precision" : "city",
        "city" : "Zurich",
        "country" : "Switzerland",
        "latitude" : 47.37,
        "longitude" : 8.55,
        "geojson" : {
            "type" : "Point",
            "coordinates" : [
                8.55,
                47.37
            ]
        },
        "ISO-3166-alpha3" : "CHE"
    }
},
{
"age": "P25Y3M2D"
}
```

Data Curation

Provide "clean and correct data" - but final verification of data from external resources lies with the user ...

- correct data is important for any type of scientific analysis
- errors in formats and values can occur during all steps between data acquisition and analysis (numerous "Excelgates"!)
- "meta"-resources and analyses are prone to erroneous data due to varying input formats and lack of source control

➡ always look for batch effects and outliers!



Geographic distribution (by corresponding author) of the **118554** genomic array, **36766** chromosomal CGH and **42105** whole genome/exome based cancer genome datasets from the **3306** listed publications. Area sizes correspond to the sample numbers reported from a given location.

Progenetix publication collection
progenetix.org/publications/list
2020-11-28

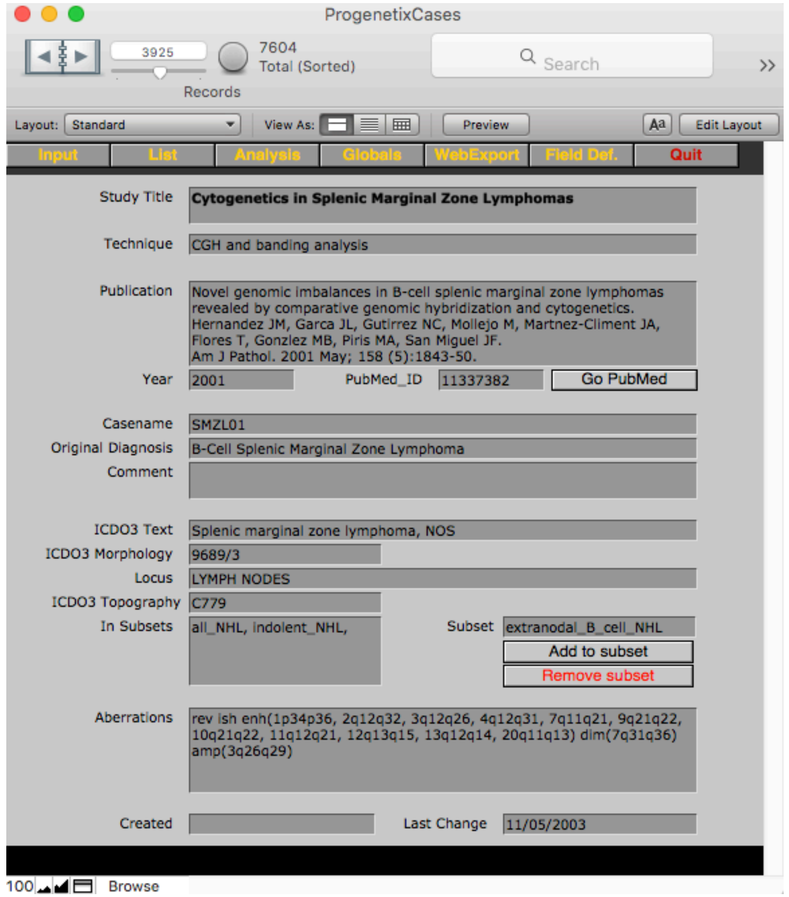
Progenetix in 2024

An oncogenomic reference resource

progenetix

Database Structure

From flat database to hierarchical object storage



Archived version of 2003 "ProgenetixCases" FMP solution

2003

- custom FileMaker database
- text-based annotations
- export & generation of static webpages and data files

2024

- non-SQL document database (MongoDB)
- different object domains connected through identifiers
- data-driven website with JavaScript based frontend and data population through API calls

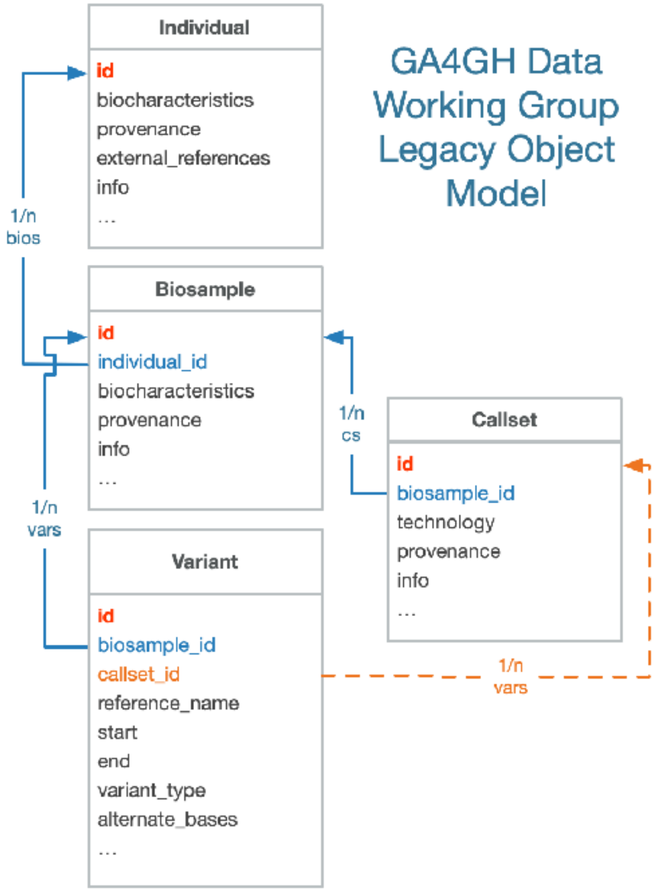

```
{
 "id" : "pgxind-kftx394x",
 "biocharacteristics" : [
  {
   "description" : "female",
   "type" : {
    "id" : "PATO:0020002",
    "label" : "female genotypic sex"
   }
  },
  {
   "description" : null,
   "type" : {
    "id" : "NCBITaxon:9606",
    "label" : "Homo sapiens"
   }
  }
 ],
 "data_use_conditions" : {
  "label" : "no restriction",
  "id" : "DUO:0000004"
 },
 "geo_provenance" : {
  "label" : "Salamanca, Spain",
  "precision" : "city",
  "city" : "Salamanca",
  "country" : "Spain",
  "latitude" : 40.43,
  "longitude" : -3.68
 },
 "info" : {
  "legacy_id" : "PGX_IND_SMZL01"
 },
 "updated" : ISODate("2018-09-26T09:51:39.775Z")
}
```

```
{
 "_id" : ObjectId("5bab583e727983b2e01255ae"),
 "callset_id" : "pgxcs-kftvv618",
 "biosample_id" : "pgxbs-kftvhcao",
 "assembly_id" : "GRCh38",
 "digest" : "7:107200000-158821424:DEL",
 "reference_name" : "7",
 "variant_type" : "DEL",
 "start" : 107200000,
 "end" : 158821424,
 "info" : {
  "cnv_value" : null,
  "cnv_length" : 51621424
 },
 "updated" : "2018-09-26 09:51:39.775397"
}
```

```
{
 "_id" : ObjectId("5bab56cd727983b2e00b0bde"),
 "id" : "pgxbs-kftvhcao",
 "description" : "Splenic Marginal Zone Lymphoma",
 "biocharacteristics" : [
  {
   "type" : {
    "id" : "UBERON:0002106",
    "label" : "spleen"
   }
  },
  {
   "type" : {
    "id" : "icdot-C42.2",
    "label" : "Spleen"
   }
  },
  {
   "type" : {
    "id" : "icdom-96893",
    "label" : "Splenic marginal zone B-cell lymphoma"
   }
  },
  {
   "type" : {
    "id" : "NCIT:C4663",
    "label" : "Splenic Marginal Zone Lymphoma"
   }
  }
 ],
 "individual_id" : "pgxind-kftx394x",
 "individual_age_at_collection" : "P67Y",
 "info" : {
  "death" : "0",
  "followup_months" : 53,
  "callset_ids" : [
   "pgxcs-kftvv618"
  ],
  "legacy_id" : "PGX_AM_BS_SMZL01"
 },
 "external_references" : [
  {
   "type" : {
    "id" : "PMID:11337382"
   }
  }
 ],
 "provenance" : {
  "material" : {
   "type" : {
    "id" : "EFO:0009656",
    "label" : "neoplastic sample"
   }
  },
  "geo" : {
   "label" : "Salamanca, Spain",
   "precision" : "city",
   "city" : "Salamanca",
   "country" : "Spain",
   "geojson" : {
    "type" : "Point",
    "coordinates" : [
     -3.68,
     40.43
    ]
   },
   "ISO-3166-alpha3" : "ESP"
  }
 },
 "data_use_conditions" : {
  "label" : "no restriction",
  "id" : "DUO:0000004"
 }
}
```

Database Structure

From flat database to hierarchical object storage



GA4GH Data Working Group Legacy Object Model

- collections in Progenetix MongoDB database reflect a consensus domain model for genomic data repositories
- flexible linking and object structure facilitates rapid change-overs
- BSON/JSON format in DB
  - equals data in JavaScript
  - "equals" objects in Python, Perl

➡ **rapid prototyping and implementation**

2024

- non-SQL document database (MongoDB)
- different object domains connected through identifiers
- data-driven website with JavaScript based frontend and data population through API calls


```
{
 "id" : "pgxind-kftx394x",
 "biocharacteristics" : [
  {
   "description" : "female",
   "type" : {
    "id" : "PATO:0020002",
    "label" : "female genotypic sex"
   }
  },
  {
   "description" : null,
   "type" : {
    "id" : "NCBITaxon:9606",
    "label" : "Homo sapiens"
   }
  }
 ],
 "data_use_conditions" : {
  "label" : "no restriction",
  "id" : "DUO:0000004"
 },
 "geo_provenance" : {
  "label" : "Salamanca, Spain",
  "precision" : "city",
  "city" : "Salamanca",
  "country" : "Spain",
  "latitude" : 40.43,
  "longitude" : -3.68
 },
 "info" : {
  "legacy_id" : "PGX_IND_SMZL01"
 },
 "updated" : ISODate("2018-09-26T09:51:39.775Z")
}
```

```
{
 "_id" : ObjectId("5bab583e727983b2e01255ae"),
 "callset_id" : "pgxcs-kftvv618",
 "biosample_id" : "pgxbs-kftvhcao",
 "assembly_id" : "GRCh38",
 "digest" : "7:107200000-158821424:DEL",
 "reference_name" : "7",
 "variant_type" : "DEL",
 "start" : 107200000,
 "end" : 158821424,
 "info" : {
  "cnv_value" : null,
  "cnv_length" : 51621424
 },
 "updated" : "2018-09-26 09:51:39.775397"
}
```

```
{
 "_id" : ObjectId("5bab56cd727983b2e00b0bde"),
 "id" : "pgxbs-kftvhcao",
 "description" : "Splenic Marginal Zone Lymphoma",
 "biocharacteristics" : [
  {
   "type" : {
    "id" : "UBERON:0002106",
    "label" : "spleen"
   }
  },
  {
   "type" : {
    "id" : "icdot-C42.2",
    "label" : "Spleen"
   }
  },
  {
   "type" : {
    "id" : "icdom-96893",
    "label" : "Splenic marginal zone B-cell lymphoma"
   }
  },
  {
   "type" : {
    "id" : "NCIT:C4663",
    "label" : "Splenic Marginal Zone Lymphoma"
   }
  }
 ],
 "individual_id" : "pgxind-kftx394x",
 "individual_age_at_collection" : "P67Y",
 "info" : {
  "death" : "0",
  "followup_months" : 53,
  "callset_ids" : [
   "pgxcs-kftvv618"
  ],
  "legacy_id" : "PGX_AM_BS_SMZL01"
 },
 "external_references" : [
  {
   "type" : {
    "id" : "PMID:11337382"
   }
  }
 ],
 "provenance" : {
  "material" : {
   "type" : {
    "id" : "EFO:0009656",
    "label" : "neoplastic sample"
   }
  },
  "geo" : {
   "label" : "Salamanca, Spain",
   "precision" : "city",
   "city" : "Salamanca",
   "country" : "Spain",
   "geojson" : {
    "type" : "Point",
    "coordinates" : [
     -3.68,
     40.43
    ]
   },
   "ISO-3166-alpha3" : "ESP"
  }
 },
 "data_use_conditions" : {
  "label" : "no restriction",
  "id" : "DUO:0000004"
 }
}
```

Progenetix in 2024

Cancer Genomics Reference Resource

- largest open resource for curated cancer genome profiles
- focus on copy number variations (CNV)
- >116'000 cancer CNV profiles, from >800 NCIt codes
- majority of data from genomic arrays with ~50% overall from SNP platforms with original data re-processing
- structured diagnostic encodings for NCIt, ICD-O 3, UBERON
- identifier mapping for PMID, GEO, Cellosaurus, TCGA, cBioPortal where appropriate
- core biosample and technical metadata annotations where accessible (TNM, genotypic sex, survival ...)
- publication database and code mapping services

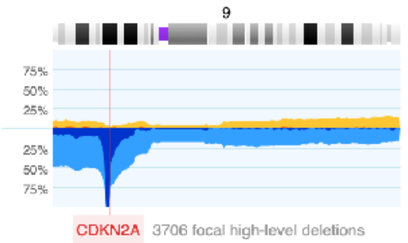
progenetix

Cancer genome data @ progenetix.org

The Progenetix database provides an overview of mutation data in cancer, with a focus on copy number abnormalities (CNV / CNA), for all types of human malignancies. The data is based on *individual sample data* of currently **136468** samples from **834** different cancer types (NCIt neoplasm classification)

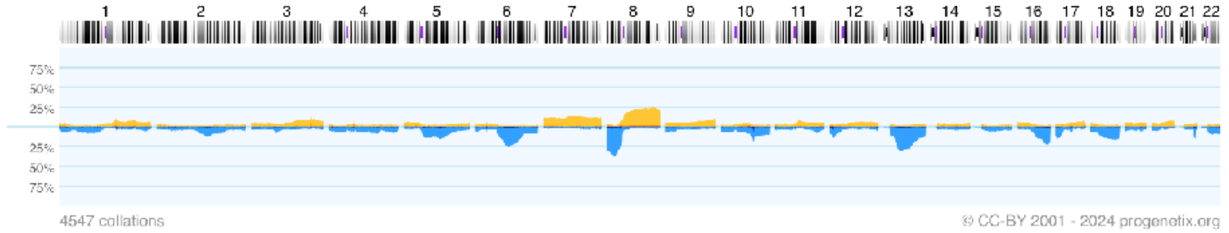
Local CNV Frequencies

A typical use case on Progenetix is the search for local copy number aberrations - e.g. involving a gene - and the exploration of cancer types with these CNVs. The [Search Page] provides example use cases for designing queries. Results contain basic statistics as well as visualization and download options.



Cancer CNV Profiles

Frequency profiles of regional genomic gains and losses for all categories (diagnostic entity, publication, cohort ...) can be accessed through the respective Cancer Types pages with visualization and sample retrieval options. Below is a typical example of the aggregated CNV data in 4547 samples in Malignant Male Reproductive System Neoplasm with the frequency of regional copy number gains (**high level**) and losses (**high level**) displayed for the 22 autosomes.



Download SVG | Go to NCIT:C8561 | Download CNV Frequencies

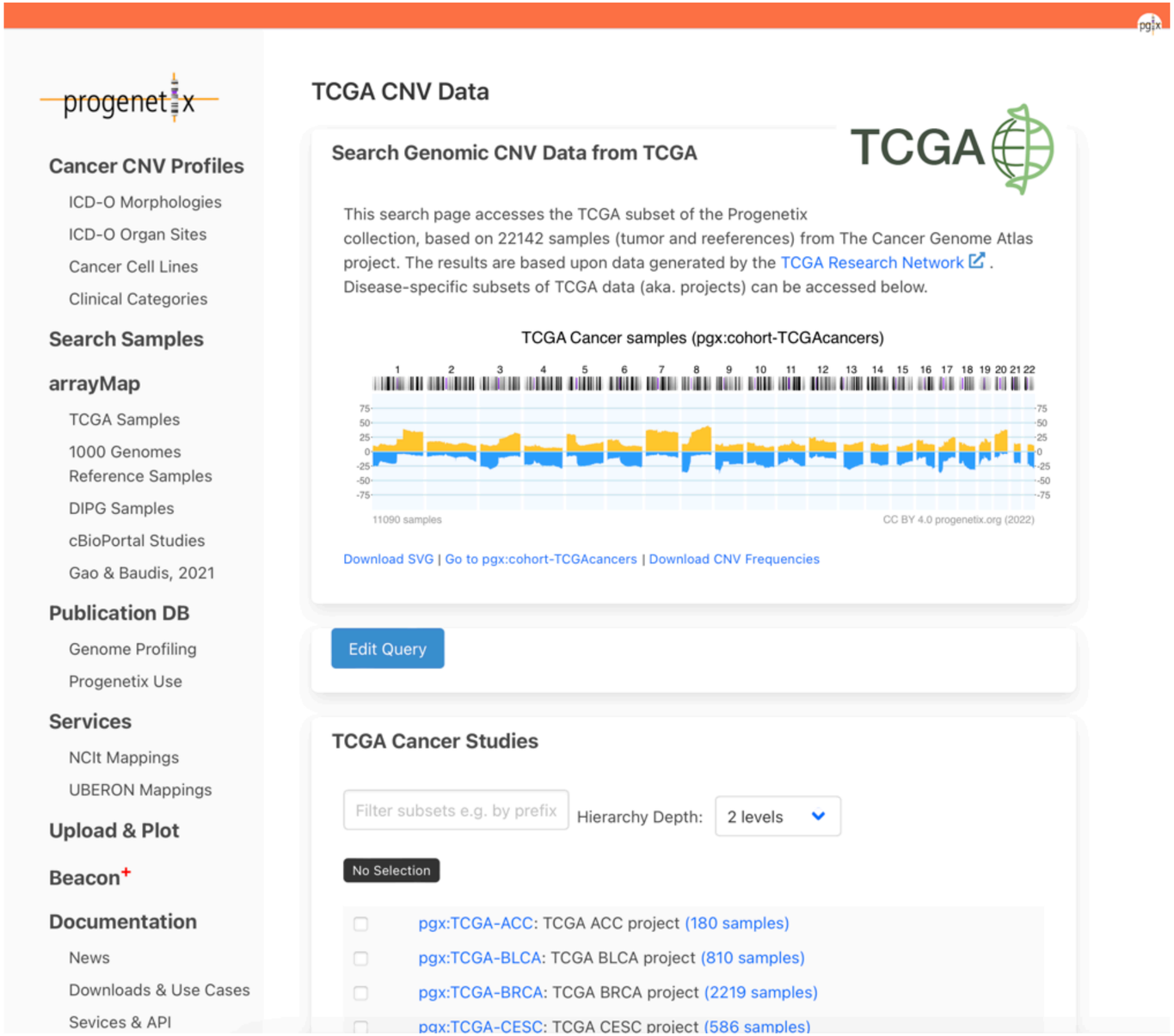
Cancer Genomics Publications

Through the [Publications] page Progenetix provides annotated references to research articles from cancer genome screening experiments (WGS, WES, aCGH, cCGH). The numbers of analyzed samples and possible availability in the Progenetix sample collection are indicated.

Progenetix in 2024

Cancer Genomics Reference Resource

- contains special data subsets, identified using the "cohorts" concept
  - TCGA CNV data
  - 1000Genomes germline CNVs (WGS)
  - Cancer cell line CNVs with upcoming addition of annotated SNV ... data
  - cBioPortal studies
  - ...

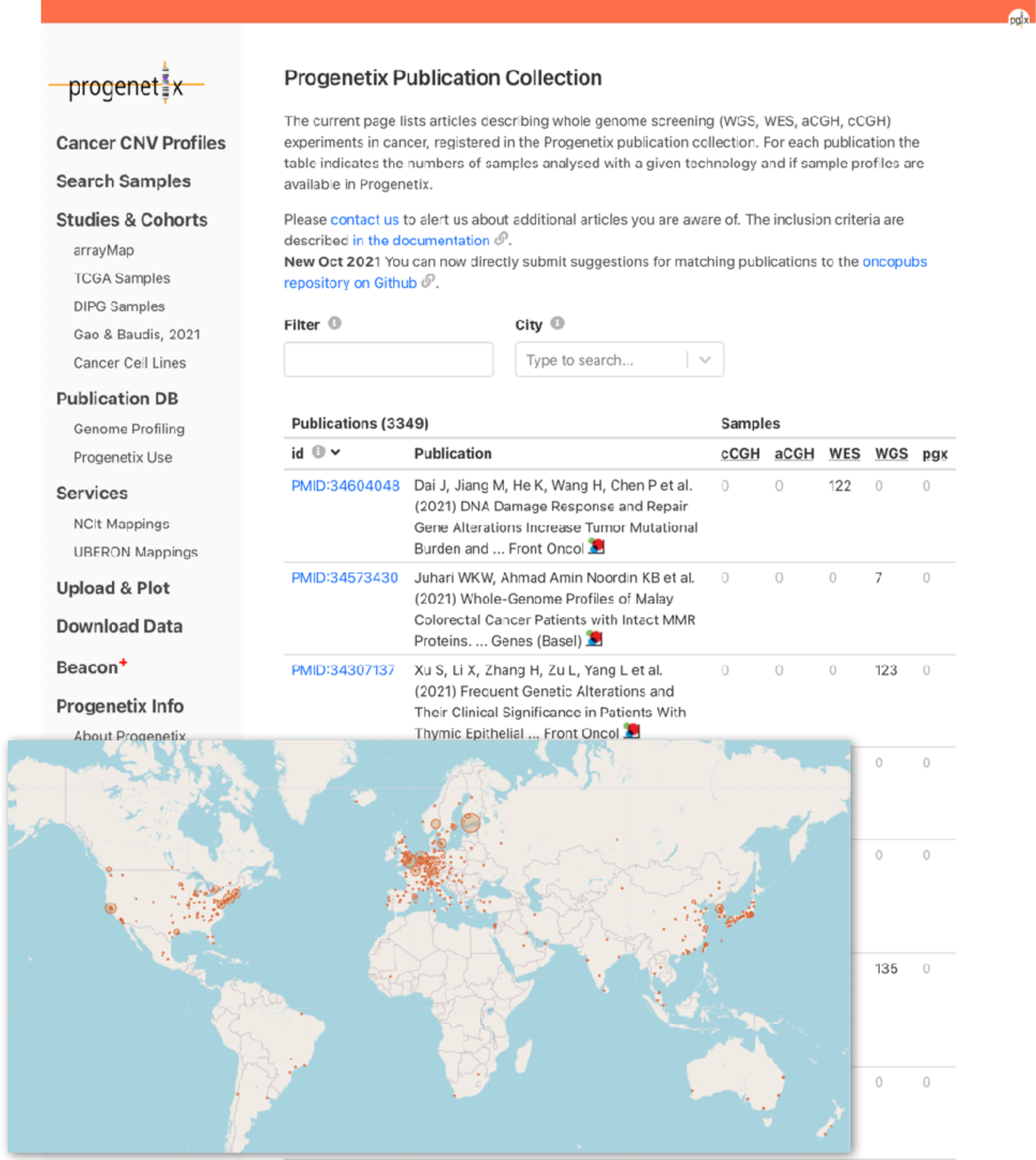


progenetix

Progenetix in 2024

Cancer Genomics Reference Resource

- open resource for curated oncogenomic profiles
- >116'000 cancer CNV profiles, from >800 types
- majority of data from genomic arrays with ~50% overall from SNP platforms with original data re-processing
- standardized encodings (e.g. NCIt, ICD-O 3)
- identifier mapping for PMID, GEO, Cellosaurus, TCGA, cBioPortal where appropriate
- core biosample and technical metadata where accessible (TNM, sex, survival ...)
- publication database and code mapping services

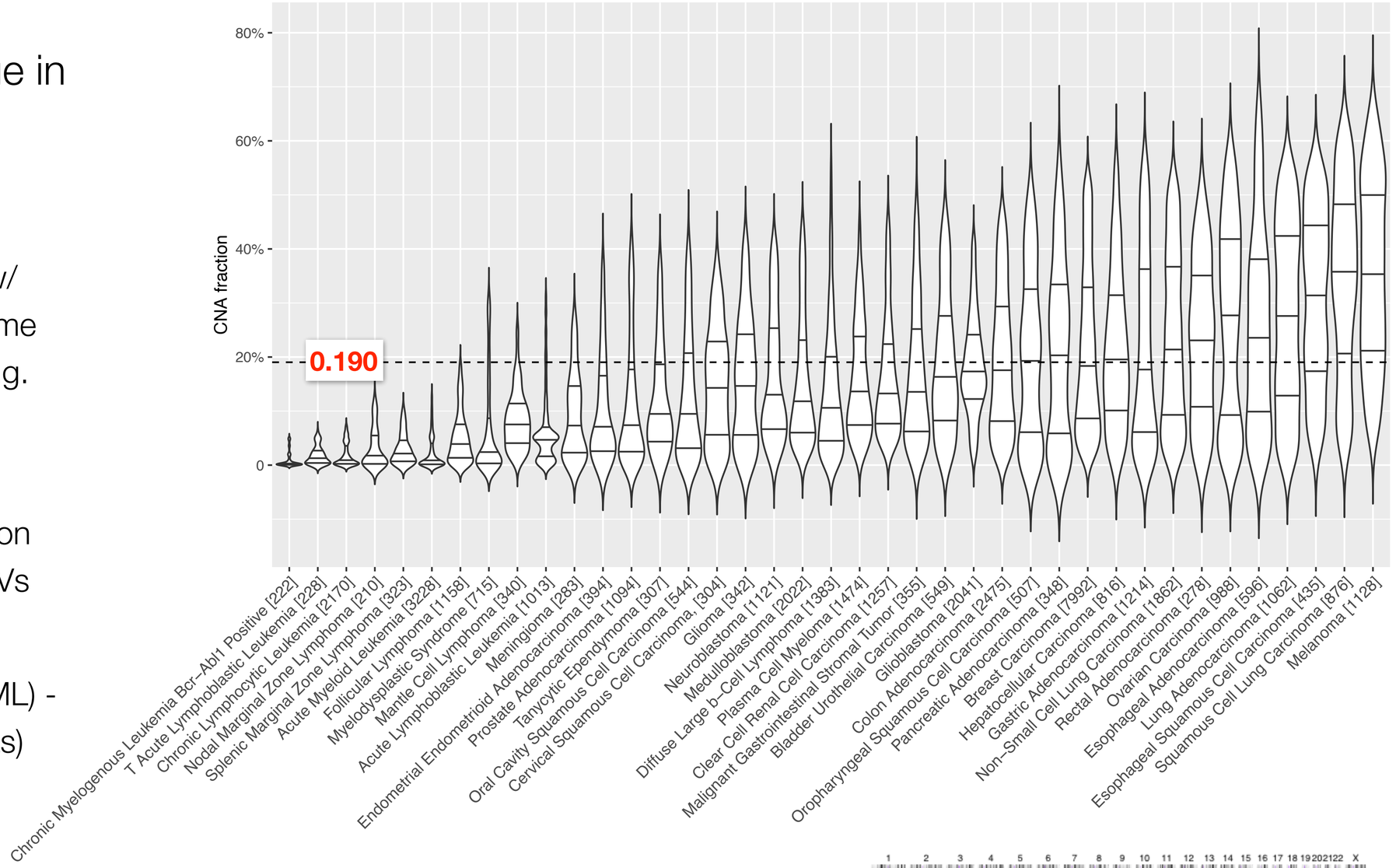


progenetix

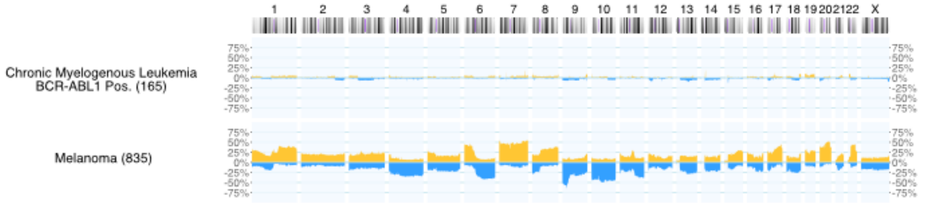
Data Use Cases

Genome CNV coverage in Cancer Classes

- 43654 out of 93640 CNV profiles; filtered for entities w/ >200 samples (removed some entities w/ high CNV rate, e.g. sarcoma subtypes)
- Single-sample CNV profiles were assessed for the fraction of the genome showing CNVs (relative gains, losses)
- range of medians 0.001 (CML) - 0.358 (malignant melanomas)



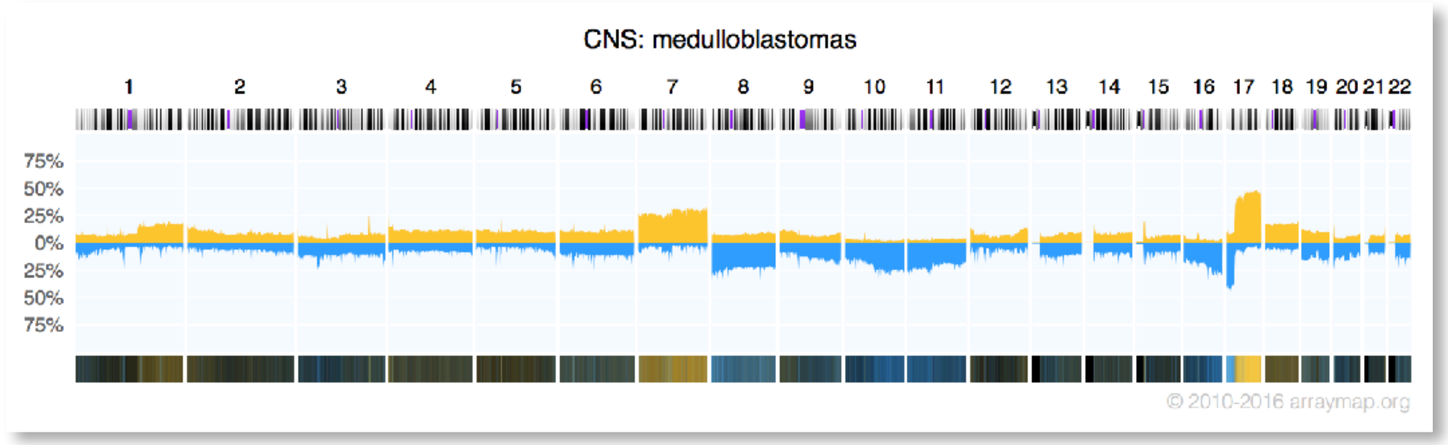
Lowest / Highest CNV fractions =>

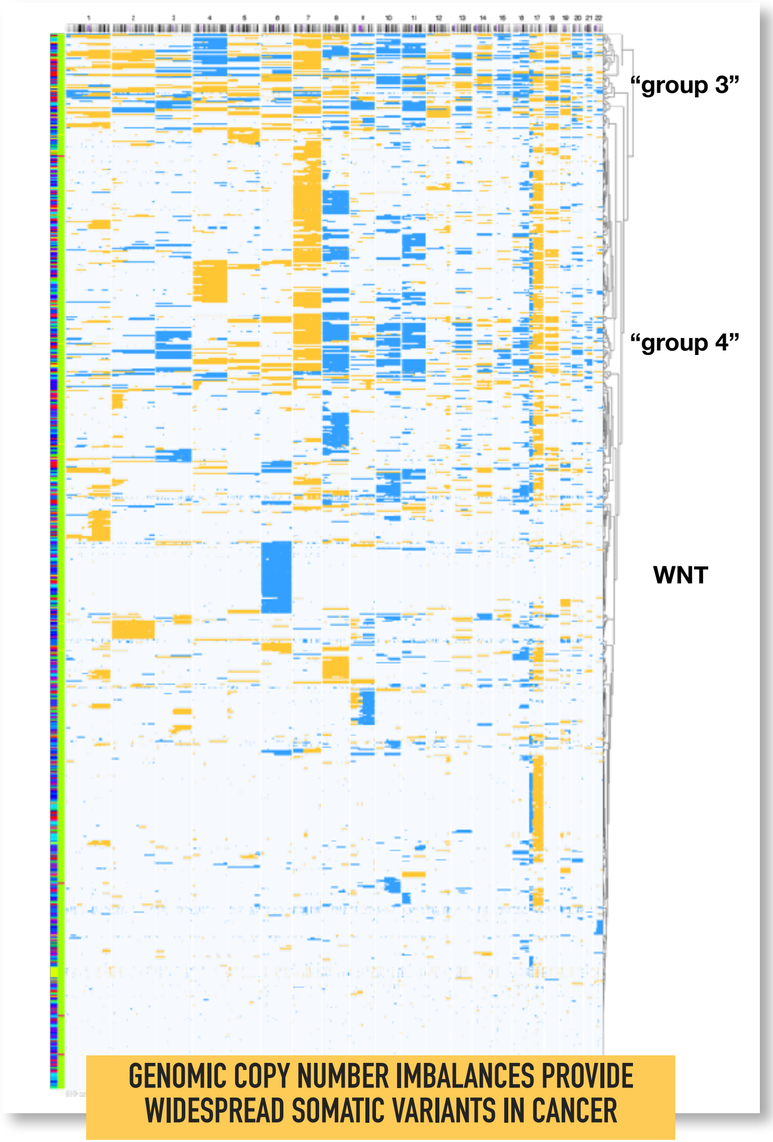


Somatic CNVs In Cancer

Recurrent mutation patterns

How can those patterns be used for classification and determination of biological mechanisms?





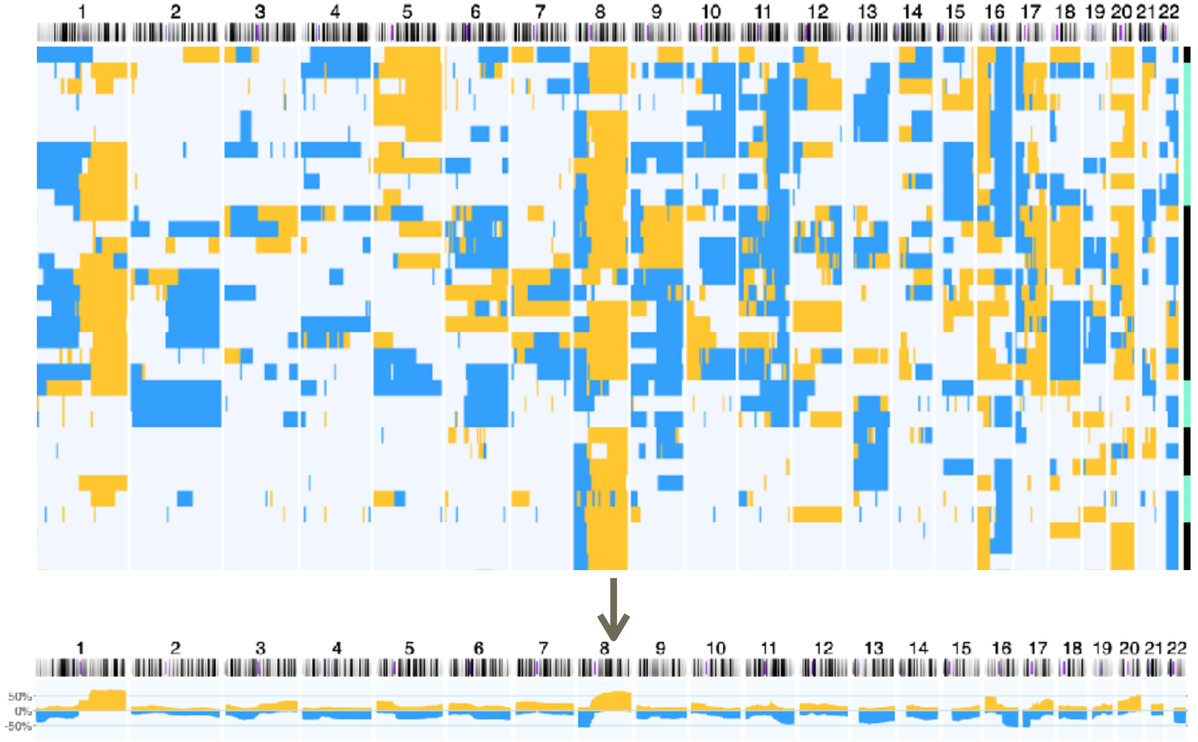
A genomic copy number histogram for malignant medulloblastomas, the most frequent type of pediatric brain tumors, displaying regions of genomic duplications and deletions. These can be decomposed into individual tumor profiles which segregate into several clusters of related mutation patterns with functional relevance and clinical

arrayMap

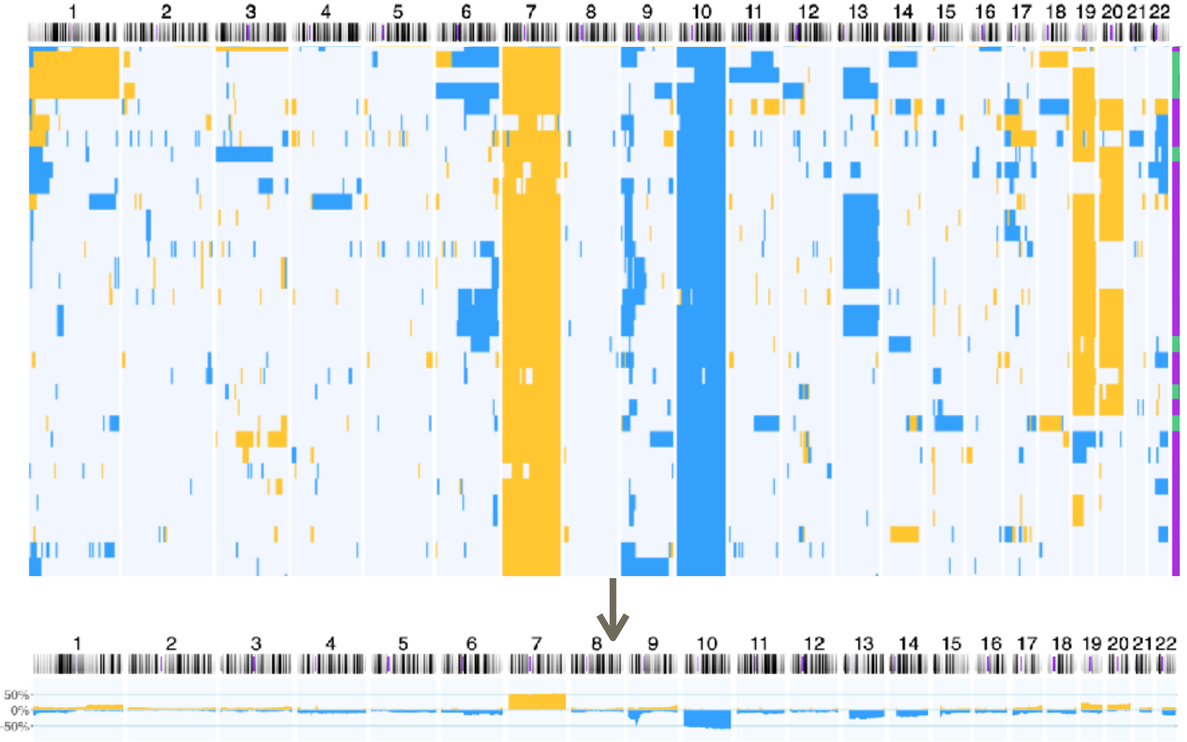
Drivers? Passengers? Markers?

Disentangling CNA Patterns

Ductal Breast Carcinoma

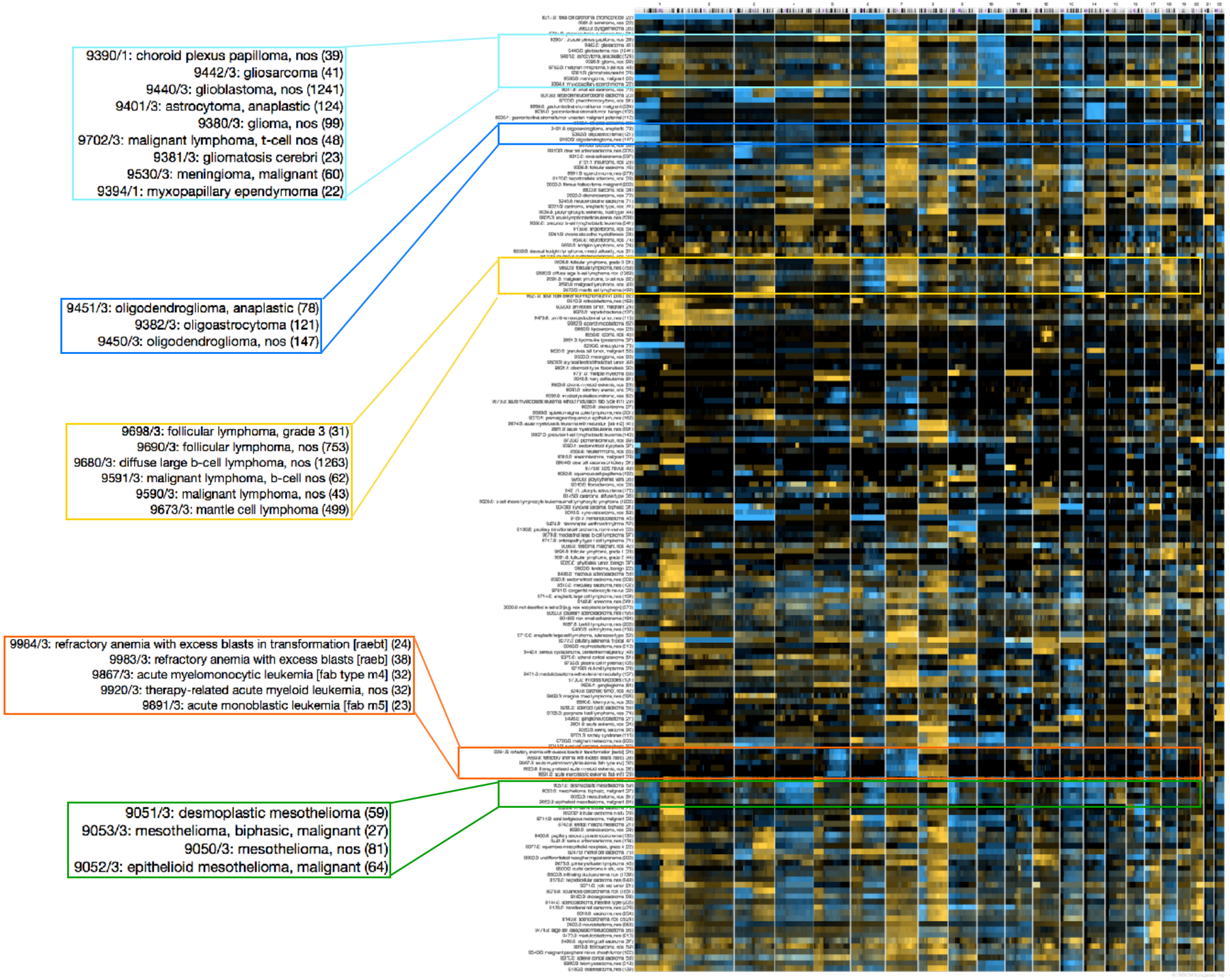


Glioblastoma



Somatic Mutations In Cancer: Patterns
Making the case for genomic classifications

Some related cancer entities show similar copy number profiles



arrayMap

frontiers in Genetics

ORIGINAL RESEARCH
published: 13 May 2021
doi: 10.3389/fgene.2021.654887

Signatures of Discriminative Copy Number Aberrations in 31 Cancer Subtypes

Bo Gao [1,2] *and Michael Baudis* [1,2*]

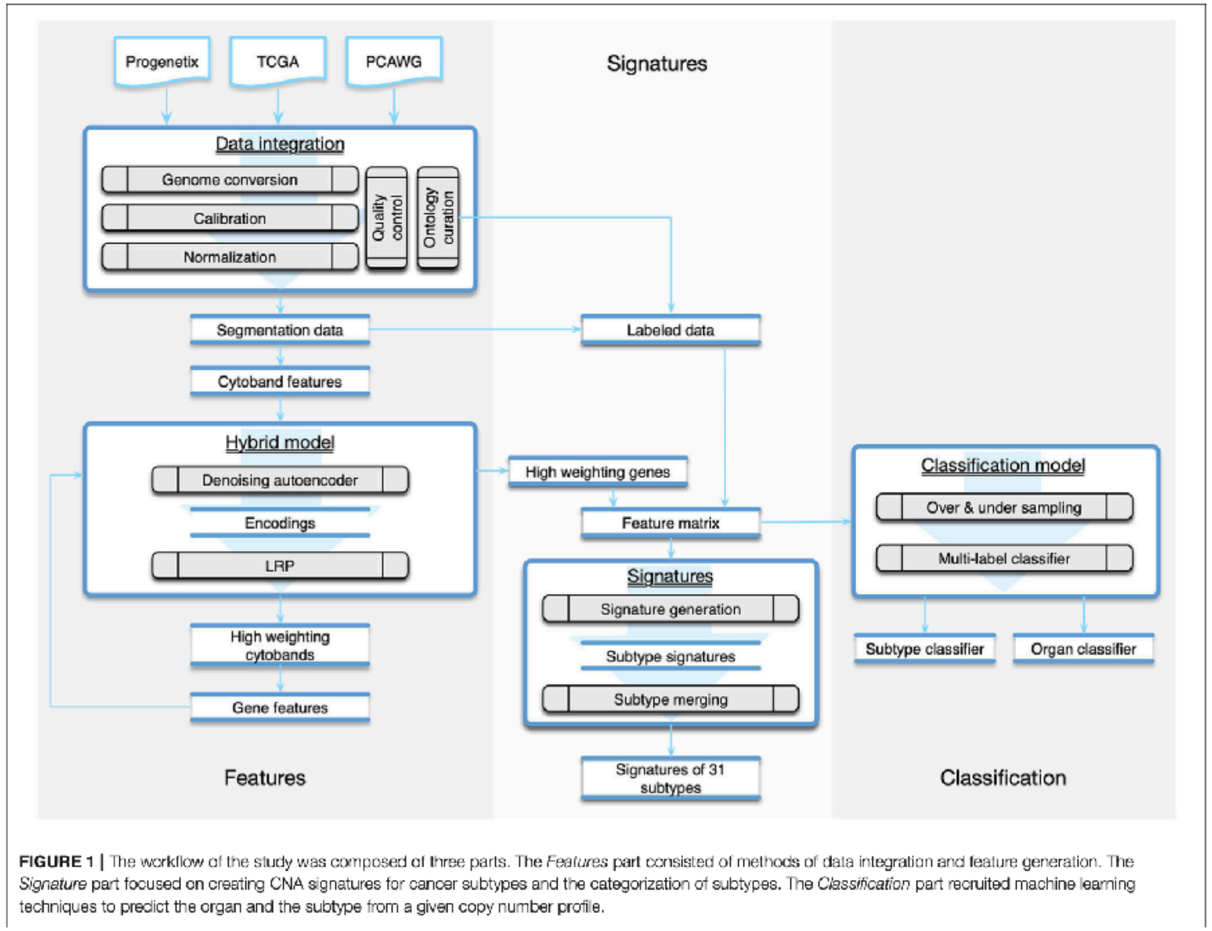


FIGURE 1 | The workflow of the study was composed of three parts. The *Features* part consisted of methods of data integration and feature generation. The *Signature* part focused on creating CNA signatures for cancer subtypes and the categorization of subtypes. The *Classification* part recruited machine learning techniques to predict the organ and the subtype from a given copy number profile.

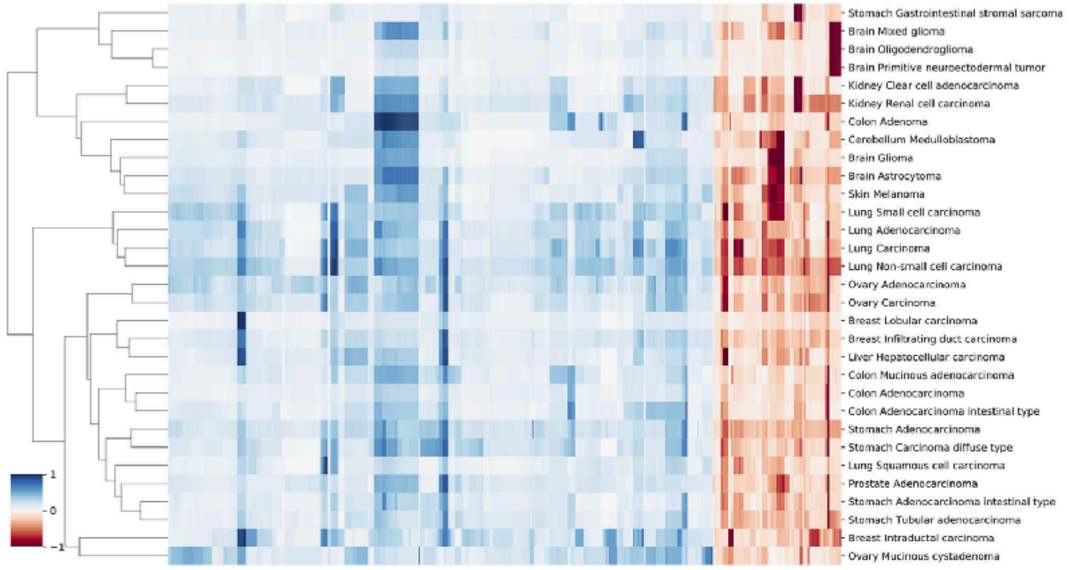


FIGURE 5 | A clustering heatmap of features in 31 signatures. Columns are normalized average CNV intensities of feature genes, where the blue colors are duplication features and red colors are deletion features. Duplication and deletion frequencies are normalized separately.

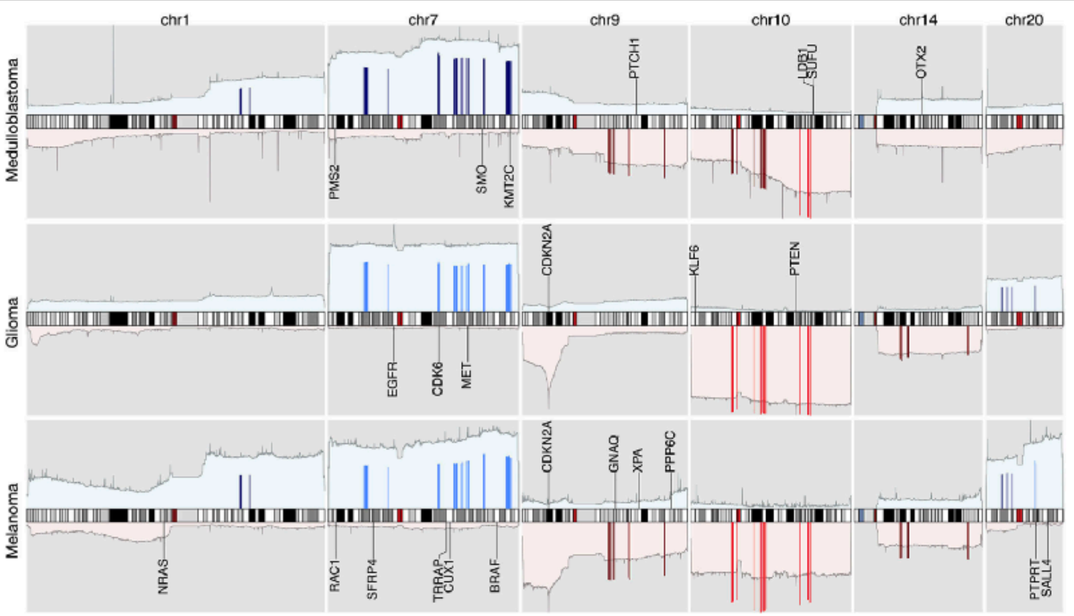


FIGURE 6 | The integrated view of the original data and the selected features, in the neural crest originating entities medulloblastoma, glioma, and melanoma. The shaded background area color illustrates the original data. Color bars illustrate the feature genes, where brighter colors indicate stronger signal intensity. The blue colors above the chromosome axis represent the average amplifications, and the red colors below the chromosome axis represent the average deletions. The amplitude of amplifications and deletions are normalized to [0,1] separately. The adjacent known driver genes are also included for each tumor type.

Population stratification in cancer samples based on SNP array data

- Despite extensive somatic mutations of cancer profiling data, consistency between germline and cancer samples reached 97% and 92% for 5 and 26 populations
- Comparison of our benchmarked results with self-reported meta-data estimated a matching rate between 88 % to 92%.
- Ethnicity labels indicated in meta-data are vague compared to the standardized output from our tool

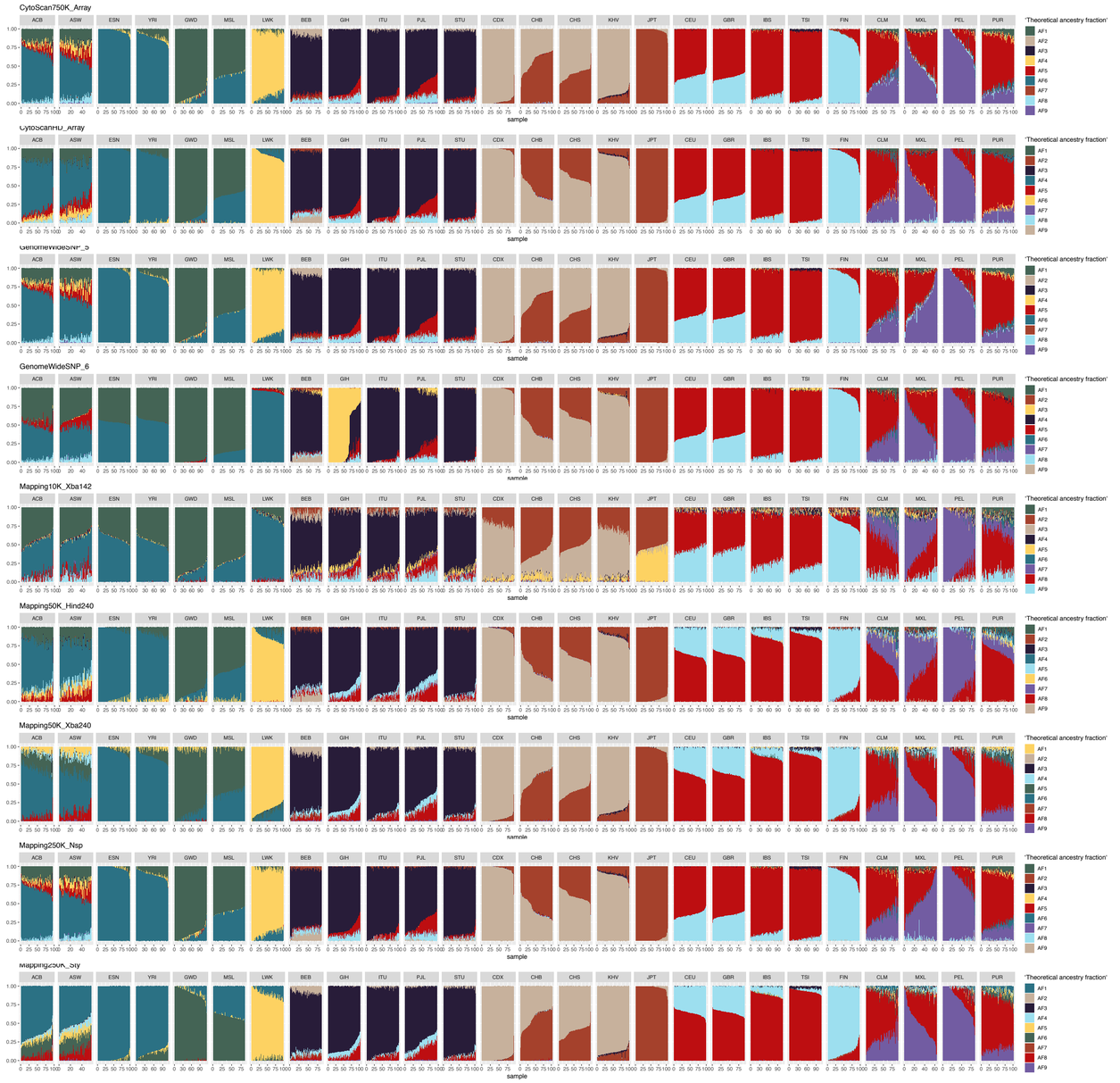


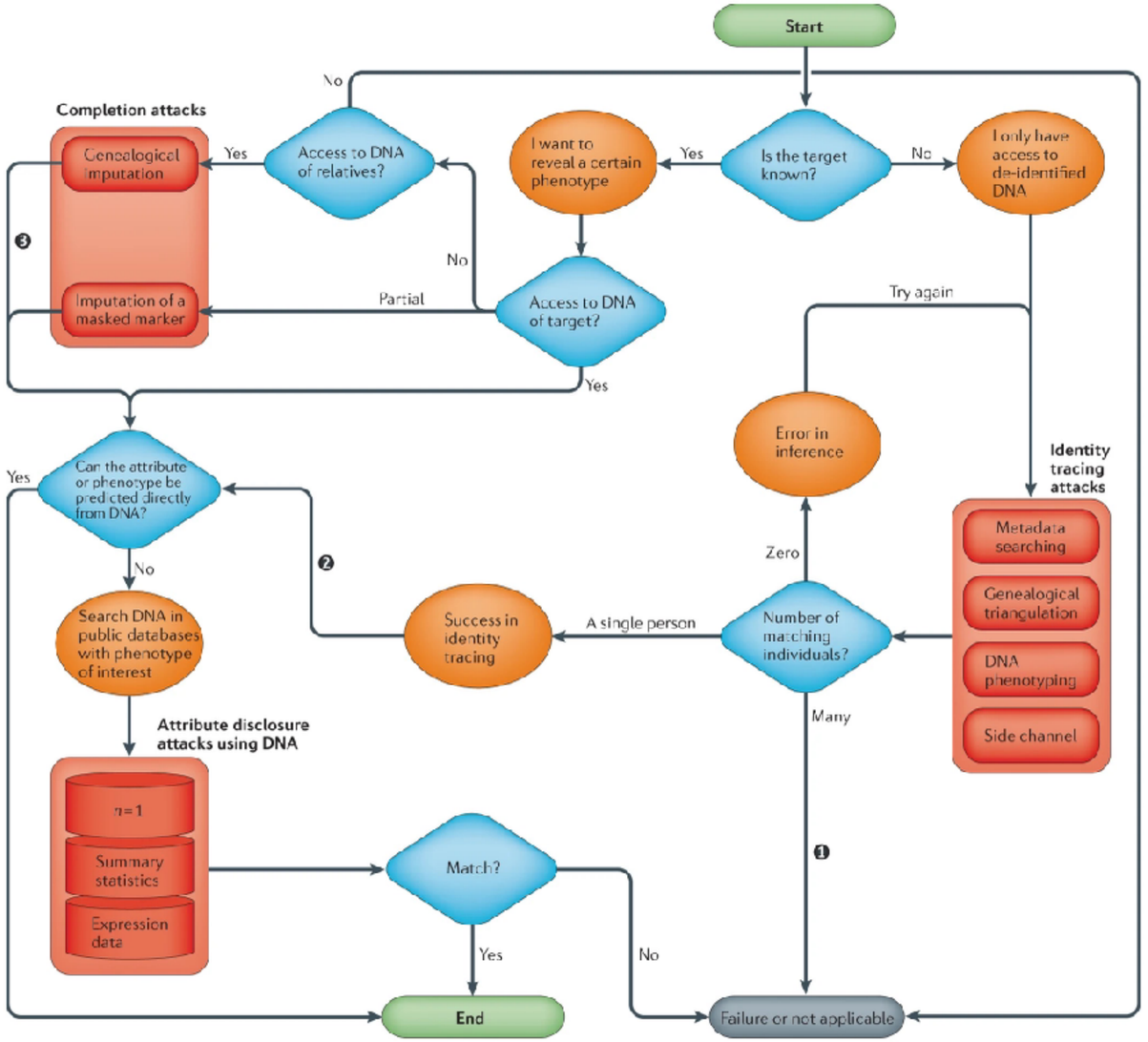
Figure S1 The fraction or contribution of theoretical ancestors (k=9) in reference individuals from 1000 Genomes Project with regard to nine SNP array platforms. The x-axis are individual samples, grouped by their respective population. Groups belonging to the same continent/superpopulation are placed neighboring to each other: AFR (1-7), SAS (8-12), EAS (13-17), EUR (18-22), AMR (23-26).

arrayMap

Qingyao Huang

How to share patient data safely?

- Long-range familiar search
  - “Golden State killer”
  - Cold cases in 1970s
  - DNA evidence led to capture in 2018
- Membership inference attack
- Reconstruction attack
- Privacy by design
  - Access control
  - Data aggregation
  - Data obfuscation



Nature Reviews | Genetics

Wan et al. 2022 Sociotechnical safeguards for genomic data privacy. NRG

Erlich and Narayanan 2014. Routes for breaching and protecting genetic privacy. NRG

Progenetix and GA4GH Beacon

Implementation driven development of a GA4GH standard

progenetix

GENOMICS

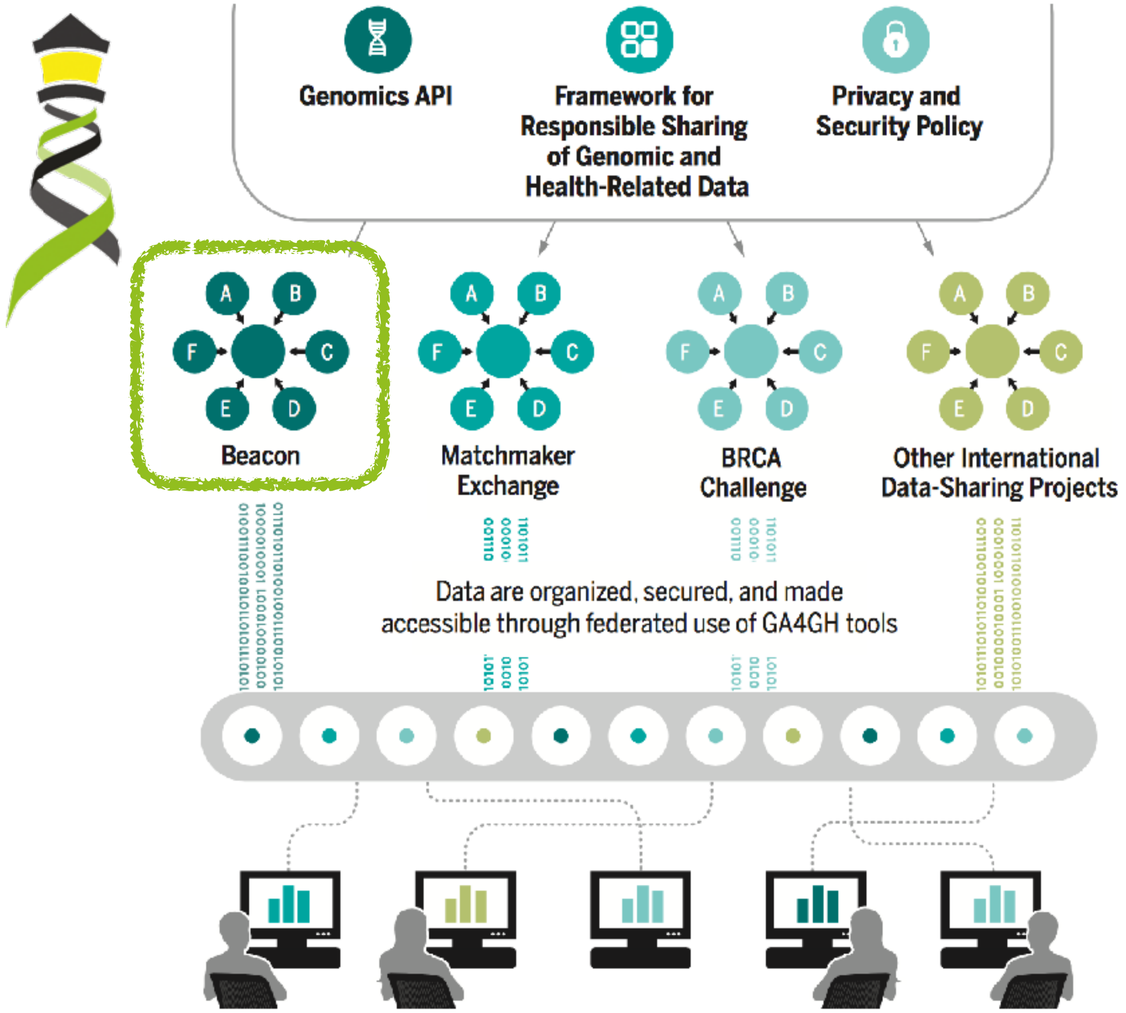
A federated ecosystem for sharing genomic, clinical data

Silos of genome data collection are being transformed into seamlessly connected, independent systems

The Global Alliance for Genomics and Health*

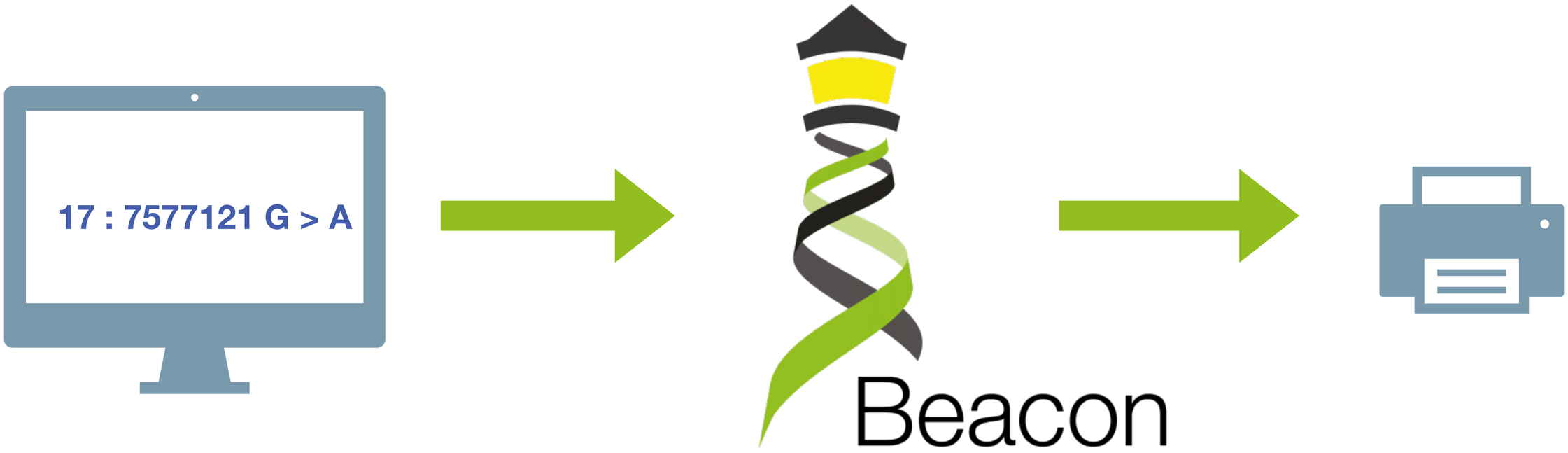
SCIENCE 10 JUNE 2016 • VOL 352 ISSUE 6291

A federated data ecosystem. To share genomic data globally, this approach furthers medical research without requiring compatible data sets or compromising patient identity.



DNASTACK

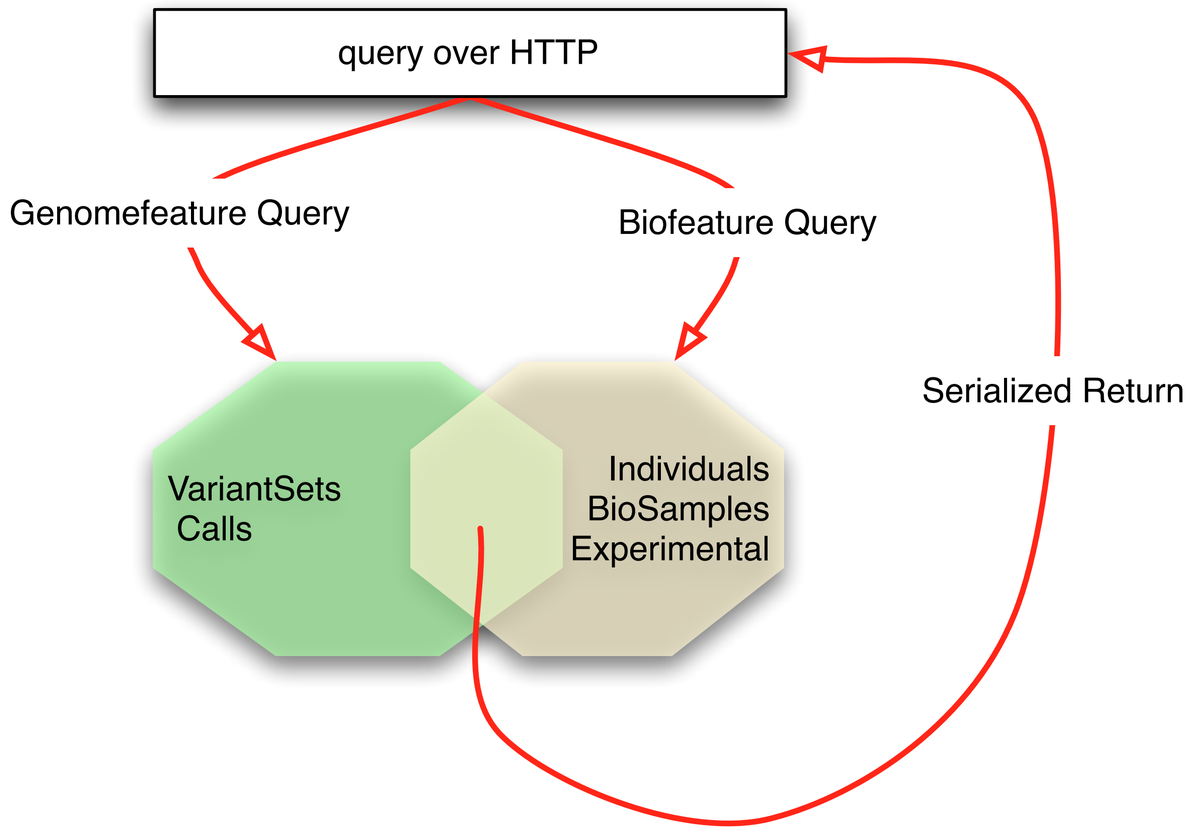
Global Alliance for Genomics & Health



A ***Beacon*** answers a query for a specific genome variant against individual or aggregate genome collections

YES | **NO** | **\0**

Minimal GA4GH query API structure



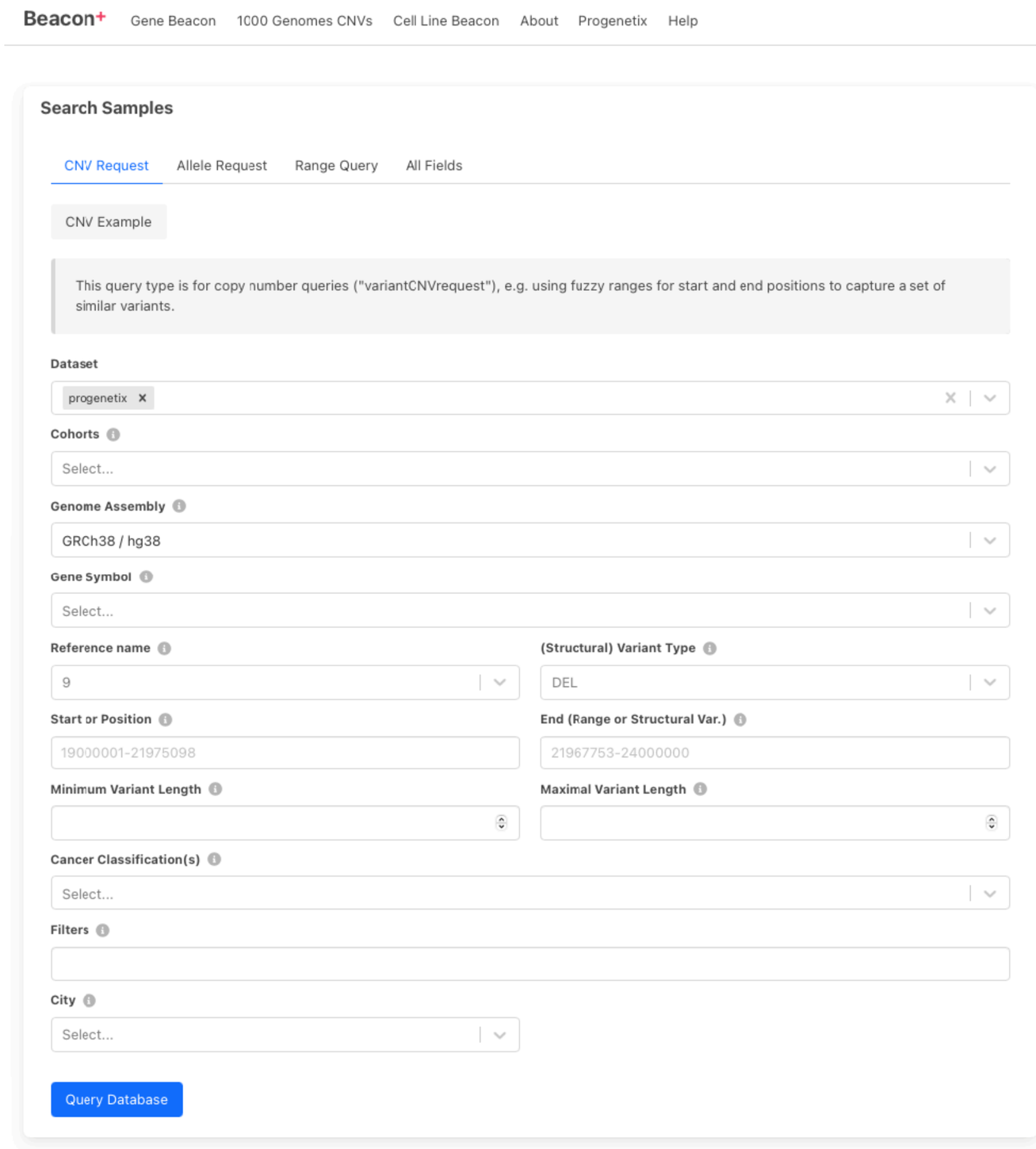
Beacon(+) “Metadata”

Beacon+ by Progenetix

From Beacon Query to Explorative Analyses of CNV Patterns

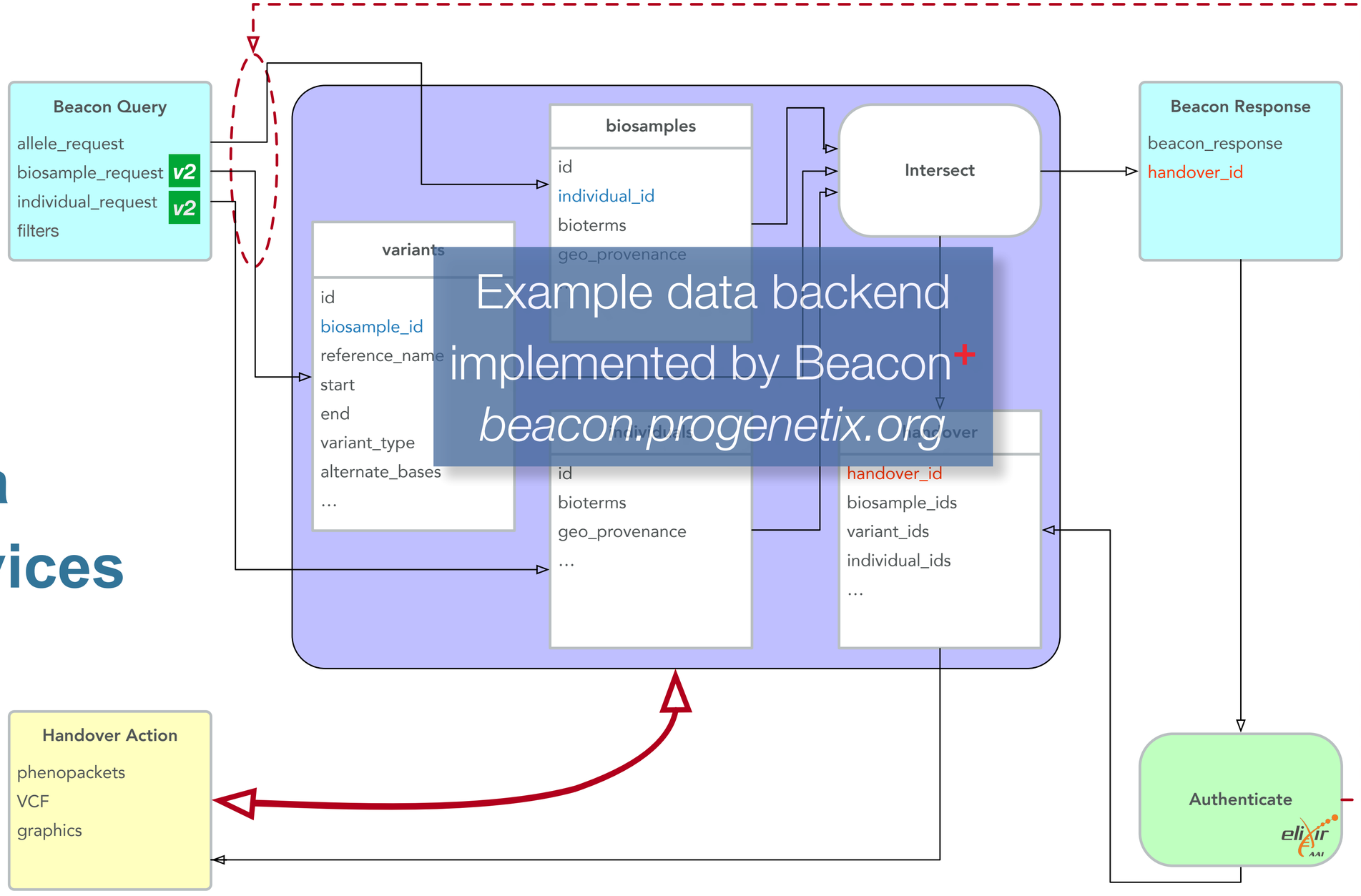
- Since 2016 the Progenetix resource has been used to model options for Beacon development
  - 138334 individual samples from 698 cancer types
- The consistent use of hierarchical diagnostic codes allows the use of Beacon "filters" for histopathological/clinically scoped queries
- Beacon's handover protocols can be utilized for data retrieval and, well, handing over to additional services, e.g.
  - downloads
  - visualization
  - use of external services (UCSC browser display...)

progenetix



Beacon & Handover

Beacons v1.1 supports **data delivery services**



Beacon I/O

Handover

Authentication

Michael Baudis

Global Alliance for Genomics & Health

SIB

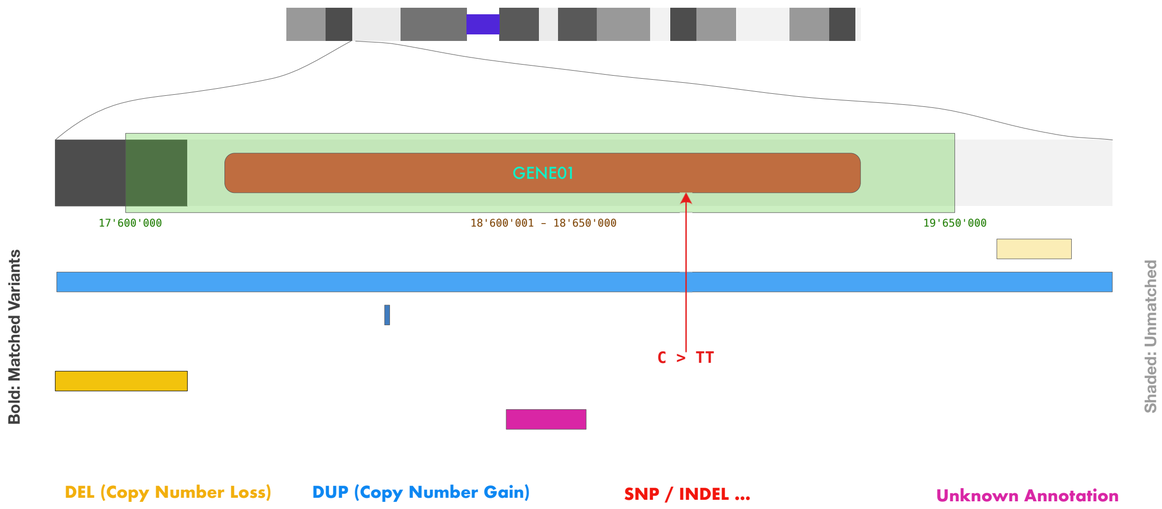
University of Zurich UZH

elixir

Beacon v2: Extended Variant Queries

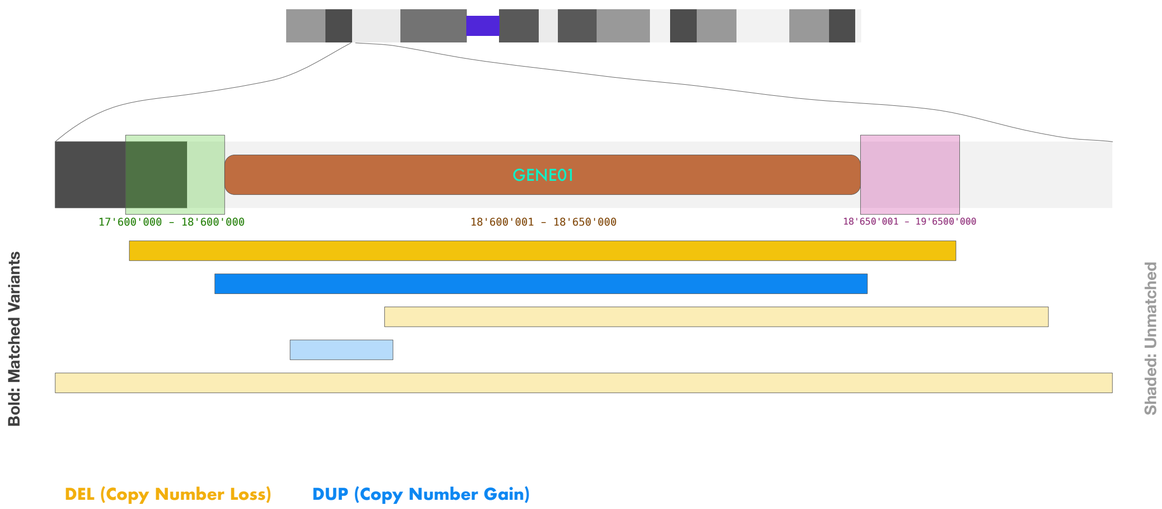
Range and Bracket queries enable positional wildcards and fuzziness

Genome Range Query
(matching variants in a region)



- Genome Range Queries provide a way to "fish" for variants overlapping an indicated region, e.g. the CDR of a gene of interest
- Additional parameters (e.g. variant type, reference or alternate bases) limit the scope of the responses
- new Beacon v2 size parameters to limit structural variants (e.g. "focal" CNVs)

Genome Bracket Query
(full match)

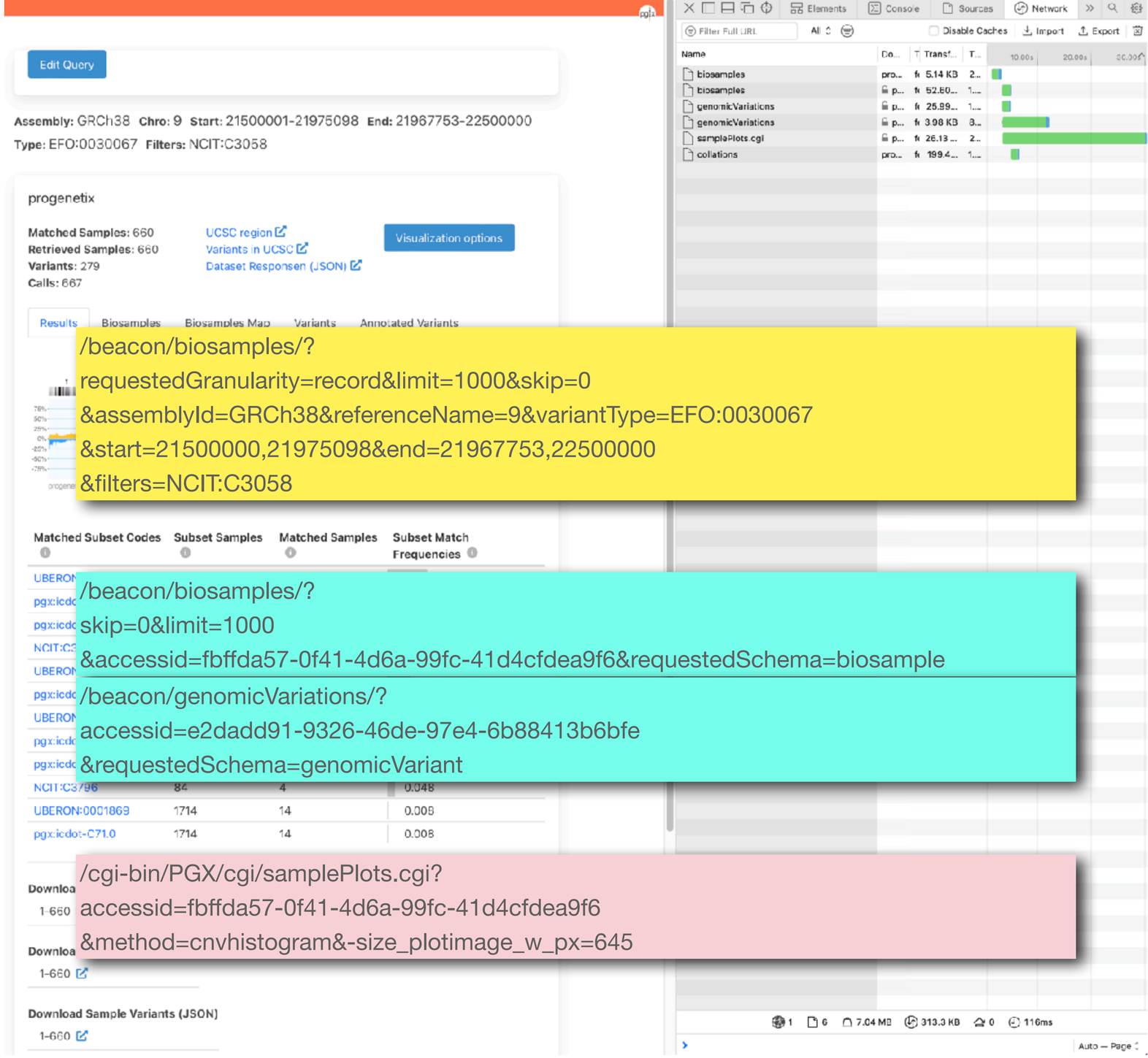


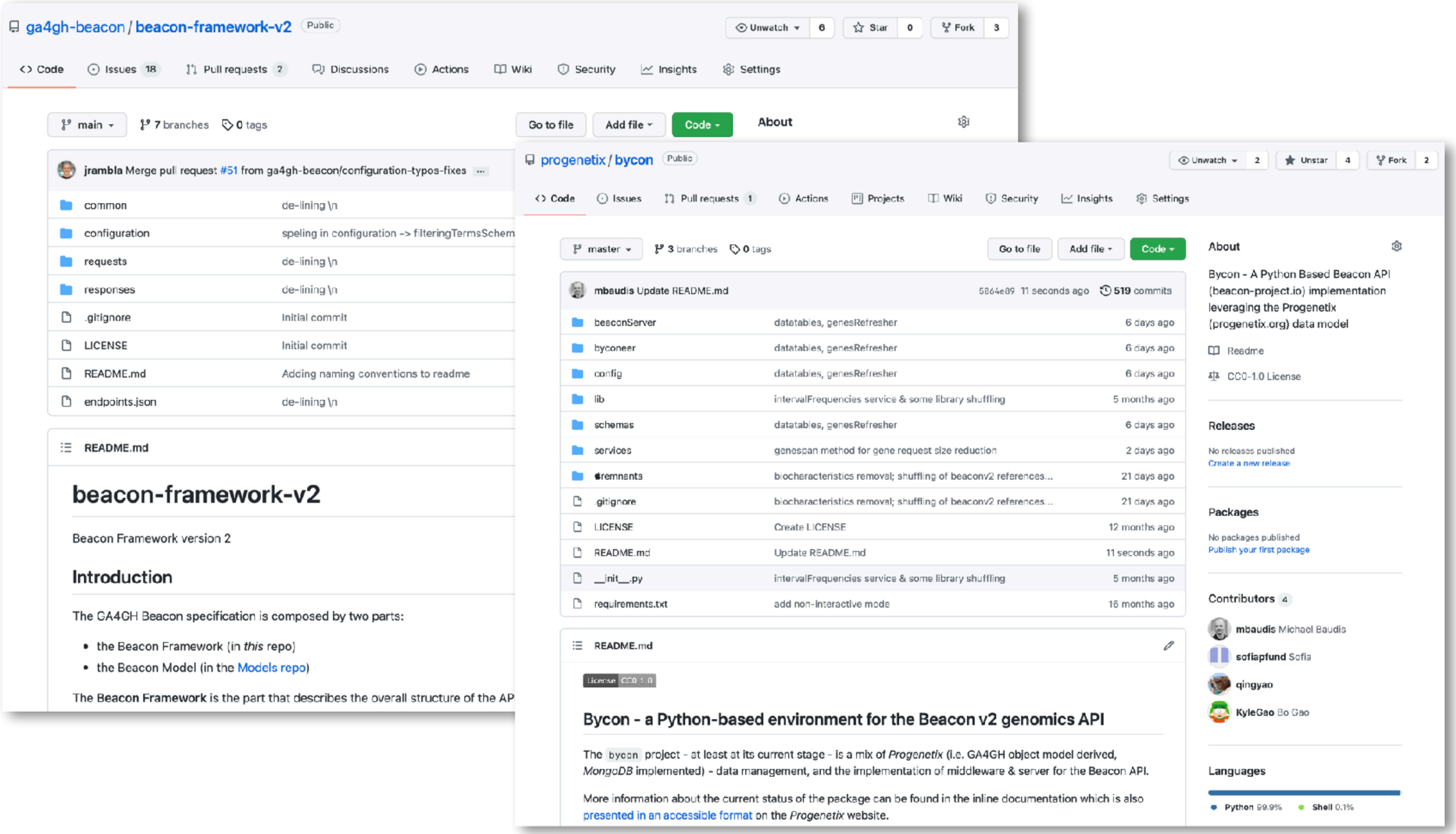
- Genome Bracket Queries allow to search for structural variants with start and end positions falling into defined sequence ranges
- allows to query any contiguous genomic variant (and in principle also can step in for range queries)
- typical use case is e.g the query for variants such as duplications covering the whole CDR of a gene, while limiting the allowed start or end regions

Progenetix

Genomic resource utilizing Beacon v2 calls

- Progenetix uses Beacon v2 queries to drive its UI
- all individuals, biosamples, variants, analyses matched by a given query are stored by their object ids
- handovers for variant purposes (e.g. to retrieve all matched variants) are returned in the original response and asynchronously retrieved




ga4gh-beacon / beacon-framework-v2
Public
Unwatch 6
Star 0
Fork 3
Code
Issues 18
Pull requests 2
Discussions
Actions
Wiki
Security
Insights
Settings
main
7 branches
0 tags
Go to file
Add file
Code
About
jrambla Merge pull request #51 from ga4gh-beacon/configuration-typos-fixes
common de-lining \n
configuration speling in configuration -> filteringTermsSchem
requests de-lining \n
responses de-lining \n
.gitignore Initial commit
LICENSE Initial commit
README.md Adding naming conventions to readme
endpoints.json de-lining \n
README.md
beacon-framework-v2
Beacon Framework version 2
Introduction
The GA4GH Beacon specification is composed by two parts:
the Beacon Framework (in this repo)
the Beacon Model (in the Models repo)
The Beacon Framework is the part that describes the overall structure of the AP
progenetix / bycon
Public
Unwatch 2
Unstar 4
Fork 2
Code
Issues
Pull requests 1
Actions
Projects
Wiki
Security
Insights
Settings
master
3 branches
0 tags
Go to file
Add file
Code
mbaudis Update README.md
5864e89 11 seconds ago
519 commits
beaconServer datatables, genesRefresher 6 days ago
byconeer datatables, genesRefresher 6 days ago
config datatables, genesRefresher 6 days ago
lib intervalFrequencies service & some library shuffling 5 months ago
schemas datatables, genesRefresher 6 days ago
services genescan method for gene request size reduction 2 days ago
tremnants biocharacteristics removal; shuffling of beaconv2 references... 21 days ago
.gitignore biocharacteristics removal; shuffling of beaconv2 references... 21 days ago
LICENSE Create LICENSE 12 months ago
README.md Update README.md 11 seconds ago
__init__.py intervalFrequencies service & some library shuffling 5 months ago
requirements.txt add non-interactive mode 16 months ago
README.md
License CC0 1.0
Bycon - a Python-based environment for the Beacon v2 genomics API
The bycon project - at least at its current stage - is a mix of Progenetix (i.e. GA4GH object model derived, MongoDB implemented) - data management, and the implementation of middleware & server for the Beacon API.
More information about the current status of the package can be found in the inline documentation which is also presented in an accessible format on the Progenetix website.
About
Bycon - A Python Based Beacon API (beacon-project.io) implementation leveraging the Progenetix (progenetix.org) data model
Readme
CC0-1.0 License
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 4
mbaudis Michael Baudis
sofiapfund Sofia
qingyao
KyleGao Bo Gao
Languages
Python 99.9%
Shell 0.1%

Onboarding

Demonstrating Compliance

- Progenetix Beacon+ has served as implementation driver since 2016
- Beacon v2 as service with protocol-driven registries for federation
- GA4GH approved Beacon v2 in April 2022

Global Alliance for Genomics & Health

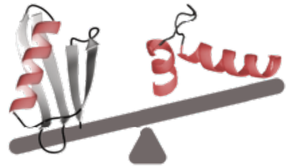
Have you seen deletions in this region on chromosome 9 in Glioblastomas from a juvenile patient, in a dataset with unrestricted access?

Beacon *v2* API

The Beacon API v2 proposal opens the way for the design of a simple but powerful "**genomics API**".

PaxDb

A Protein abundance reference resource



paxdb

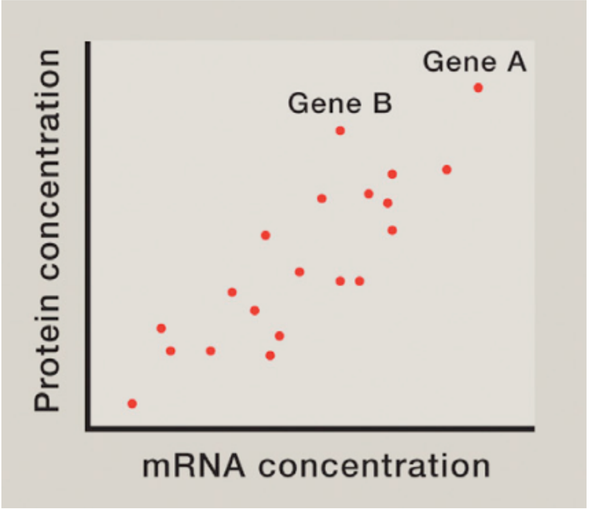
PaxDb

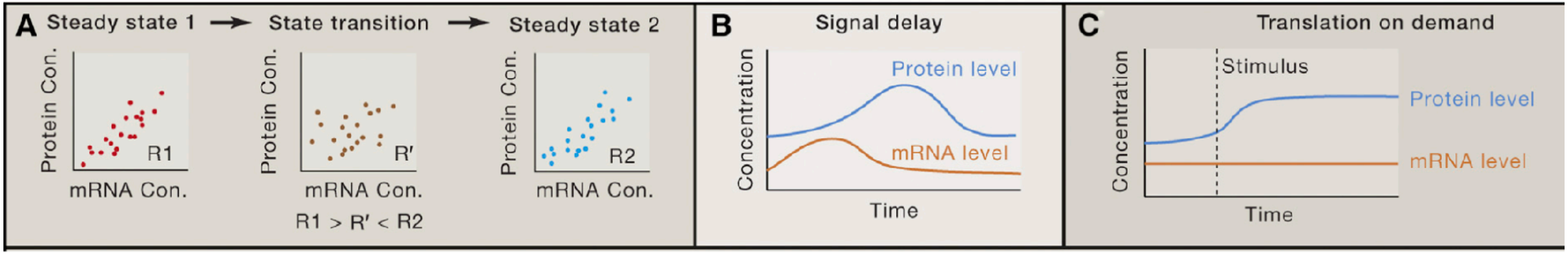
- Motivation for building a DB
  - Relevance
- What is the quantitative data and how is it represented?
  - Protein abundances
  - Orthology relationships
  - Techniques
- What is the metadata?
  - species, tissue, protein ID, ortholog
  - publication, experimental condition
- How to use the resources?
  - Web browsing, bulk download, upload own data

PaxDb

Motivation

- At Steady State, mRNA Levels Primarily Explain Protein Levels
- Buffering of Excess mRNA Variation at Multiple Levels
  - Transcriptional variation
  - Translational variation
  - Post-translational variation



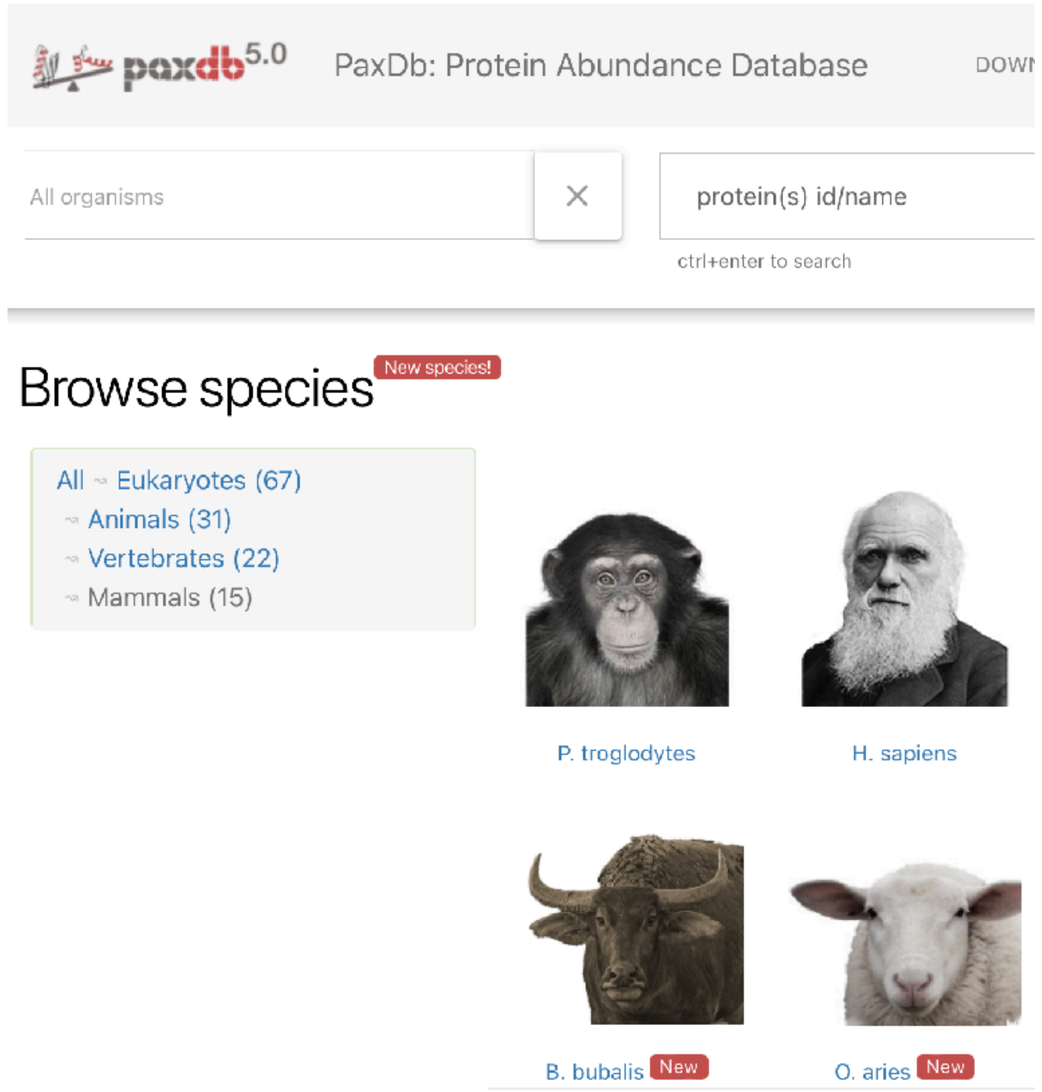


Liu et al. 2016 On the Dependency of Cellular Protein Levels on mRNA Abundance.Cell

PaxDb

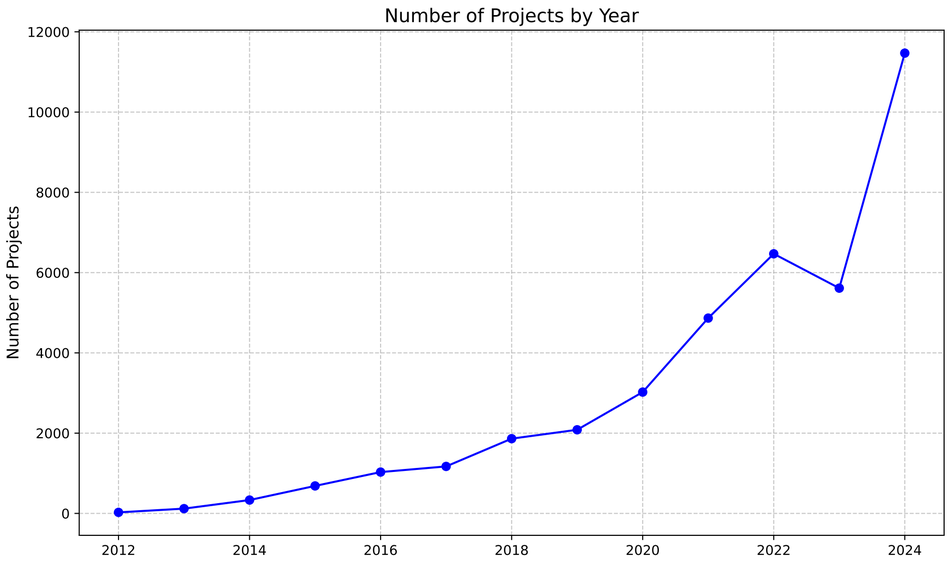
Motivation

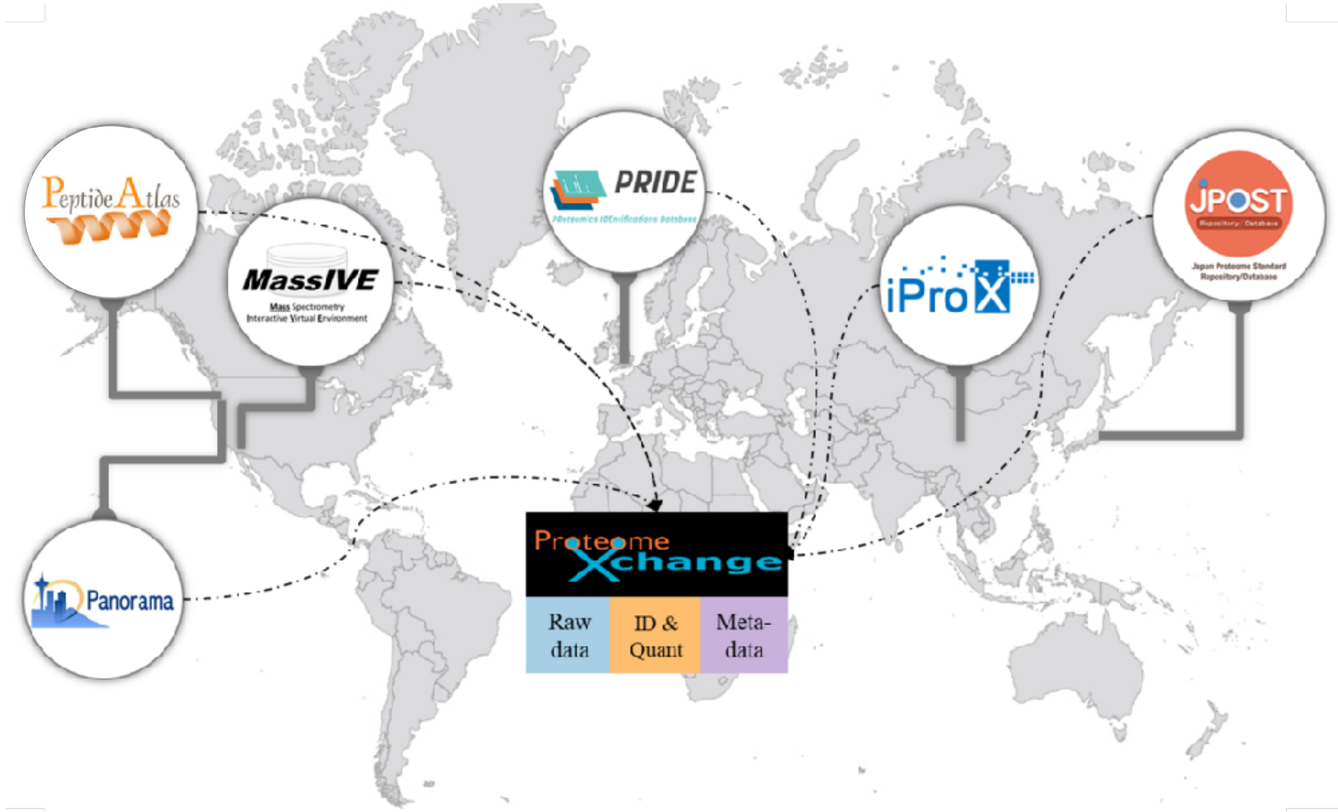
- Protein abundance across organisms
  - highly regulated at the functional level
  - conserved across species
  - fundamental to evolution, translation and folding
- Proteomics datasets are large and difficult to process and compare
- Need a reference for common and rare species
- Need integration on datasets of same type



ProteomeXchange

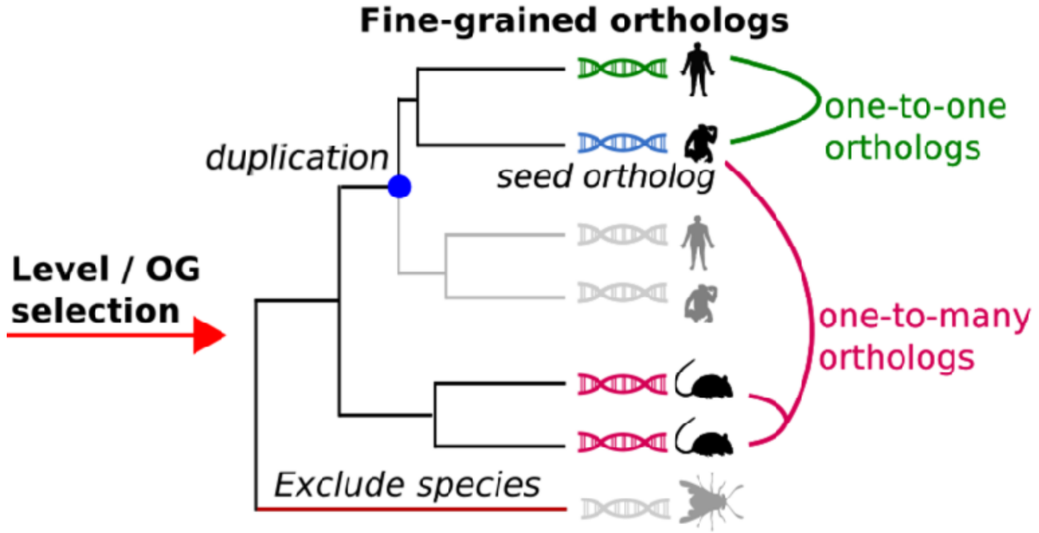
- Data registry (consortium) of multi-regional data repositories
- At least raw data but also processed data





Ortholog relation

- In the course of evolution, proteins continue to change after speciation events
- Orthologs typically retain the same function
- At higher taxonomic level, the clusters of orthologous genes (Cogs) are larger and orthologs are more distant.
- All species' proteomes have orthologs mapped to all levels up to Luca.



Quantitative Proteomics with LC-MS/MS

Protein
P

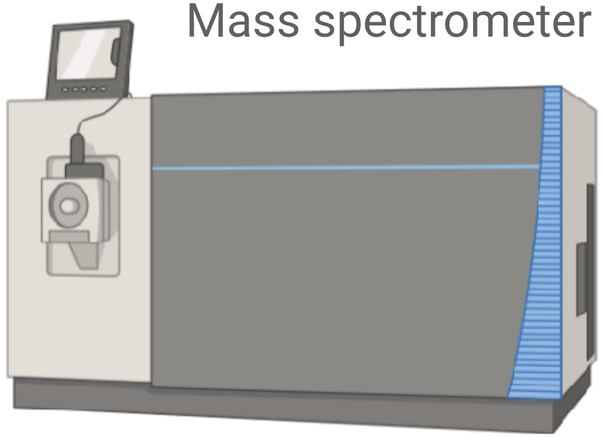
Trypsin

Pep2

Pep1

Liquid chromatography

Mass spectrometer



MS1

Z
Retention
Time (RT)

Y
Intensity

Top 5

X
M/z

MS2

a
c
b
d

MS2

a
c
b

Y
Int

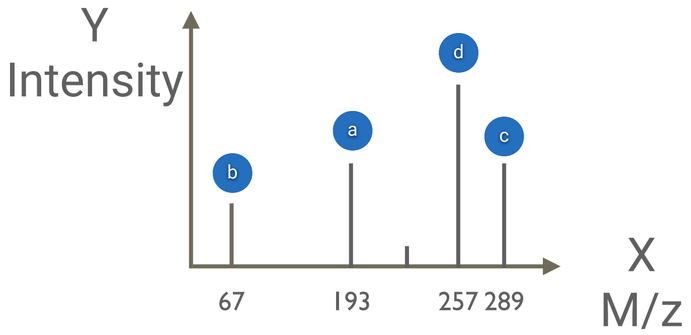
Peptide spectrum match
PSM

Int

X
M/z

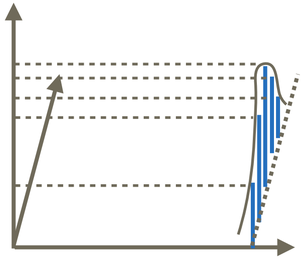
Data acquisition by Mass spectrometry

Peptide spectrum match (PSM)



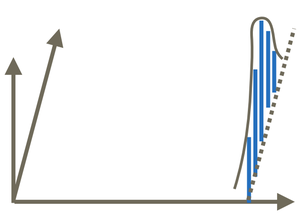
1. Peptide sequence from database search

2. Sum of MS1 Intensity



Intensity-based

2. count of peptide appearing as top peaks



count-based

- Sequence + intensity
- Sequence + count
- Protein ID + intensity
- Protein ID + count

Data collection

Manually downloaded

Filename	Description
nph17756-sup-0006-TableS5.xlsx Excel 2007 spreadsheet , 1.4 MB	**Table S5** Full data list of wheat grain protein turnover rates during grain development.

Annotation					
First ID	Protein gorup	Protein name	Intensity_R1	Intensity_R2	Intensity_R3
TraesCS1A01G00	TraesCS1A01G00	Nucleic acid-bind	341240	194450	426600
TraesCS1A01G00	TraesCS1A01G00	Paired amphipath	431440	393670	207190
TraesCS1D01G00	TraesCS1D01G00	Transcription init	57589	56370	29279
TraesCS1A01G00	TraesCS1A01G00	E3 ubiquitin-prot	1099200	1120300	1462500
TraesCS1A01G00	TraesCS1A01G00	Gamma-gliadin	670790	622010	945570
TraesCS1A01G00	TraesCS1A01G00	Peptidyl-prolyl ci	186530	167950	194550
TraesCS1A01G00	TraesCS1A01G00	Low molecular w	1524600	929110	1274100
TraesCS1A01G00	TraesCS1A01G00	MICOS complex s	231330	181940	222810
TraesCS1D01G00	TraesCS1D01G00	Ankyrin repeat fa	290850	421780	305910
TraesCS1A01G01	TraesCS1A01G01	Defensin	271140	263810	267260
TraesCS5D01G13	TraesCS5D01G13	V-type proton ATI	1726900	1476500	2310400

↓

Normalization

Downloaded in bulk from repos

Project Files

FTP GLOBUS

Name	Type	Size (M)	Download
VELOS16664.raw	RAW	451	FTP
VELOS16644.raw	RAW	449	FTP
VELOS16646.raw	RAW	455	FTP

+

Name	Fraction	Experiment
VELOS16626		ZS1
VELOS16627		ZS1
VELOS16629		TiO1
VELOS16630		TiO1

↓

Mass spectrometry pipeline

Quality evaluation based on protein interaction

interacting proteins often have roughly similar abundances:

origin recognition complex

ORC1: 8.6 ppm	ORC4: 12.3 ppm
ORC2: 1.4 ppm	ORC5: 2.7 ppm
ORC3: 3.2 ppm	ORC6: 6.4 ppm

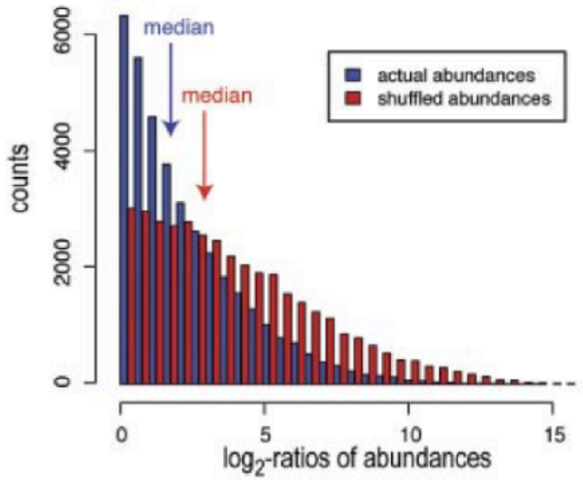
5.7 ± 4.2 ppm

replication factor A

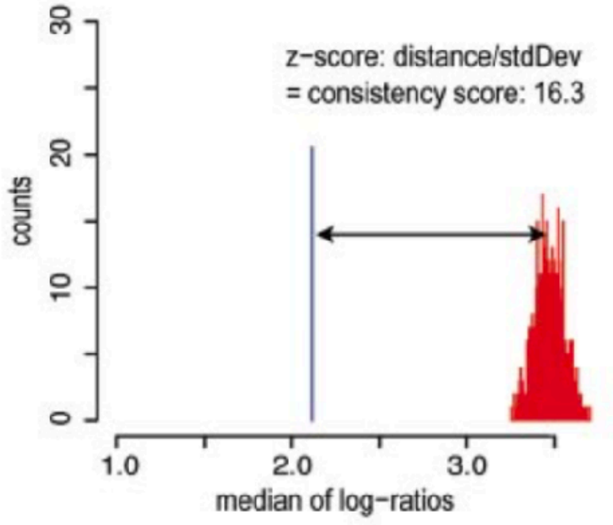
RFA1: 57 ppm
RFA2: 97 ppm
RFA3: 123 ppm

92 ± 33 ppm

pairwise comparisons of all interacting proteins in yeast

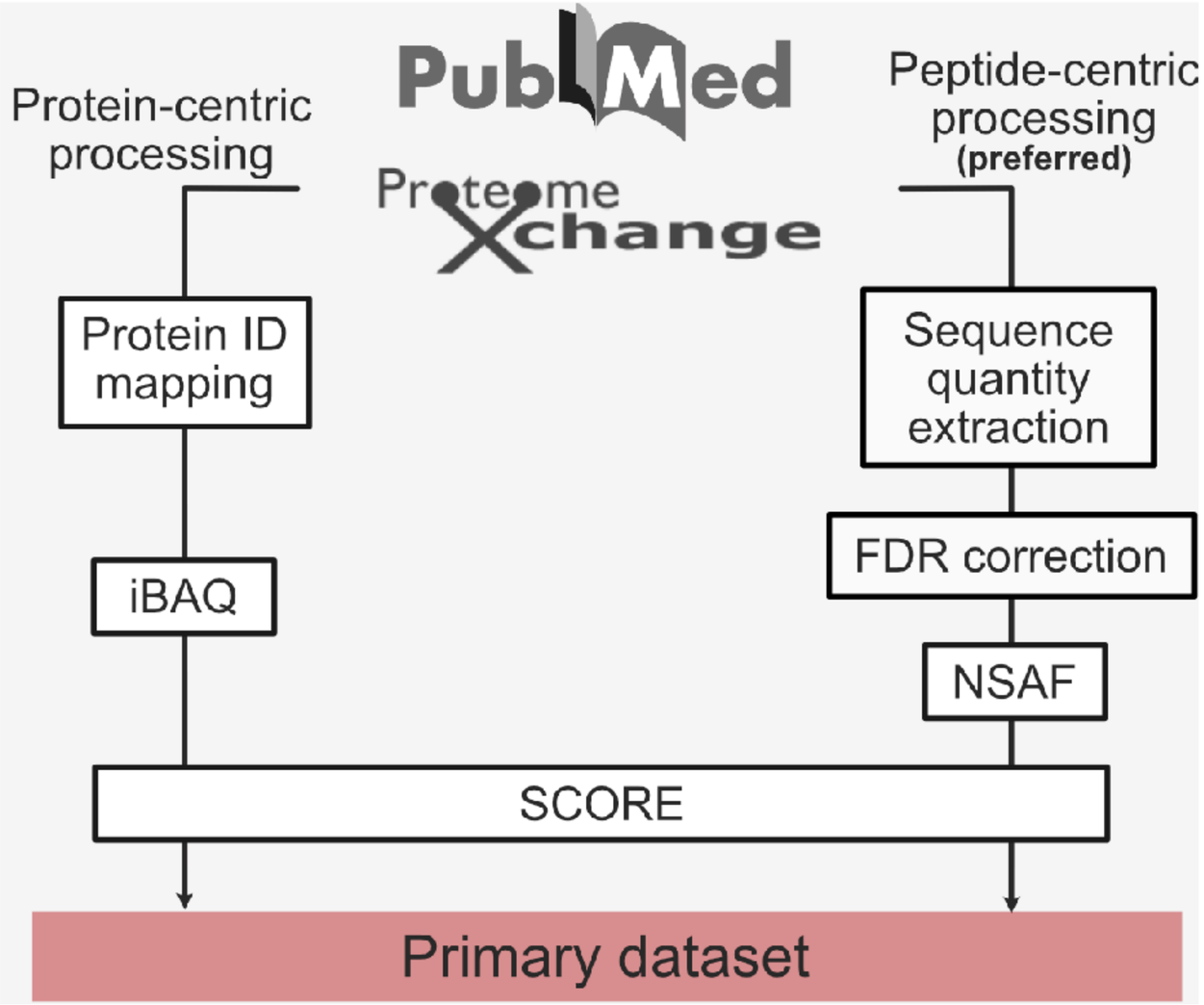


Consistency with protein interactions

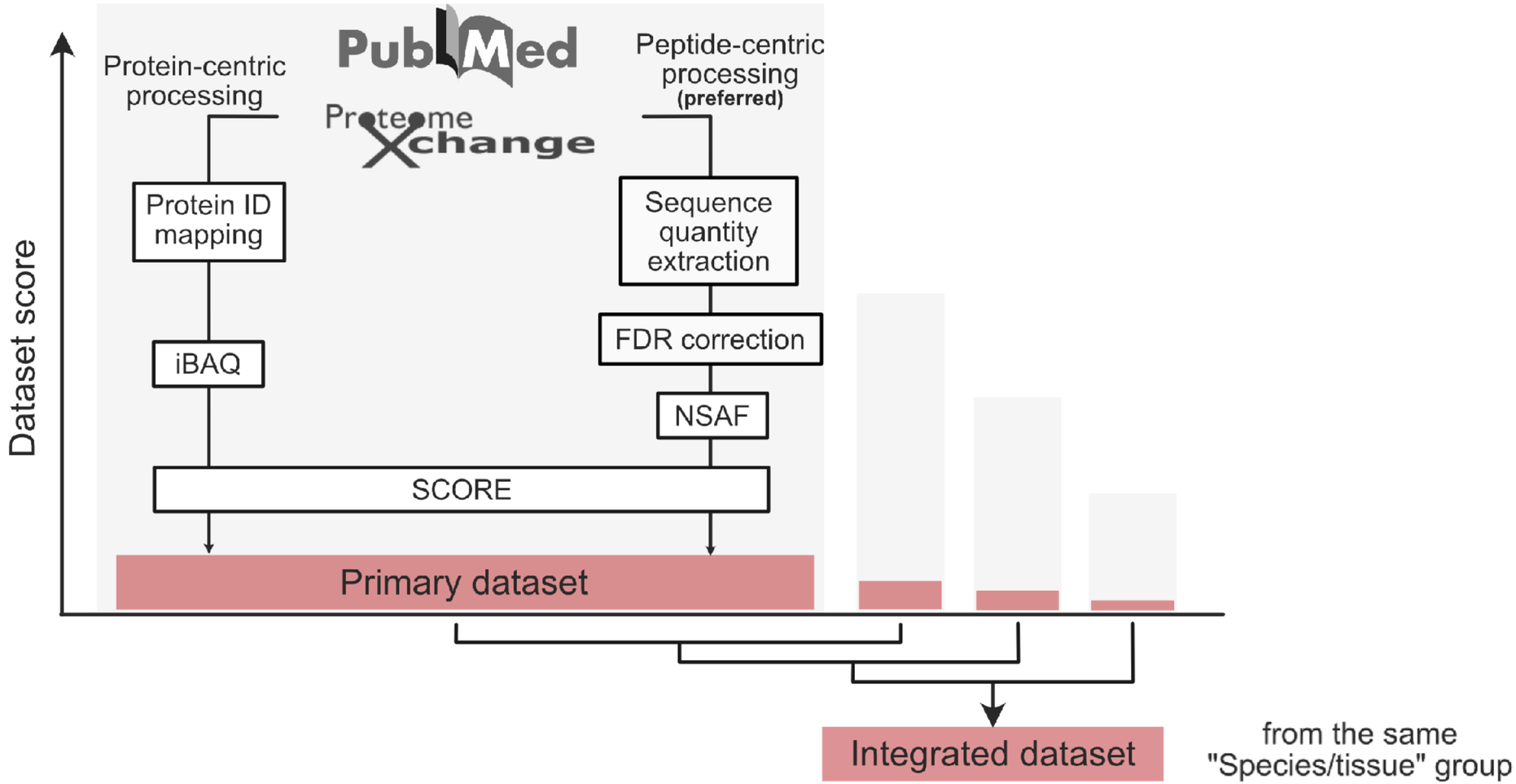


Wang et al, MCP 2012

Computation pipeline



Data integration pipeline



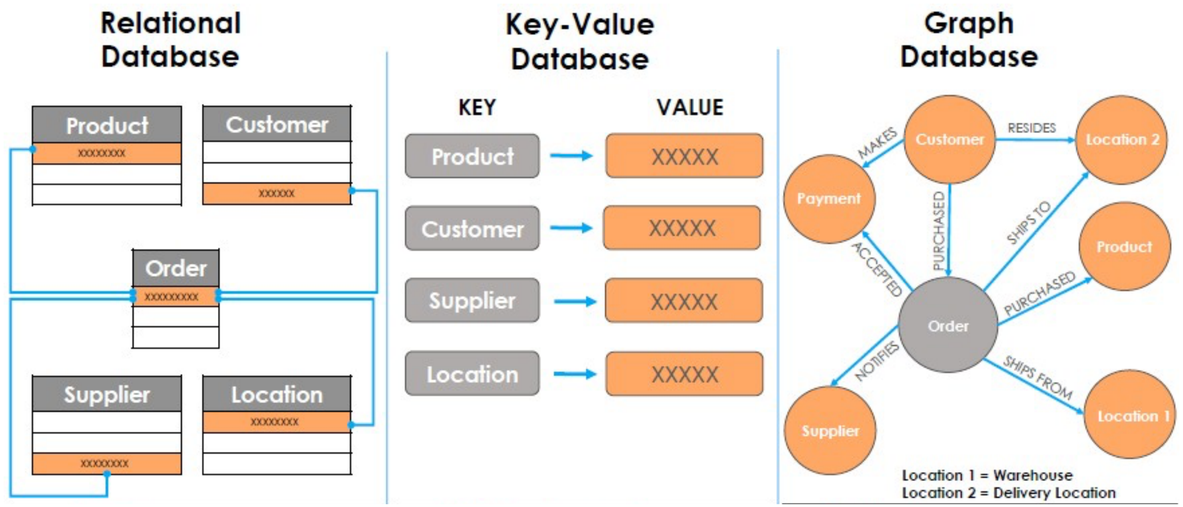
Metadata

- Ontology
  - Species name → taxonomical ID
    - Homo sapiens → 9606
  - Tissue / organ → Uberon, Plant ontology …
    - THYROID_GLAND → UBERON:0002046
    - PERICARP → PO:0009084
  - Protein name → Protein ID
    - APOA2 → ENSP00000356969

Metadata

- Ortholog
  - opisthokonta → 33154
  - primata → 9443
- eggNOG orthologs mapping
  - APOA2 → 9443.ENOG504MJTR (primate level)
  - APOA2 → 33208.ENOG503BTNZ (metazoa level)
- Publication
  - PaxDb 5.0… → PMID:37659604, 2023, Mol Cell Proteomics
- Experimental condition (free text)
  - Spectral counting, SILAC, DIA

Orthology data stored in a Graph database neo4j



e.g. SQL Redis neo4j

Search “friends of friends”…

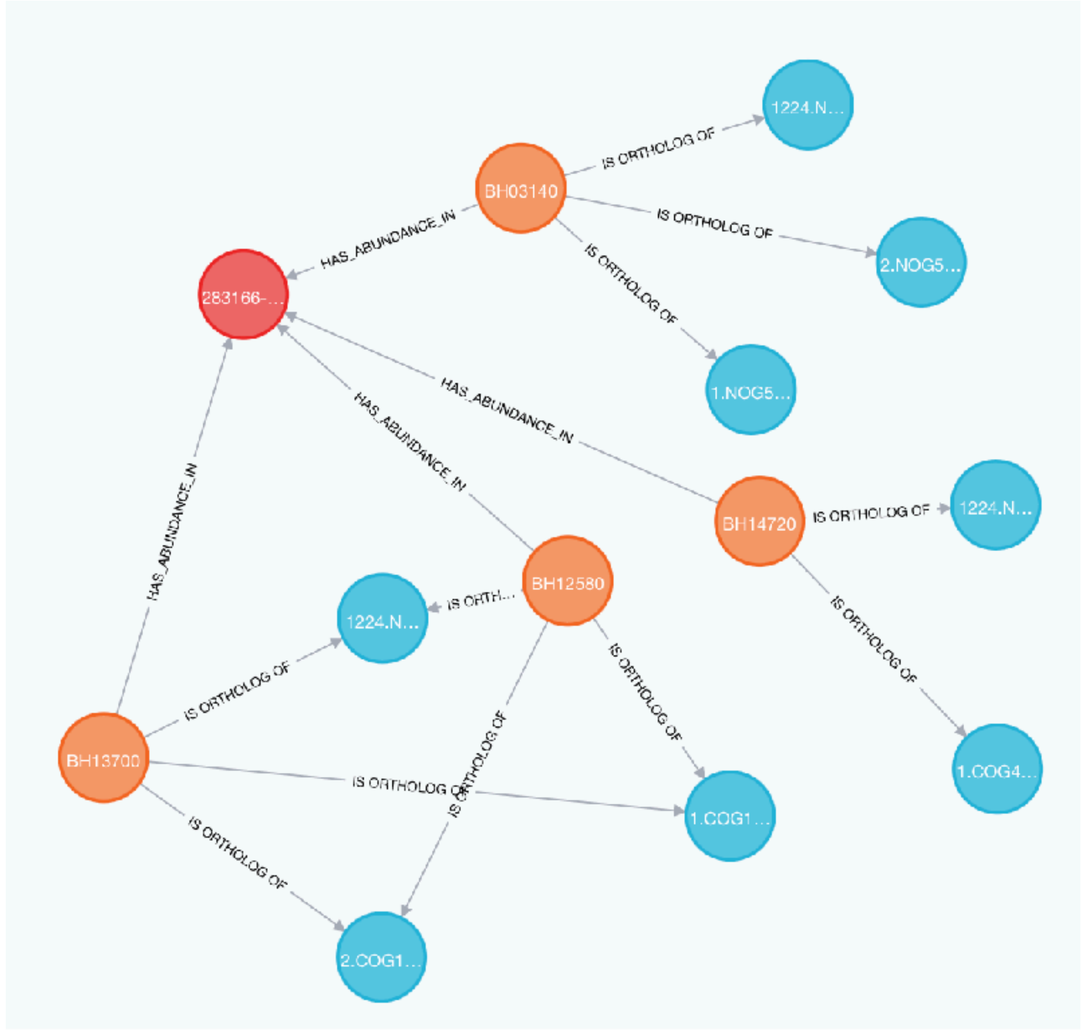
Depth	Execution Time – MySQL	Execution Time –Neo4j	Faster by…
2	0.016	0.010	60%
3	30.267	0.168	180x
4	1,543.505	1.359	1134x
5	Not Finished in 1 Hour	2.132	

Jonas Partner and Aleksa Vukotic. Neo4j in Action, 2014

Performant (comparison with SQL)

https://www.nextplatform.com/2018/09/19/the-graph-database-poised-to-pounce-on-the-mainstream/

Database structure



2,625,001 nodes and 11,824,389 edges

Protein(4) **NOG**(22) **Dataset**(1)

HAS_ABUNDANCE_IN(4) **IS ORTHOLOG OF**(11)

NOG **<id>:** 2254040 **level:** BACTERIA **levelId:** 2 **name:** 2.NOG52678

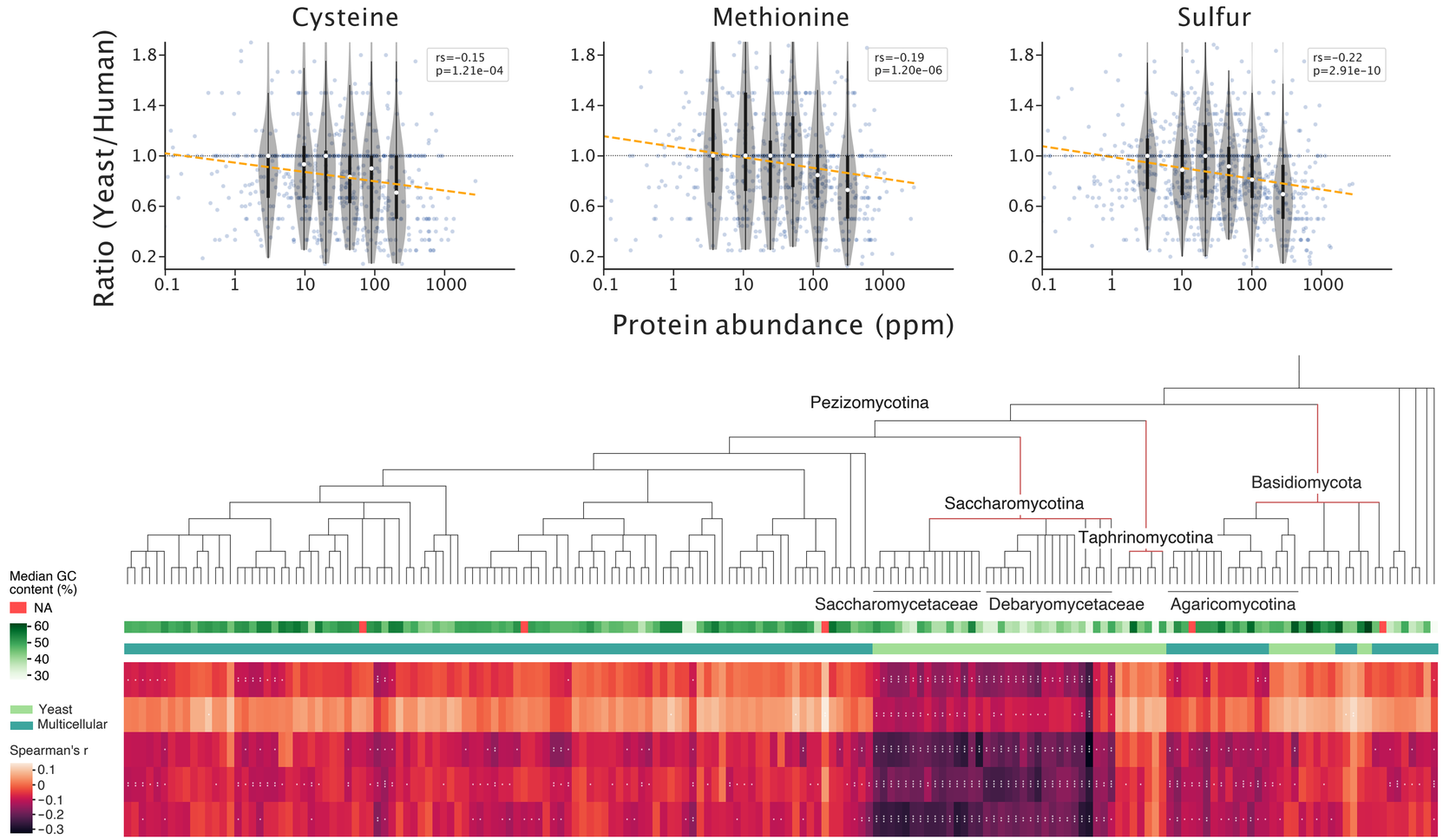
Protein **<id>:** 1087193 **eid:** 283166.BH14720 **iid:** 12091534 **name:** BH14720

Dataset **<id>:** 2624041 **coverage:** 86 **filename:** 283166-Bhenselae_Albrethsen_2013 **iid:** 534428800 **integrated:** false **organ:** WHOLE_ORGANISM **score:** 8.9

HAS_ABUNDANCE_IN **<id>:** 6874388 **ppm:** 590.0 **rank:** 306/1275

Data Use Cases

Protein abundance data reveals Evolutionary signal



Other uses

Model protein turnover and half life

Predict codon bias

Predict binding affinity

Reference for stoichiometry

Verify own proteomics experiments

Access the data

Bulk download

- https://pax-db.org/downloads/

Index of /downloads/latest/

../		
datasets/	23-Jul-2023 06:53	-
paxdb-mapped_peptides-v5.0/	14-Feb-2023 15:00	-
paxdb-orthologs-v5.0/	09-Feb-2023 22:11	-
paxdb-protein-sequences-v5.0/	14-Feb-2023 13:53	-
paxdb-uniprot-links-v5.0/	14-Feb-2023 13:57	-
paxdb-mapped_peptides-v5.0.zip	14-Feb-2023 15:04	201795666
paxdb-orthologs-v5.0.zip	14-Feb-2023 13:59	25308621
paxdb-protein-sequences-v5.0.zip	14-Feb-2023 14:04	522039256
paxdb-uniprot-crossreferences.txt	03-Jun-2024 08:04	54403288
paxdb-uniprot-links-v5.0.zip	14-Feb-2023 13:58	10860175

Access the data

Individual dataset download by filtering

Datasets

heart

Name	Tissue type	Interaction consistency score	Coverage	Download
H.sapiens - Heart (Integrated)	Heart	33.6	68%	Download
H.sapiens - Heart, Fetal, SC (Kim,nature,2014)	Heart	26.7	51%	Download
H.sapiens - Heart, SC (Wangetal,molsystbiol2019)	Heart	17.1	47%	Download
H.sapiens - Heart, SC (Peptideatlas,aug,2014)	Heart	24.2	40%	Download
H.sapiens - Heart, SC (Kim,nature,2014)	Heart	30.5	33%	Download
H.sapiens - Heart, SC (Aye,mol_bio_syst,2010)	Heart	14.2	17%	Download
H.sapiens - Heart, SC (Kline,j.proteome_res,2008)	Heart	9.3	17%	Download
H.sapiens - Heart, normalized data APEX (Aye,mol_bio_syst,2010)	Heart	13	11%	Download

Microservice APIs

Ortholog API

For human protein FABP1,

- what taxonomy groups does it belong to?
- what orthologs does it have at primate level?
- In what tissues does at least one ortholog have abundance values at primate level?
- what are the primate-level orthologs' abundances in the liver?

- https://orthologs-api.pax-db.org/protein/9606.ENSP00000295834/ortholog_groups/
- https://orthologs-api.pax-db.org/protein/9606.ENSP00000295834/ortholog_groups/PRIMATES/list_orthologs
- https://orthologs-api.pax-db.org/protein/9606.ENSP00000295834/ortholog_groups/PRIMATES/list_tissues
- https://orthologs-api.pax-db.org/protein/9606.ENSP00000295834/ortholog_groups/PRIMATES/LIVER

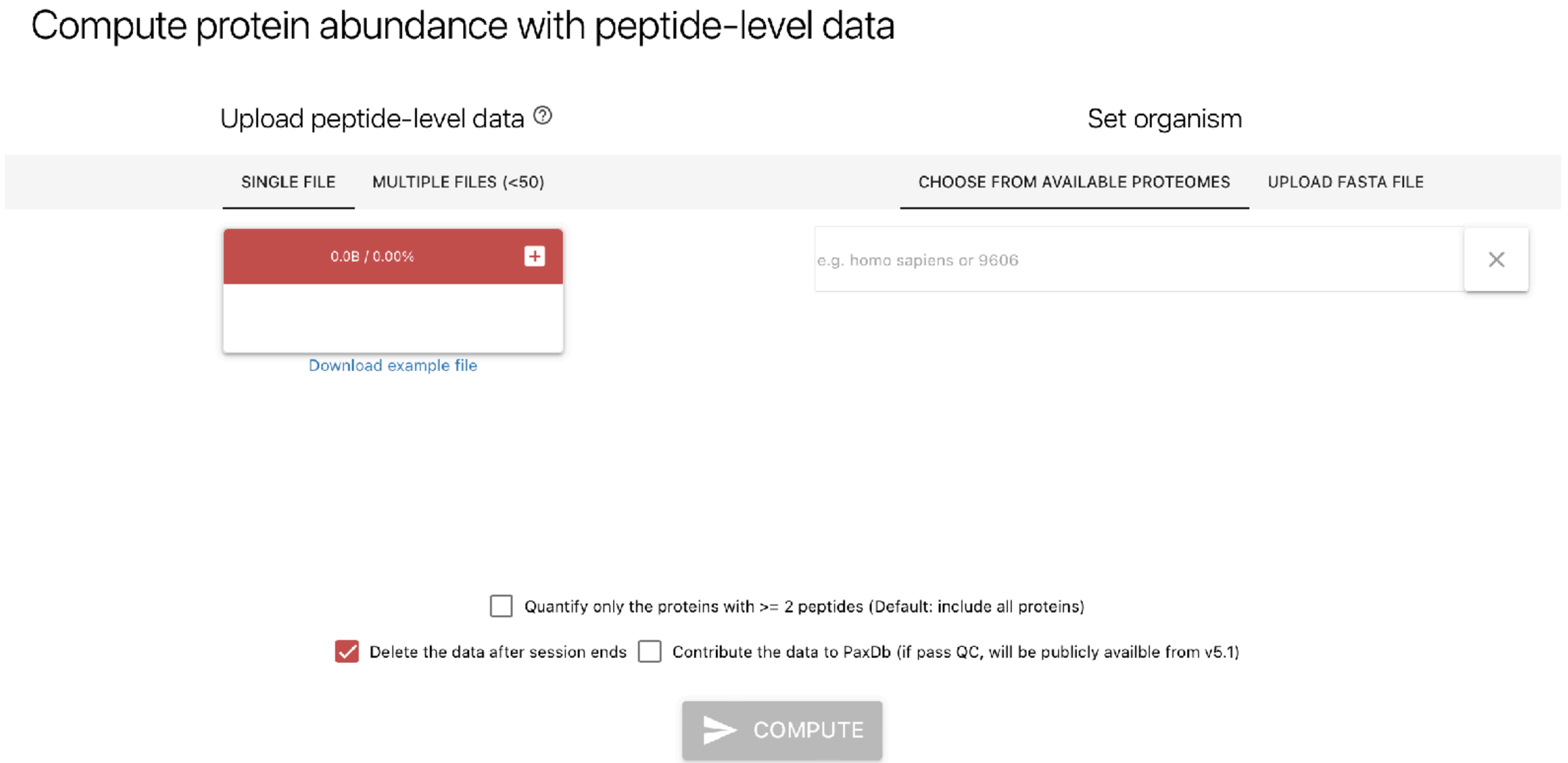
Microservice APIs

Data API

1. Show info of all datasets of *Arabidopsis thaliana*.
2. What are all the information about dataset xxx?
3. What are all protein abundance and annotation in the dataset xxx?
4. How is the protein abundance distrubution of dataset xxx?
5. Where does the protein xxx stand in the distribution?
6. What are all abundances of the protein by string ID xxx?
7. What are all abundances of the protein by Ubiprot ID xxx?
8. What are abundances of multiple proteins x, y, z … across all datasets?

1. https://api.pax-db.org/species/3702
2. https://api.pax-db.org/dataset/9606/986013392/abundances
3. https://api.pax-db.org/dataset/9606/986013392/
4. https://api.pax-db.org/dataset/986013392/histogram/
5. https://api.pax-db.org/dataset/986013392/histogram/?highlightProteinId=ENSP000003700
6. https://api.pax-db.org/protein/string/9606.ENSP00000295897
7. https://api.pax-db.org/protein/uniprot/Q851P9_ORYSJ
8. https://api.pax-db.org/proteins?ids=9606.ENSP00000269305,9606.ENSP00000258149

Upload own data



Components of an Online Bioinformatics Resource

Going Full Stack?

Components of an Online Bioinformatics Resource

A Stack to work with/through

- dedicated server or cloud storage
- own domain | institutional sub-domain or fixed address | cloud service sub-domain
  - ➡ progenetix.org, pax-db.org | mls.uzh.ch/en/research/baudis | baudisgroup.github.io
- database
  - ➡ SQL databases such as PostGres, MySQL
  - ➡ document databases such as MongoDB, CouchDB ...
  - ➡ hierarchical file system & index files
- webserver gateway for server-side generated, active content delivery
  - ➡ Perl CGI, Python, PHP ...
- active front-end (JavaScript environment)?

(Bio)informatics Skill Set

What has been needed to develop & maintain progenetix.org?

- Scripting and application development using Python, Perl and JavaScript
- Data analysis and plotting in R, Python and Perl
- Regular expressions for data entry an (programmatic) identifier matching
- JSON, YAML, tab-delimited text as file formats; some binary source files (.CEL)
- non-SQL database (MongoDB) for flexibility and document structure
- web development with Perl, Python, JS, React and Apache server; Cloudflare
- No proprietary software involved (some OpenOffice Calc / Google Sheets spreadsheets for data cleanup)

progenetix

(Bio)informatics Skill Set

What has been needed to develop & maintain progenetix.org?

GitHub

Perl

regular expressions

s/knowledge/mastery/

text mining

React

mongoDB

M↓

APACHE HTTP SERVER PROJECT

R

MkDocs

Project documentation with Markdown.

python™

HTML CSS JS

array & sequencing pipelines

(Bio)informatics Skill Set

What has been needed to develop & maintain pax-db.org?

text mining

Mass spectrometry pipelines

Last but NOT Least…

Documentation is, actually, rather important

Documentation Strategies

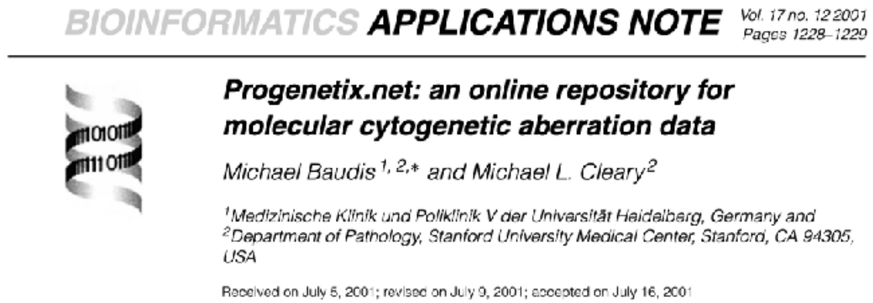
(Not so) Best Practices

```
f_d = f_d_s[c_t]
r = {}
for k in res_schema.keys():
    if k in f_d:
        r.update({k:f_d[k]})
```

- What is documentation? I'll remember this! ¯_(ツ)_/¯
- Just email me if help is needed, unexpectedly
- We had money for a chat bot.
- Clean code documents itself - Just use explicit variable/function names.
- Clean code documents itself - Never use explicit variable/function names.
- Perl POD it is. There is a command to show the notes in your terminal…
- I wrote a paper about the resource. In 2001.
- Haven't you found the GoogleGroups account?
- Documentation? StackOverflow, whelp!

mbaudis@netscape.net

```
normalize_variant_values_for_export(v, byc, drop_fields=None):
```



BIOINFORMATICS APPLICATIONS NOTE Vol. 17 no. 12 2001
Pages 1228–1229

Progenetix.net: an online repository for molecular cytogenetic aberration data

Michael Baudis[1, 2,*] and Michael L. Cleary[2]

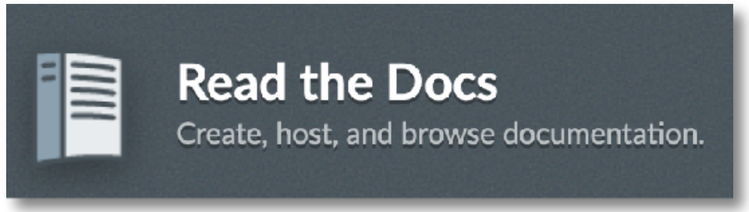
[1]Medizinische Klinik und Poliklinik V der Universität Heidelberg, Germany and [2]Department of Pathology, Stanford University Medical Center, Stanford, CA 94305, USA

Received on July 5, 2001; revised on July 9, 2001; accepted on July 16, 2001

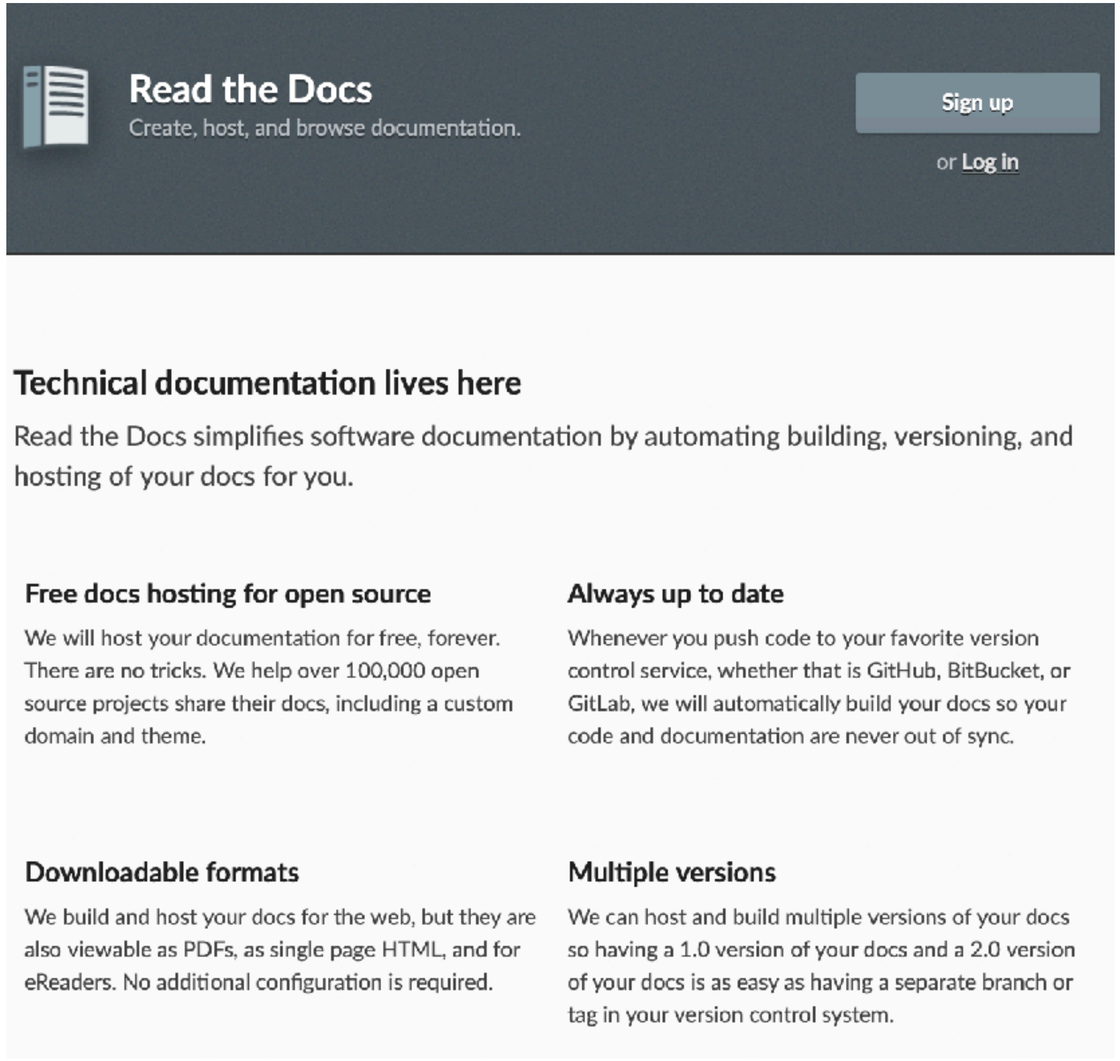
Documentation Strategies

Currently en Vogue

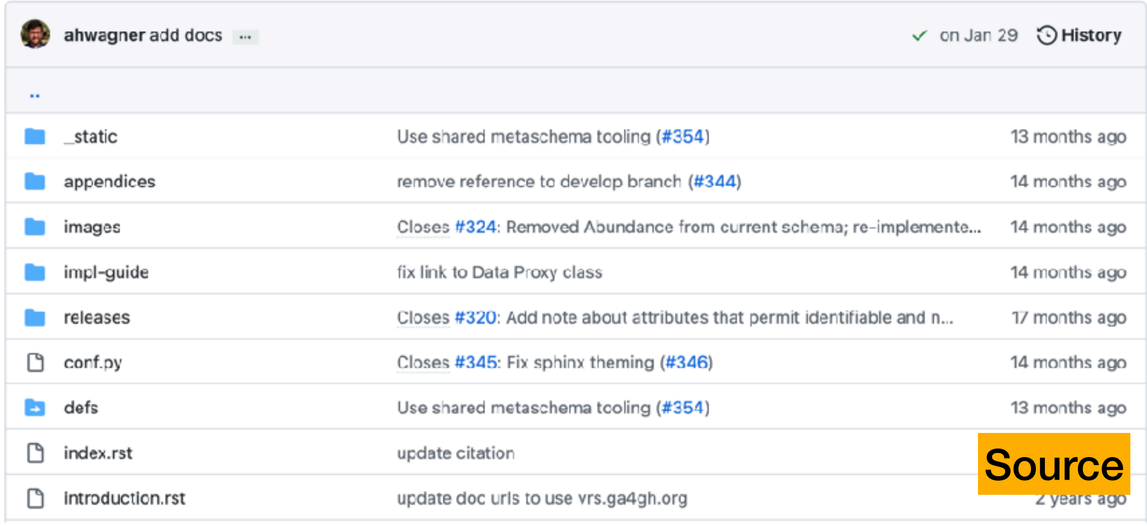
- Cloud-based documentation systems with online compilation
- written in simplified markup languages
  - ➡ Markdown (Yeah!)
  - ➡ Restructured Text (Meeh...)
- local and/or service based compilation and hosting
- build systems & output hosting
  - ➡ ReadTheDocs
    - direct building from .rst document tree or MkDocs based
  - ➡ Github Pages
    - direct using Jekyll or over MkDocs through GH actions



Documentation Strategies



Example: GA4GH Variation Representation Standard ->

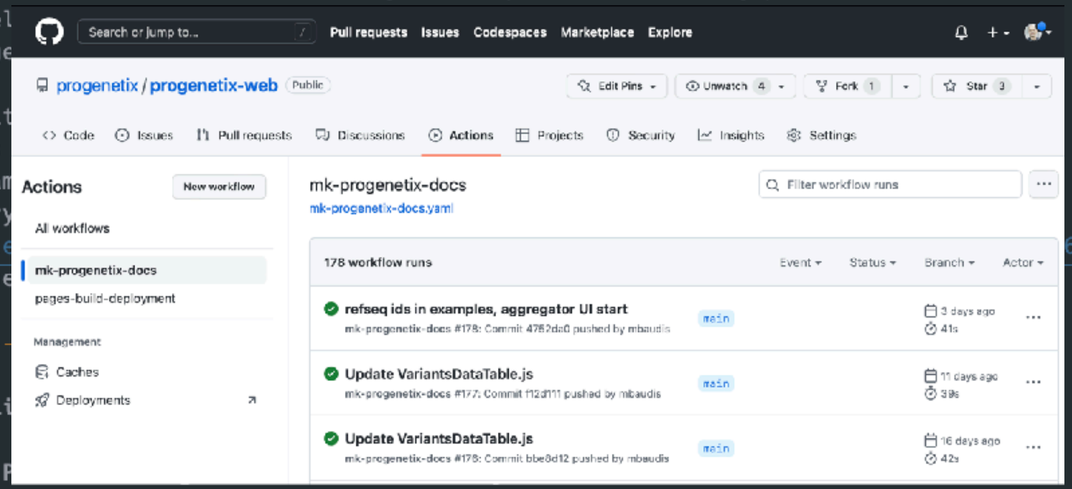


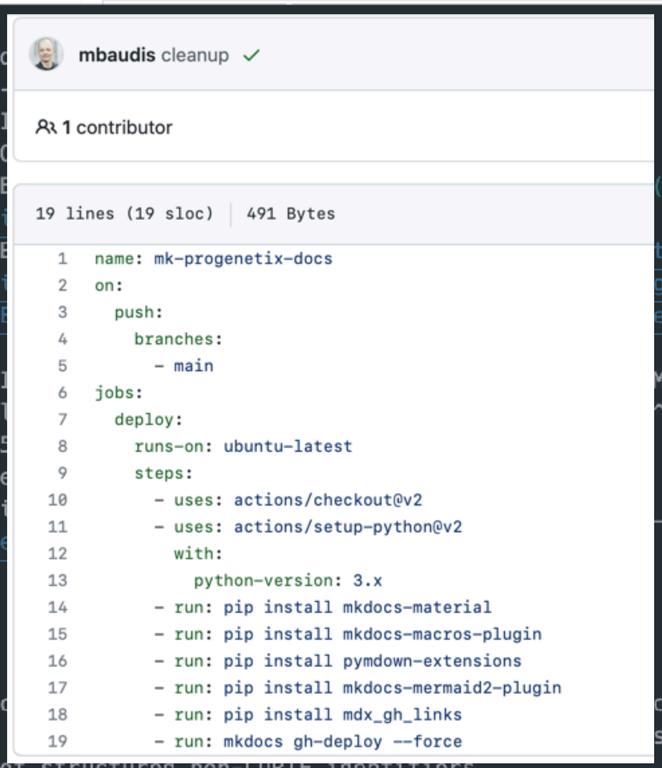
FOLDERS
progenetix-web
.github
.next
docs
css
img
javascripts
news
beaconplus.md
changelog.md
classifications-an
CNAME
index.md
progenetix-data-r
progenetix-websi
publication-collec
services.md
technical-notes.m
ui.md
use-cases.md
extra
node_modules
out
public
src
.babelrc
.env.development
.env.production
.eslintrc.json
.gitignore
.prettierrc
jest.config.js
mkdocs.yaml
next.config.js
package-lock.json
package.json
README.md
mkdocs.yaml
site_name: Progenetix Documentation
site_description: 'Documentation for the Progenetix oncogen
site_author: Michael Baudis
copyright: '© Copyright 2022, Michael Baudis and proge
repo_name: 'progenetix-web'
repo_url: https://github.com/progenetix/progenetix-web
nav:
- Documentation Home: index.md
- News & Changes: news
- Publication Collection: publication-collection
- Data Review: progenetix-data-review
- Technical Notes: technical-notes
- Progenetix Website Builds: progenetix-website-builds
- Progenetix Data ↗: http://progenetix.org
- Baudisgroup @ UZH ↗: http://info.baudisgroup.org
markdown_extensions:
- toc:
toc_depth: 2-3
permalink: true
- admonition
- attr_list
- footnotes
- md_in_html
- pymdownx.critic
- pymdownx.caret
- pymdownx.details
- pymdownx.keys
- pymdownx.magiclink:
hide_protocol: true
- pymdownx.mark
- pymdownx.tilde
- pymdownx.saneheaders
classifications-and-ontologies.md
Classifications, Ontologies and Standards
The Progenetix resource utilizes standardized diagnostic coding systems, with a move towards hierarchical ontologies. As part of the coding process we have developed and provide several code mapping resources through repositories, the Progenetix website and APIs.
Additionally to diagnostic and other clinical concepts, Progenetix increasingly uses hierarchical terms and concepts for the annotation and querying of technical parameters such as platform technologies. Overall, the Progenetix resource uses a query syntax based around the [Beacon v2 "filters"](https://beacon-project.io/v2/filters.html) concept with a [CURIE](https://www.w3.org/TR/2010/NOTE-curie-20101216/) -based syntax
MkDocs & Material for MkDocs & Github Actions
Public Ontologies with CURIE-based syntax
CURIE prefix	Code/Ontology	Examples
NCIT	NCIt Neoplasm[^1]	NCIT:C27676
HP	HPO[^2]	HP:0012209
PMID	NCBI Pubmed ID	[PMID:18810378](http://progenetix.org/services/ids/PMID:18810378)
geo	NCBI Gene Expression Omnibus[^3]	[geo:GPL6801](http://progenetix.org/services/ids/geo:GPL6801), [geo:GSE19399](http://progenetix.org/services/ids/geo:GSE19399), [geo:GSM491153](http://progenetix.org/services/ids/geo:GSM491153)
arrayexpress	EBI ArrayExpress[^4]	arrayexpress:E-MEXP-1008
cellosaurus	Cellosaurus - a knowledge resource on cell lines [^5]	cellosaurus:CVCL_1650
UBERON	Uberon Anatomical Ontology[^6]	UBERON:0000992
cbioportal	cBioPortal[^9]	[cbioportal:msk_impact_2017](http://progenetix.org/services/ids/cbioportal:msk_impact_2017)
Private filters
Since some classifications cannot directly be referenced, and in accordance with the upcoming Beacon v2 concept of "private filters", Progenetix uses additionally a set of structured non-CURIE identifiers.

Local Testing

```
→  progenetix-web git:(main) mkdocs serve
INFO     -  Building documentation...
INFO     -  [macros] - Macros arguments: {'module_name': 'main',
            'modules': [], 'include_dir': '', 'include_yaml': [],
            'j2_block_start_string': '', 'j2_block_end_string': '',
            'j2_variable_start_string': '', 'j2_variable_end_string': '',
            'on_undefined': 'keep', 'on_error_fail': False, 'verbose': False}
INFO     -  [macros] - Extra variables (config file):
            ['excerpt_separator', 'blog_list_length', 'social']
INFO     -  [macros] - Extra filters (module): ['pretty']
INFO     -  MERMAID2  - Initialization arguments: {}
INFO     -  MERMAID2  - Using javascript library (8.8.0):
               https://unpkg.com/mermaid@8.8.0/dist/mermaid.min.js
INFO     -  Cleaning site directory
INFO     -  The following pages exist in the docs directory, but are not
            included in the "nav" configuration:
              - beaconplus.md
              - changelog.md
              - classifications-and-ontologies.md
              - progenetix-data-review.md
              - progenetix-website-builds.md
              - publication-collection.md
INFO     -  MERMAID2  - Found superfences config: {'custom_fences': [{'name':
            'mermaid', 'class': 'mermaid', 'format': <function fence_mermaid at
            0x104075ab0>}]}
INFO     -  MERMAID2  - Page 'Technical Notes': found 2 diagrams, adding scripts
INFO     -  Documentation built in 0.83 seconds
INFO     -  [09:05:32] Watching paths for changes: 'docs', 'mkdocs.yaml'
INFO     -  [09:05:32] Serving on http://127.0.0.1:8000/
INFO     -  [09:05:33] Browser connected:
            http://127.0.0.1:8000/classifications-and-ontologies/
```

Web Deployment (Github)





Progenetix Documentation
Documentation Home
News & Changes
Pages & Forms
Services API
Beacon+ API & bycon
Use Case Examples
Classifications, Ontologies & Standards
Publication Collection
Data Review
Technical Notes
Progenetix Website Builds
Progenetix Data ↗
Baudisgroup @ UZH ↗

Classifications, Ontologies and Standards

The Progenetix resource utilizes standardized diagnostic coding systems, with a move towards hierarchical ontologies. As part of the coding process we have developed and provide several code mapping resources through repositories, the Progenetix website and APIs.

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List of filters recognized by different query endpoints

Public Ontologies with CURIE-based syntax

CURIE prefix	Code/Ontology	Examples
NCIT	NCIt Neoplasm[1]	NCIT:C27676

Table of contents
List of filters recognized by different query endpoints
Public Ontologies with CURIE-based syntax
Private filters
Diagnoses, Phenotypes and Histologies
NCIt coding of tumor samples
ICD coding of tumor samples
UBERON codes
Genomic Variations (CNV Ontology)
Geolocation Data
Provenance and use of geolocation data

Documentation Strategies

Best Practices

- start early
- update often
- sometimes try to follow your own guide
- balance between inline documentation & doc system
- use Markdown
- plan for contingencies
  - ➡ cloud providers disappear | cancel services | change terms

https://en.wikipedia.org/wiki/List_of_defunct_social_networking_services

https://en.wikipedia.org/wiki/List_of_search_engines#Defunct_or_acquired_search_engines